# ZERO-FREE CRITERIA FROM SIGNED LIOUVILLE ENERGIES

DEEP BHATTACHARJEE AND USHASHI BHATTACHARYA

Abstract. Signed Liouville energies characterize zero-free half-planes for the zeta function. We derive a quantitative weighted-energy certificate and its pole-order boundary cost, together with exact decomposition and Abel-obstruction results. The critical energy upper bound needed for RH and the corresponding Dirichlet GRH bounds remain unproved.

## Contents

1. The question and the revision 2
2. Finite identities and the maximal hypothesis 3
3. The maximal large sieve on a common support 6
4. Obstructions from signs and from prime data 7
5. An endpoint obstruction for Liouville itself 11
6. A signed energy and its exact boundary terms 12
7. The zero-free exponent 14
7.1. Simultaneous pole contributions to the Fejér energy 15
7.2. A two-frequency positive weight 18
8. The energy and the construction 20
9. Rounding prime weights and computing the Euler product 21
10. Summatory bounds and the exact energy exponent 23
11. The Abel zero and its leading coefficient 24
12. Quantitative limits of local Abel data 25
12.1. The unresolved upper estimate and the Mellin step 29
13. What the recent results give 29
13.1. Refinements after the seven-eighths manuscript 30
14. Proof attempts without Chowla 31
14.1. A single smooth weight that detects every right-hand zero 31
14.2. Reciprocal dilates with an explicit error rate 32

*Date*: 8 November 2026. Version 2.8.1. Research revision.

2020 *Mathematics Subject Classification.* Primary 11M26. Secondary 11N37, 11N35, 11F72.

*Key words and phrases.* Liouville function, Riemann hypothesis, Abel derivatives, Fejér energy, completely multiplicative functions.

Deep Bhattacharjee is the corresponding author. ORCID 0000-0003-0466-750X. Ushashi Bhattacharya's ORCID is 0009-0002-2254-3914.

Repository `https://github.com/creelie/Openproblemsforfieldsmedal1`. Research release v2.8.1. Version DOI 10.5281/zenodo.24045618.

14.3. Why the explicit family stops at the critical norm 33
14.4. A coefficient cost for adaptive approximants 37
14.5. A scale-dependent logarithmic taper 39
14.6. The tail and quantitative tests at a zero 41
14.7. What the quasi-Riemann proof transfers 43
15. The character version and GRH 44
16. Smoothing across Dirichlet conductors 45
16.1. Orthogonality and square divisors 47
16.2. Why increasing the modulus retains conductor one 48
17. A direct local-energy construction 48
17.1. Reconstruction with a logarithmic cost 49
17.2. The exact signed term 51
17.3. One primitive correlation and a one-sided target 52
17.4. Why a prime Gram argument does not close the estimate 54
17.5. The sharp norm of the signed dilation operator 54
17.6. An explicit finite-prime model 56
17.7. The large-prime cancellation that remains 57
18. A weighted energy certificate and its boundary cost 59
Appendix A. From a fixed zero-free half-plane to summatory cancellation 61
Appendix B. The ordinary-dilate model has a different limit 62
Appendix C. Mellin positivity and one-sided pole detection 63
Appendix D. Elementary prime bounds and exact checks 66
Appendix E. Finite Abel cancellations do not control the exponential sum 67
Appendix F. Continuity with version 1 and the role of the literature 70
F.1. The analytic context retained from the original paper 70
F.2. Funding and availability 71
F.3. Ethics and interests 71
Appendix G. Exact normalization and rounding controls 71
Appendix H. Reproduction and provenance 72
References 72

## 1. The question and the revision

Let $\lambda(n) = (-1)^{\Omega(n)}$ and $L(x) = \sum_{n \le x} \lambda(n)$, where $\Omega$ counts prime factors with multiplicity. Version 1 [10] investigated the sum $L(x)$ through the correlations $\sum_{n \le y} \lambda(n)\lambda(n+h)$. It took absolute values across shifts and maximized over $y$ separately for each shift. That operation makes a spectral estimate convenient to state. It also changes the scale one can reasonably expect the resulting energy to have.

We keep the maximal large sieve argument, with a common support for the fixed-endpoint input and an explicit logarithmic factor. The original bound would control every additive twist uniformly and imply the Dirichlet

generalized Riemann hypothesis. The bound with $\varepsilon = 0$ fails for every sign sequence. Even square-root cancellation of every additive twist and decay at every fixed lag do not give the proposed maximal estimate for a general sign sequence.

Our second approach averages squared sums over sliding intervals of length $N$ in the truncated sequence on $[1, N]$. The average has a positive Fourier representation, but its expansion in shifts is signed. We determine its growth exponent in terms of the rightmost zeta zeros. The calculation gives a criterion for a zero-free half-plane and an obstruction at any zero outside it. The remaining analytic problem is to bound the energy at exponent $1 + \varepsilon$ for Liouville. The argument does not establish that bound.

Section 17 constructs the local reconstruction kernel and the finite-prime comparison. Appendix C derives the pole constraints, and Appendix E tests the proposed transfer from Abel cancellation to exponential smoothing by an explicit bounded-coefficient construction. These new results require no external zero-free theorem. They do not give the Liouville-specific remainder estimate needed to prove RH.

## 2. Finite identities and the maximal hypothesis

Write $\mathrm{e}(t) = e^{2\pi i t}$. Throughout, a sequence $a = (a_n)$ is real with $|a_n| \le 1$, unless specified otherwise. Put

$$A_N(\alpha) = \sum_{n=1}^{N} a_n \mathrm{e}(n\alpha), \qquad C_{a,h}(Y) = \sum_{n=1}^{Y} a_n a_{n+h}.$$

Here $N, h, Y$ are integers, $h \ge 1$, and $C_{a,h}(0) = 0$. At $1 \le h < N$ let $C^*_{a,h}(N) = \max_{0 \le Y \le N-h} |C_{a,h}(Y)|$, and set it to zero at $h \ge N$. Define

$$\mathcal{E}_a(N, H) = \sum_{H < h \le 2H} C^*_{a,h}(N)^2, \qquad H = 1, 2, 4, \dots.$$

The dyadic ranges cover $h \ge 2$ once each. The shift $h = 1$ is always handled separately.

**Lemma 2.1** (Square and twist identities)**.** *For every real $\alpha$,*

$$|A_N(\alpha)|^2 = \sum_{n \le N} a_n^2 + 2\,\mathrm{Re} \sum_{h=1}^{N-1} \mathrm{e}(h\alpha) C_{a,h}(N - h). \tag{2.1}$$

*On any finite set of shifts $J$, the $\ell^2$ norm of $C^*_{a,h}(N)$ equals*

$$\max_{(Y_h)} \sup_{\sum_{h \in J} |\beta_h|^2 \le 1} \left| \sum_{h \in J} \beta_h C_{a,h}(Y_h) \right|,$$

*with $0 \le Y_h \le N - h$ for $h < N$ and the out-of-range coordinates set to zero.*

*Proof.* Group the terms in $A_N(\alpha)\overline{A_N(\alpha)}$ by the signed difference of their indices. For fixed endpoints, Cauchy–Schwarz gives the asserted dual norm,

with equality at $\beta_h = \overline{C_{a,h}(Y_h)}/\|C\|_2$ when $C \neq 0$. Each maximum over $Y_h$ is attained in a finite set, and the choices are independent. □

**Proposition 2.2** (The terminal-range form of the original criterion)**.** *Fix $0 < \eta < 1$. Suppose that, for every $\delta > 0$,*

$$\mathcal{E}_a(N,H) \ll_{\eta,\delta} N^{2+\delta}/H \quad (H \le N^{1-\eta},\ H \text{ dyadic}),$$
$$\sum_{N^{1-\eta}<h<N} C^*_{a,h}(N) \ll_{\eta,\delta} N^{1+\delta}.$$

*Then $\sup_\alpha |A_N(\alpha)| \ll_{\eta,\varepsilon} N^{1/2+\varepsilon}$ for every $\varepsilon > 0$. In particular, for $a = \lambda$ this implies RH.*

*Proof.* The term $h = 1$ is at most $N$. In a dyadic block beginning at $H \le N^{1-\eta}$, Cauchy–Schwarz gives

$$\sum_{H<h\le 2H} C^*_{a,h}(N) \le \sqrt{H}\,\mathcal{E}_a(N,H)^{1/2} \ll_{\eta,\delta} N^{1+\delta/2}.$$

There are $O(\log(2N))$ such blocks. Include the entire last block, even if its upper end exceeds $N^{1-\eta}$. All shifts not covered by these blocks or by $h = 1$ exceed that threshold, so their sum is bounded by the assumed terminal sum. Equation (2.1) therefore bounds the squared exponential sum by $O_{\eta,\delta}(N^{1+\delta/2}\log(2N) + N^{1+\delta})$. Taking $0 < \delta < \varepsilon$, enlarging the constant for bounded $N$, and taking square roots proves the claim. The implication for $\lambda$ follows from Theorem 7.1. □

The proposition retains the terminal-range formulation of version 1. Its hypotheses concern the signed Liouville sequence when that sequence is substituted; they are not established here. Theorem 2.3 below gives the stronger consequence of assuming the collar bound over the entire dyadic range.

**Theorem 2.3** (Consequence of the full maximal collar)**.** *Suppose that for every $\varepsilon > 0$,*

$$\mathcal{E}_\lambda(N,H) \ll_\varepsilon N^{2+\varepsilon}/H \qquad (1 \le H < N,\ H \text{ dyadic}). \tag{2.2}$$

*Then $\sup_{\alpha\in\mathbb{R}} |A_N(\alpha)| \ll_\varepsilon N^{1/2+\varepsilon}$ for $a = \lambda$. In particular, the generalized Riemann hypothesis holds for all primitive Dirichlet L-functions.*

*Proof.* Each dyadic block contributes at most $\sqrt{H}\,\mathcal{E}_\lambda(N,H)^{1/2} \ll_\varepsilon N^{1+\varepsilon/2}$ to the sum of the absolute values in (2.1). There are $O(\log(2N))$ blocks. The diagonal and the contribution of $h = 1$ are $O(N)$. Absorb the logarithm by choosing a smaller positive exponent in (2.2).

For a Dirichlet character $\chi$ mod $q$, use its finite Fourier expansion $\chi(n) = \sum_{r \bmod q} \widehat{\chi}(r)\mathrm{e}(rn/q)$. The normalization $\widehat{\chi}(r) = q^{-1}\sum_{b \bmod q} \chi(b)\mathrm{e}(-rb/q)$

gives $\sum_r |\widehat{\chi}(r)|^2 \le 1$. Thus $\sum_{n\le N}\lambda(n)\chi(n) \ll_\varepsilon \sqrt{q}\,N^{1/2+\varepsilon}$. Partial summation gives a holomorphic function

$$F_\chi(s) = s\int_1^\infty \left(\sum_{n\le x}\lambda(n)\chi(n)\right) x^{-s-1}\,\mathrm{d}x \quad (\operatorname{Re} s > 1/2).$$

For $\operatorname{Re} s > 1$, complete multiplicativity gives $L(s,\chi)F_\chi(s) = L(2s,\chi^2)$ by the Euler products. The identity holds meromorphically in $\operatorname{Re} s > 1/2$. The right side has a convergent, nonvanishing Euler product there. A zero of $L(s,\chi)$ in that half-plane would contradict the identity. The functional equation for primitive characters excludes nontrivial zeros to the left of the line by applying the same argument to $\bar{\chi}$. □

**Proposition 2.4** (Exact coefficient-uniform maximum)**.** *For all $N, H$,*

$$\sup_{|a_n|\le 1}\mathcal{E}_a(N,H) = \sum_{H<h\le\min(2H,N-1)}(N-h)^2.$$

*Both $a_n = 1$ and $a_n = (-1)^n$ attain the maximum.*

*Proof.* Use $|C_{a,h}(Y)| \le Y \le N-h$. For the two stated choices $a_n a_{n+h}$ is constant in $n$ and has modulus one. At $2H \le N/2$ the maximum is between $HN^2/4$ and $HN^2$. In particular, $H \asymp N^{1/4}$ realizes the power gap found in version 1. □

**Corollary 2.5** (The explicit power gap of version 1)**.** *Fix $0 < \eta < 3/4$, and let $H$ be the largest dyadic integer at most $N^{1/4}$. For sufficiently large $N$,*

$$H \le N^{1-\eta}, \qquad \mathcal{E}_{\mathbf{1}}(N,H) \ge \frac{1}{16}N^{9/4}, \qquad \frac{N^{2+1/8}}{H} \le 2N^{15/8}.$$

*Consequently no constant independent of the bounded coefficient sequence and of $N$ can make the proposed collar bound hold at $\varepsilon = 1/8$. The ratio of the displayed lower bound to the proposed scale grows at least as $N^{3/8}/32$.*

*Proof.* The choice gives $N^{1/4}/2 < H \le N^{1/4}$. The condition on $\eta$ places this block in the stated range. For large $N$, $2H \le N/2$; hence Proposition 2.4 gives $\mathcal{E}_{\mathbf{1}}(N,H) \ge HN^2/4 > N^{9/4}/8$. We weaken the constant to retain the original displayed bound. Also $N^{17/8}/H < 2N^{17/8-1/4} = 2N^{15/8}$. Subtracting the exponents gives $9/4 - 15/8 = 3/8$. □

**Lemma 2.6** (Square divisors and the signed correlation)**.** *For every positive integer $n$,*

$$\lambda(n) = \sum_{r^2\mid n}\mu(n/r^2).$$

*Consequently, for integers $Y \ge 1$ and $h \ge 1$,*

$$C_{\lambda,h}(Y) = \sum_{r\le\sqrt{Y}}\sum_{s\le\sqrt{Y+h}}\sum_{\substack{1\le n\le Y\\ r^2\mid n,\ s^2\mid n+h}}\mu(n/r^2)\mu((n+h)/s^2). \tag{2.3}$$

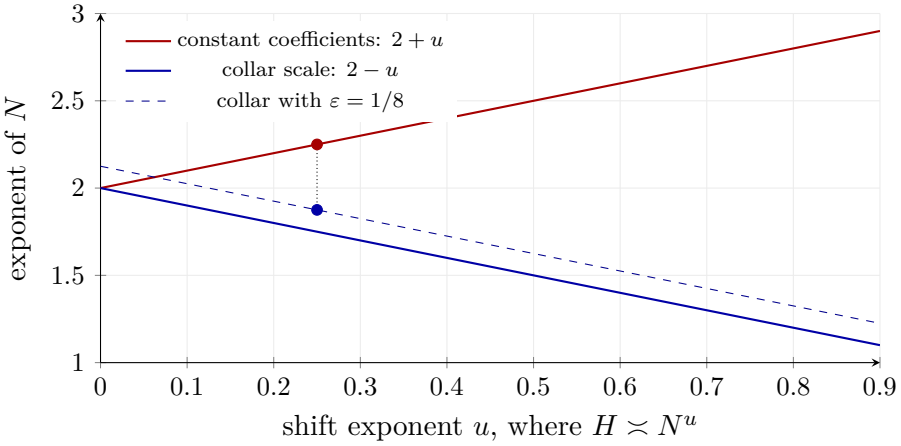



FIGURE 1. The original exponent-gap comparison, with the $\varepsilon = 1/8$ line made explicit. At $u = 1/4$ the marked exponents are $9/4$ and $15/8$. For each fixed $0 < u < 1$, the exact maximum has exponent $2 + u$. The point $u = 1$ is excluded: the block $H = N$ contains no admissible shifts.

*The inner sum is empty unless* $\gcd(r,s)^2 \mid h$. *The summand indexed by* $r = s = 1$ *is exactly* $\sum_{n \le Y} \mu(n)\mu(n+h)$.

*Proof.* Both sides of the first identity are multiplicative. At $p^e$, only $e-2j = 0$ or 1 contributes to $\sum_{2j \le e} \mu(p^{e-2j})$, giving $(-1)^e$. Insert the identity into each factor of the correlation. All sums are finite, so rearrangement is exact. The two congruences $n \equiv 0 \pmod{r^2}$, $n \equiv -h \pmod{s^2}$ are compatible exactly when $\gcd(r^2, s^2) \mid h$, by the generalized Chinese remainder theorem. This gcd is $\gcd(r,s)^2$. The final assertion follows by selecting $r = s = 1$. □

This identity preserves the multiplicative signs. The $r = s = 1$ contribution is one stratum of the exact sum; it does not approximate the whole correlation without an estimate for the other strata. Replacing each Möbius coefficient by its absolute value would instead return to the coefficient-uniform obstruction just proved.

## 3. The maximal large sieve on a common support

The maximal argument only needs a Bessel bound for all coefficient vectors on a single index set. We state it in that form. In the spectral application one uses the usual coefficient weights and the fixed-endpoint bound for the chosen spectrum, with the cutoff tied to the support of the coefficients. We make no shift-uniform assertion about short intervals placed at arbitrarily large indices.

**Theorem 3.1** (Maximal Bessel bound)**.** *Let $(\Omega, \mu)$ be a positive measure space. Suppose $\rho_1, \dots, \rho_M$ satisfy*

$$\int_\Omega \left| \sum_{n=1}^{M} b_n \rho_n \right|^2 \, \mathrm{d}\mu \le B \sum_{n=1}^{M} |b_n|^2 \quad \text{for every } b \in \mathbb{C}^M.$$

*With $K = \lceil \log_2 M \rceil$,*

$$\int_\Omega \max_{0 \le Y \le M} \left| \sum_{n \le Y} b_n \rho_n \right|^2 \, \mathrm{d}\mu \le (K+1)^2 B \sum_{n=1}^{M} |b_n|^2.$$

*Proof.* Extend by zero to length $2^K$. For each $Y$, its prefix is the disjoint union of at most one aligned interval of each dyadic length. Cauchy–Schwarz bounds the square of its sum by $K+1$ times the sum of the squared block sums. Enlarge to all aligned dyadic intervals and integrate. At each level the intervals partition the support, and the Bessel bound applies to each masked vector. The sum of the squared coefficient norms on that level is $\sum_n |b_n|^2$. There are $K+1$ levels. □

For example, take $M = 2N$ and coefficients supported on $N < n \le 2N$ in a spectral large sieve with fixed bound $B \ll_\varepsilon (T^2 + 2N)(2TN)^\varepsilon$. The maximal bound has the same power exponents, since $(1 + \log_2(2N))^2$ is absorbed by a smaller choice of $\varepsilon$ in the fixed estimate. This is the Rademacher–Menshov argument applied to a spectral inequality of the Deshouillers–Iwaniec type [11, 13]. It does not identify the coefficient vector needed for the signed Liouville problem.

**Proposition 3.2** (An additive Fourier benchmark)**.** *For any $a$ as above,*

$$\mathcal{E}_a(N, H) \le (1 + \lceil \log_2 N \rceil)^2 \|A_N\|^2_{L^\infty(\mathbb{R}/\mathbb{Z})} \sum_{n \le N} |a_n|^2.$$

*In particular Davenport's bound yields $\mathcal{E}_\lambda(N, H) \ll_A N^3/(\log(2N))^A$ for every $A > 0$, uniformly in $H$.*

*Proof.* Extend $a\mathbf{1}_{[1,N]}$ by zero to $\mathbb{Z}$. For a dyadic block $I \subset [1, 2^{\lceil \log_2 N \rceil}]$, set $a_I = a\mathbf{1}_I$. The Fourier transform of the full correlation $\sum_n a_I(n) a(n+h)$ in $h$ is $A_N \overline{\widehat{a_I}}$, up to the Fourier sign convention. Parseval bounds its squared norm by $\|A_N\|^2_\infty \sum_{n \in I} |a_n|^2$. Use the same dyadic decomposition as in Theorem 3.1, and sum over all integer $h$ to majorize the chosen shifts. For the Liouville estimate apply Davenport's uniform additive bound in the form quoted in [20], with an increased logarithmic exponent. □

## 4. Obstructions from signs and from prime data

We give two deterministic models with different properties and a probabilistic comparison. The first deterministic model retains the arithmetic data at any finite set of primes. The second retains the strength of the additive Fourier estimate that would follow from (2.2).

**Theorem 4.1** (Finite prime data)**.** *For any finite set $P$ of primes and signs $\epsilon_p$ on $P$, there exists a completely multiplicative sign function $f$ with $f(p) = \epsilon_p$ for $p \in P$ and*

$$\left|\sum_{n\le N} f(n)\right| \ll_P (1+\log N)^{|P\cup\{3\}|}.$$

*There is a positive integer $D$ depending only on $P$ such that $\mathcal{E}_f(N,H) \ge HN^2/(32D)$ whenever $2D \le H \le N/4$.*

*Proof.* Let $S = P \cup \{3\}$ and let $\chi_3$ be the nonprincipal character modulo 3. Specify the given values on $P$, use $f(3) = 1$ if $3 \notin P$, and set $f(p) = \chi_3(p)$ for $p \notin S$. Extend completely multiplicatively. Write $n = am$ with $a$ supported on $S$ and $(m, \prod_{p\in S} p) = 1$. Then $f(m) = \chi_3(m)$. The sum of $\chi_3(m)\mathbf{1}_{(m,\prod_{p\in S} p)=1}$ over a complete residue system modulo $\prod_{p\in S} p$ vanishes. The admissible residues with $m \equiv 1$ and $m \equiv 2 \pmod 3$ have the same cardinality by the Chinese remainder theorem. Each inner sum over $m \le N/a$ is therefore bounded by that period. There are at most $\prod_{p\in S}(1 + \lfloor \log N/\log p\rfloor)$ choices for $a$.

Choose $K$ with $\sum_{p\in S} p^{-K} \le 1/4$ and put $D = \prod_{p\in S} p^K$. If $D \mid h$ and $p^K \nmid n$ for all $p \in S$, then $v_p(n+h) = v_p(n)$ for all $p \in S$. The factor $a$ is unchanged and $h/a$ is divisible by 3. The two unit factors agree modulo 3, so $f(n+h) = f(n)$. Among $n \le Y$ the exceptional set has size at most $Y/4$. Thus $C_{f,h}(Y) \ge Y/2$ for $D \mid h$. There are at least $H/(2D)$ multiples of $D$ in $(H, 2H]$ under the stated conditions, and $Y = N - h \ge N/2$ for those shifts. Squaring and summing gives the bound. □

**Theorem 4.2** (Additive cancellation and fixed shifts)**.** *There is a sign sequence $r$ such that, with absolute constants,*

$$\sup_{\alpha}\left|\sum_{n\le N} r_n \mathrm{e}(n\alpha)\right| \ll \sqrt{N}, \qquad |C_{r,h}(Y)| \ll \sqrt{hY} \quad (h, Y \ge 1).$$

*Nevertheless $\mathcal{E}_r(3H, H) \ge H^2$ for every dyadic $H$.*

*Proof.* Use the Rudin–Shapiro polynomials $P_0 = Q_0 = 1$ and

$$P_{m+1} = P_m + z^{2^m} Q_m, \qquad Q_{m+1} = P_m - z^{2^m} Q_m.$$

The prefixes of the $P_m$ are consistent, and $r_n = [z^{n-1}]P_m$ once $n \le 2^m$. The complementary identity $|P_m(z)|^2 + |Q_m(z)|^2 = 2^{m+1}$ on $|z| = 1$ follows by induction. Every aligned dyadic block of length $2^j$ in the infinite sequence is the coefficient vector of $\pm P_j$ or $\pm Q_j$. This follows by taking recursive halves in a large enough $P_m$. In the aligned dyadic partition of a prefix of length $N$, each block contributes at most $2^{(j+1)/2}$, and the sum over its lengths is $O(\sqrt{N})$.

If $h > Y$, the fixed-shift estimate follows from the trivial bound $Y$. Otherwise take $B = 2^k$ with $\sqrt{hY} \le B < 2\sqrt{hY}$. Partition the starting indices

into full aligned blocks of length $2B$ and a final remainder. Each full block consists of $P_k$ and $Q_k$, up to signs. The two sums for which both indices stay in the same half cancel by the $h$-th Fourier coefficient of the complementary identity. At most $h$ starting indices cross the midpoint and at most $h$ cross the endpoint of each block. The remainder contributes at most $2B$. This bounds the sum by $2h\lfloor Y/(2B)\rfloor + 2B \ll \sqrt{hY}$.

Finally $P_{m+2}$ has four successive blocks $P_m, Q_m, P_m, -Q_m$. At $H = 2^m$, the first and third blocks give $C_{r,2H}(H) = H$, an admissible endpoint for $N = 3H$. This one term already gives the lower bound. □

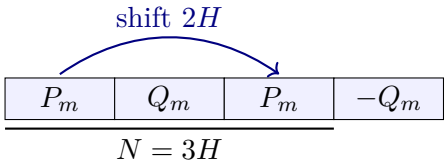



FIGURE 2. The first and third Rudin–Shapiro blocks agree entry by entry. Taking the endpoint $Y = H$ sees this agreement before the fourth block can contribute.

**Proposition 4.3** (Independent signs)**.** *For independent fair signs $\epsilon_n$, almost surely $\mathcal{E}_\epsilon(3H, H) \geq H^2/2$ for every sufficiently large dyadic $H$.*

*Proof.* Put $T_H = \sum_{H<h\leq 2H} |C_{\epsilon,h}(H)|^2$. On expanding, the diagonal gives $H^2$. The other terms are $2\epsilon_n\epsilon_m\epsilon_{n+h}\epsilon_{m+h}$ with $1 \leq n < m \leq H$. Each set of four indices determines the term uniquely: the two indices in $[1, H]$ give $n, m$, and the smaller remaining index gives $n + h$. The products for distinct sets of indices are orthogonal in probability. Thus $\mathbf{E}T_H = H^2$ and $\mathrm{Var}(T_H) = 4H\binom{H}{2} \leq 2H^3$. Chebyshev bounds the probability of $T_H < H^2/2$ by $8/H$. This is summable on the dyadic scales. Apply Borel–Cantelli and use $\mathcal{E}_\epsilon(3H, H) \geq T_H$. □

**Proposition 4.4** (Random completely multiplicative signs)**.** *Give the primes independent fair signs and extend them completely multiplicatively to $f$. Then almost surely*

$$\mathcal{E}_f(3H, H) \gg H^2/\log H$$

*for infinitely many dyadic $H$.*

*Proof.* Condition on all prime signs up to $2H$. A composite integer at most $3H$ has all prime factors at most $3H/2$. Thus the only fresh signs in the vector $C_{f,h}(H)$, $H < h \leq 2H$, are those of primes $p \in (2H, 3H]$ in the second factor. Write the vector as $u + \sum_p A_p\epsilon_p$, where $u$ is fixed under the conditioning and $A_p(h) = f(p - h)$ if $1 \leq p - h \leq H$, and zero otherwise. Let $T = \sum_p \|A_p\|_2^2$. Each prime in $(2H, 5H/2]$ contributes at least $H/2$ nonzero coordinates for dyadic $H \geq 2$, so the prime number theorem gives $T \gg H^2/\log H$.

For $Z = \|u + \sum_p A_p \epsilon_p\|_2^2$, the conditional mean is $m = \|u\|_2^2 + T$. Orthogonality of the independent linear and quadratic sign monomials gives

$$\mathrm{Var}(Z) = 4\sum_p |\langle u, A_p\rangle|^2 + 4\sum_{p<q} |\langle A_p, A_q\rangle|^2 \le 4\|u\|_2^2 T + 2T^2 \le 2m^2.$$

Since $\mathbf{E}[Z\mathbf{1}_{Z\ge m/2}] \ge m/2$, Cauchy–Schwarz and $\mathbf{E}[Z^2] \le 3m^2$ give conditional probability at least $1/12$ of $Z \ge m/2 \ge T/2$. All earlier dyadic events depend only on primes at most $3H/2$, hence are measurable under the conditioning at scale $H$. By iterative conditioning, the probability of no successes in any block of $r$ consecutive large dyadic scales is at most $(11/12)^r$. Let $r \to \infty$ and take the countable union over starting scales. The conclusion follows because $\mathcal{E}_f(3H, H) \ge Z$. □

The complementary identities used in Theorem 4.2 belong to the classical Rudin–Shapiro construction. Peak aperiodic correlations of the full polynomials have a separate literature [15]. The prefix maximum in $\mathcal{E}_a$ is essential to the block example.

**Theorem 4.5** (A deterministic logarithmic obstruction)**.** *For every sign sequence $a$ and every $B \ge 0$, the number of dyadic $H$ with*

$$\mathcal{E}_a(3H, H)/H \le B$$

*is $O(B+1)$, with an absolute constant. Consequently $\mathcal{E}_a(3H,H)/H \to +\infty$ along all dyadic $H$. In each range $J/2 \le j \le J$ of dyadic exponents, a positive proportion of the $H = 2^j$ satisfy $\mathcal{E}_a(3H,H) \gg H\log H$, uniformly for sufficiently large $J$.*

*Proof.* Restrict first to one residue class of $j$ modulo 3, so distinct scales increase by a factor of at least 8. Set $M = \lceil 64(B+1)\rceil$. Suppose there are $K = 2M$ scales $H_i \ge M$, numbered in increasing order, with the stated bound. Put $I_i = (3H_i/2, 7H_i/4] \cap \mathbb{Z}$. Since $H_i$ is divisible by 4, $|I_i| = H_i/4$. For $n \in I_i$ let $v_n = (a_{n+1}, \dots, a_{n+M})^{\mathsf{T}}$ and form

$$B_i = |I_i|^{-1} \sum_{n\in I_i} v_n v_n^{\mathsf{T}}.$$

These $M \times M$ matrices are positive semidefinite with trace $M$, so $\mathrm{tr}(B_i^2) \le M^2$. For $i < j$, $n \in I_i$ and $m \in I_j$, the gap $h = m - n$ lies in $(H_j, 2H_j]$, using $H_i \le H_j/8$. Also $m + M \le 3H_j$. Thus

$$|v_n^{\mathsf{T}} v_m| = |C_{a,h}(n+M) - C_{a,h}(n)| \le 2C^*_{a,h}(3H_j).$$

At a fixed $h$ there are at most $|I_i|$ pairs $(n,m)$. On expanding the cross trace this gives $\mathrm{tr}(B_iB_j) \le 4\mathcal{E}_a(3H_j, H_j)/|I_j| \le 16B$. On the other hand, Cauchy–Schwarz on the $M$ eigenvalues of $\sum_i B_i$ gives $\mathrm{tr}((\sum_i B_i)^2) \ge K^2M$. The diagonal and cross bounds imply

$$K^2M \le KM^2 + 16BK(K-1).$$

With $K = 2M$, this forces $M \le 32B$, a contradiction.

There are therefore fewer than $2M$ such scales above $M$ in each residue class. The number of omitted smaller scales is $O(1+\log M)$. This proves the bound on the number of exceptions. To obtain the last claim take $B = cJ$ with a sufficiently small absolute $c > 0$, and apply the bound to the $J/2 + O(1)$ scales in the range. □

This argument applies to $\lambda$ without a correlation theorem. It also proves that a fixed bound on the normalized collar cannot hold along any infinite sequence of dyadic scales for any sign sequence. The next result records an exact weighted second-moment asymptotic for the actual Liouville function.

## 5. An endpoint obstruction for Liouville itself

The examples in Section 4 concern other sequences. To obtain a statement about $\lambda$ itself we use Tao's logarithmically averaged two-point Chowla theorem [25]. We need only the consequence

$$\sum_{t\le T} \frac{\lambda(t)\lambda(t+d)}{t} = o_d(\log T) \qquad (d \ge 1 \text{ fixed}). \tag{5.1}$$

**Theorem 5.1** (Failure of the absolute endpoint)**.** *The Liouville collar satisfies*

$$\lim_{J\to\infty} \frac{1}{J} \sum_{j=0}^{J-1} \frac{\mathcal{E}_\lambda(3\cdot 2^j, 2^j)}{2^j} = +\infty.$$

*In particular, $\mathcal{E}_\lambda(3H,H)/H$ is unbounded along dyadic $H$.*

*Proof.* Fix an integer $M \ge 1$. In

$$|C_{\lambda,h}(M)|^2 = \sum_{1\le n,m\le M} \lambda(n)\lambda(m)\lambda(n+h)\lambda(m+h)$$

the diagonal contributes $M$. For $n < m$, changing $t = h + n$ changes the harmonic weights $1/h$ to $1/t$ with total error $O_M(1)$, since $\sum_{h\ge 1} n/(h(h+n)) < \infty$. Changing the endpoints costs $O_M(1)$. By (5.1), the logarithmic mean of this off-diagonal term vanishes. The terms with $m < n$ have the same treatment. Hence

$$\sum_{h\le T} \frac{|C_{\lambda,h}(M)|^2}{h} = M\log T + o_M(\log T). \tag{5.2}$$

When $H \ge M$ and $H < h \le 2H$, the endpoint $M$ is admissible at $N = 3H$, and $1/h \le 1/H$. Thus the contribution of that block to (5.2) is at most $\mathcal{E}_\lambda(3H,H)/H$. Sum over all dyadic $H < 2^J$, omitting the finitely many with $H < M$ and the initial $h = 1$. After dividing by $J$, the lower limit of the displayed average is at least $M\log 2$. Since $M$ was arbitrary, the limit is $+\infty$. □

The theorem disproves the estimate with a fixed constant at exponent $\varepsilon = 0$. The argument takes $M$ fixed before $J \to \infty$ and supplies no positive

power of $H$. One can see the role of uniformity directly from the matrix in (5.2).

**Proposition 5.2** (A power saving in ordinary correlations would obstruct the collar)**.** *Suppose there are $\delta, \eta > 0$ and $K < \infty$ such that for all sufficiently large dyadic $H$ and all $1 \le n < m \le H^\eta$,*

$$\left| \sum_{H<h\le 2H} \lambda(h+n)\lambda(h+m) \right| \le KH^{1-\delta}.$$

*For every $0 < \kappa < \min(\delta, \eta, 1)$, one then has $\mathcal{E}_\lambda(3H, H) \gg H^{1+\kappa}$ for large dyadic $H$. Thus (2.2) would fail for $\varepsilon < \kappa$.*

*Proof.* Use $M = \lfloor H^\kappa \rfloor$ and expand $\sum_{H<h\le 2H} |C_{\lambda,h}(M)|^2$. The diagonal is $HM$, and the absolute value of the sum of off-diagonal terms is at most $KM(M-1)H^{1-\delta}$. Its ratio to $HM$ tends to zero. Use $M \le H$ to bound the collar below by the resulting second moment. □

For comparison, ordinary two-point Chowla at every *fixed* shift would imply $\mathcal{E}_\lambda(3H, H)/H \to \infty$ without Cesàro averaging. For fixed $M$, the same expansion and subtraction of partial sums at $H$ and $2H$ give

$$\frac{1}{H} \sum_{H<h\le 2H} |C_{\lambda,h}(M)|^2 \to M.$$

This conclusion uses the stated ordinary correlation input. The recent OpenAI manuscript [24] gives fixed-affine-form estimates with logarithmic savings. Such savings do not supply the uniform power input in Proposition 5.2. Theorem 5.1 uses only the published logarithmic theorem [25].

## 6. A signed energy and its exact boundary terms

Extend the truncation $a_1, \dots, a_N$ by zero to the integers, and write

$$S_{N,H}(t) = \sum_{t<n\le t+H} a_n \mathbf{1}_{[1,N]}(n), \quad \mathcal{Q}_a(N, H) = \frac{1}{H} \sum_{t\in\mathbb{Z}} |S_{N,H}(t)|^2.$$

The sum over $t$ has finite support. The factor $H^{-1}$ normalizes the diagonal. Here $H \ge 1$ is any integer. We distinguish this $H$, the window length, from a dyadic shift range in $\mathcal{E}_a$.

**Proposition 6.1** (Fejér identity)**.** *Let $F_H(\alpha) = H^{-1} |\sum_{j=1}^{H} \mathrm{e}(j\alpha)|^2$. Then*

$$\mathcal{Q}_a(N, H) = \int_0^1 |A_N(\alpha)|^2 F_H(\alpha)\, \mathrm{d}\alpha \tag{6.1}$$

$$= \sum_{n\le N} a_n^2 + 2 \sum_{h=1}^{\min(H-1, N-1)} (1 - h/H) C_{a,h}(N-h). \tag{6.2}$$

*For $L_a(j) = \sum_{n\le j} a_n$ one has*

$$\mathcal{Q}_a(N, H) \ge \frac{H}{N+H-1} |L_a(N)|^2. \tag{6.3}$$

*If $H \geq N$, one has the exact formula*

$$\mathcal{Q}_a(N,H) = \frac{H-N+1}{H}|L_a(N)|^2 + \frac{1}{H}\sum_{j=1}^{N-1}\bigl(|L_a(j)|^2 + |L_a(N)-L_a(j)|^2\bigr). \tag{6.4}$$

*Proof.* Two indices $m,n \in [1,N]$ lie together in exactly $(H-|m-n|)_+$ integer windows, where $u_+ = \max(u,0)$. Expand the squares and group by $m-n$. This gives (6.2), and the same weights are the Fourier coefficients of $F_H$, proving (6.1). There are at most $N+H-1$ windows meeting $[1,N]$. Each index lies in $H$ windows, so $\sum_t S_{N,H}(t) = HL_a(N)$. Cauchy–Schwarz proves (6.3). For $H \geq N$, exactly $H-N+1$ windows contain $[1,N]$, and each proper nonempty prefix and suffix occurs in one window. This gives (6.4). □

At $H = N$, let $Q_a(N) = \mathcal{Q}_a(N,N)$. Completing the square in (6.4) gives

$$Q_a(N) = \frac{N+1}{2N}|L_a(N)|^2 + \frac{2}{N}\sum_{j=1}^{N-1}|L_a(j) - \tfrac{1}{2}L_a(N)|^2. \tag{6.5}$$

In particular, $\frac{1}{2}|L_a(N)|^2 \leq Q_a(N) \leq 5\max_{j\leq N}|L_a(j)|^2$, with the upper bound following directly from (6.4). The two inequalities compare a single positive energy to the sum at its own endpoint and to the maximal prefix sum. They show why the square-root-scale Fejér question is neither an unsigned collar estimate nor a positivity statement alone.

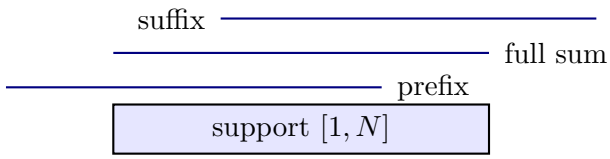



FIGURE 3. Windows of length $N$ in the zero-extended sequence. The boundary terms in (6.4) are exactly one copy of each proper prefix and suffix.

For independent fair signs, $\mathbf{E}Q_\epsilon(N) = N$ by (6.2), although the same model violates (2.2). For alternating signs, $Q_a(N) = 1$ for every $N$. When $N$ is odd the full window contributes 1, and each proper prefix and its complementary suffix contribute 1 between them. When $N$ is even the full window vanishes, and exactly $N/2$ pairs of odd prefix and suffix contribute 2 each. For constant signs the boundary formula gives $Q_{\mathbf{1}}(N) = (N^2 + 2\sum_{j<N} j^2)/N = (2N^2+1)/3$. Thus alternating and constant signs have the same maximal absolute energy and different orders of signed energy.

One may shorten the window at a known cost. If $H = \lfloor N^\alpha \rfloor$ for $0 < \alpha \leq 1$ and $\mathcal{Q}_\lambda(N,H) \ll_\varepsilon N^{1+\varepsilon}$, then (6.3) implies $L(N) \ll_\varepsilon N^{1-\alpha/2+\varepsilon}$. A sequence of such estimates with $\alpha$ arbitrarily close to 1 would suffice for RH. Estimates

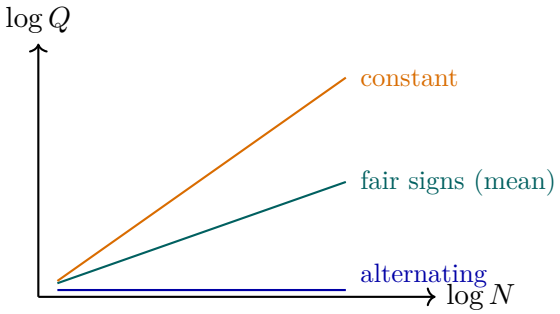



FIGURE 4. Orders of the signed energy in the three models. The slopes are $0, 1, 2$, respectively, from the exact formulas. The lines show the exponents schematically.

confined to windows much shorter than $N$ do not give that conclusion from this inequality.

## 7. The zero-free exponent

We write $Q_\lambda(N) = \mathcal{Q}_\lambda(N, N)$ and use $\theta$ for a real number in $[1/2, 1]$.

**Theorem 7.1** (Equivalent zero-free bounds)**.** *The following conditions are equivalent.*

(1) $\zeta(s) \neq 0$ *for* $\operatorname{Re} s > \theta$.
(2) $L(x) \ll_{\theta,\varepsilon} x^{\theta+\varepsilon}$ *for every* $\varepsilon > 0$.
(3) $Q_\lambda(N) \ll_{\theta,\varepsilon} N^{2\theta+\varepsilon}$ *for every* $\varepsilon > 0$.

*In particular, the third condition at* $\theta = 1/2$ *is equivalent to RH.*

*Proof.* The inequalities following (6.5) give the equivalence of the second and third conditions by rescaling $\varepsilon$. For the implication from the second to the first, partial summation defines

$$F(s) = s \int_1^\infty L(x) x^{-s-1} \, \mathrm{d}x$$

holomorphically on $\operatorname{Re} s > \theta$. The convergence is locally uniform by taking $\varepsilon$ smaller than the distance of a compact set from the boundary. For $\operatorname{Re} s > 1$, $F(s) = \sum_{n \geq 1} \lambda(n) n^{-s} = \zeta(2s)/\zeta(s)$. This follows from the local factor $(1 + p^{-s})^{-1} = (1 - p^{-s})/(1 - p^{-2s})$. The identity $\zeta(s)F(s) = \zeta(2s)$ extends meromorphically. The right side has no zero on $\operatorname{Re} s > \theta$. This excludes a zero of the factor $\zeta(s)$ there.

The reverse implication is classical. Appendix A supplies the reciprocal growth bound and the contour calculation with their dependence on the fixed distance from $\operatorname{Re} s = \theta$. At $\theta = 1$ the conclusion about $L(x)$ is trivial. At $\theta = 1/2$, apply the functional equation to exclude nontrivial zeros to the left of the line. □

**Corollary 7.2** (Exact growth exponent)**.** *Let $\Theta$ be the supremum of the real parts of the nontrivial zeros of $\zeta$. Then*

$$\limsup_{N\to\infty} \frac{\log(1+Q_\lambda(N))}{\log N} = 2\Theta.$$

*Proof.* The functional equation and the existence of nontrivial zeros give $1/2 \le \Theta \le 1$. Theorem 7.1 at $\theta = \Theta$ gives the upper bound. If $\Theta > 1/2$ and the limit superior were strictly smaller than $2\Theta$, choose $\theta \in [1/2, \Theta)$ strictly above half that limit superior. Theorem 7.1 would exclude all zeros to the right of $\theta$, contrary to the definition of $\Theta$. If $\Theta = 1/2$ and the limit superior were less than 1, then (6.5) would give $L(x) \ll x^\sigma$ with $\sigma < 1/2$. The Mellin integral would be holomorphic near $1/2$, whereas the identity $\zeta(s)F(s) = \zeta(2s)$ has a pole there on the right side and no pole on the left side. This is impossible. □

**Corollary 7.3** (A boundary constraint at the square-root scale)**.** *If $Q_\lambda(N) \ll N(\log(2N))^B$ for some $B \ge 0$, then RH holds and every nontrivial zero has multiplicity at most $1 + B/2$. In particular, if $B < 2$ all nontrivial zeros are simple.*

*Proof.* Theorem 7.1 gives RH. The same bound and (6.5) give $L(x) \ll x^{1/2}(\log(2x))^{B/2}$. At a nontrivial zero $\rho = 1/2 + i\gamma$, the numerator $\zeta(2\rho)$ is finite and nonzero by the classical nonvanishing on $\operatorname{Re} s = 1$ and $\gamma \neq 0$. For $u \downarrow 0$, the Mellin integral at $s = \rho + u$ is $O_\rho(u^{-1-B/2})$ after the change of variable $v = \log x$. A zero of multiplicity $m$ gives a pole of order $m$ in $F$ at $\rho$, which forces $m \le 1 + B/2$. □

7.1. **Simultaneous pole contributions to the Fejér energy.** The signed Fejér energy also gives quantitative information at a fixed exponent. The following bound records the contributions of any finite set of poles on the same vertical line.

**Proposition 7.4** (A sum of pole contributions)**.** *Put*

$$M(s) = \frac{\zeta(2s)}{s\zeta(s)}, \qquad K(s) = 1 + \frac{2}{1+2\operatorname{Re} s} - 2\operatorname{Re}\frac{1}{1+s}.$$

*Fix $\beta \ge 1/2$ and an integer $r \ge 1$. For a finite set $\mathcal{P}$ of distinct poles $\rho = \beta + i\gamma$ of $M$ of order exactly $r$, write $A_\rho = \lim_{s\to\rho}(s-\rho)^r M(s)$. Then*

$$\limsup_{N\to\infty} \frac{Q_\lambda(N)}{N^{2\beta}(\log N)^{2r-2}} \ge \sum_{\rho\in\mathcal{P}} K(\rho) \left|\frac{A_\rho}{(r-1)!}\right|^2. \tag{7.1}$$

*If the limit superior is finite, $M$ has no pole in $\operatorname{Re} s > \beta$ and every pole on $\operatorname{Re} s = \beta$ has order at most $r$.*

*Proof.* Take $L(X) = 0$ on $0 < X < 1$. In the complex Hilbert space $\mathcal{H} = L^2((0,1);\mathbb{C}^2)$, put

$$U_X(u) = (L(uX), L(X) - L(uX)), \qquad w_s(u) = (u^s, 1-u^s).$$

For an integer $N$, the value $L(uN) = L(j)$ occupies the interval $[j/N, (j+1)/N)$ for $0 \le j < N$. By (6.4), this gives $\|U_N\|_{\mathcal{H}}^2 = Q_\lambda(N)$. For $X = N + a$, $0 \le a < 1$, the substitution $v = uX$ gives the exact interpolation

$$\|U_X\|_{\mathcal{H}}^2 = \frac{NQ_\lambda(N) + a|L(N)|^2}{X}. \tag{7.2}$$

Since $|L(N)|^2 \le 2Q_\lambda(N)$ by (6.5), the numerator in (7.1) can be replaced by $\|U_X\|_{\mathcal{H}}^2$ and the limit superior taken over real $X$, with the denominator at $X$. Denote the resulting limit superior by $C$.

For $\operatorname{Re} s > 1$, $\|U_X\|_{\mathcal{H}} \ll X$ ensures absolute convergence of the Hilbert space integral

$$\int_1^\infty U_X X^{-s} \frac{\mathrm{d}X}{X} = M(s) w_s. \tag{7.3}$$

For the first coordinate, substitute $v = uX$, and use $L(v) = 0$ below 1. For the second, subtract the first coordinate from the transform of $L(X)$. The scalar transform of $L$ is $M(s)$ by partial summation and the Euler product. The map $s \mapsto w_s$ is holomorphic into $\mathcal{H}$ on $\operatorname{Re} s > -1/2$, since on compact subsets the derivatives of $u^s$ are dominated in $L^2$ by a power of $u$ times a power of $|\log u|$. Direct integration gives $\|w_s\|_{\mathcal{H}}^2 = K(s) > 0$ there.

If $C = \infty$, (7.1) is immediate. Otherwise put $F(t) = e^{-\beta t} U_{e^t}$ for $t \ge 0$. The definition of $C$ gives $\|F(t)\|_{\mathcal{H}} \ll 1 + t^{r-1}$. Consequently the integral in (7.3) is holomorphic for $\operatorname{Re} s > \beta$. Meromorphic continuation identifies it with the displayed right side there. As $w_s \neq 0$, this excludes poles of $M$ in that open half-plane. At a boundary point $\rho$, the same bound on $F$ gives $\|M(\rho + \delta) w_{\rho+\delta}\|_{\mathcal{H}} = O(\delta^{-r})$ as $\delta \downarrow 0$, which excludes a pole of order greater than $r$.

For each $\delta > 0$, equip the space of $\mathcal{H}$-valued functions on $[0, \infty)$ with the $L^2$ norm for the measure $e^{-\delta t}\,\mathrm{d}t$. Denote its inner product by $\langle \cdot, \cdot \rangle_\delta$, linear in its first argument, and put $G_r = \Gamma(2r-1)$. For every $\varepsilon > 0$, the bound $\|F(t)\|_{\mathcal{H}}^2 \le (C + \varepsilon) t^{2r-2}$ holds for all sufficiently large $t$. Its integral over a fixed bounded interval remains bounded as $\delta \downarrow 0$. Thus Euler's gamma integral gives

$$\limsup_{\delta \downarrow 0} \delta^{2r-1} \|F\|_\delta^2 \le C G_r. \tag{7.4}$$

For each $\rho = \beta + i\gamma \in \mathcal{P}$, take the test function

$$g_{\rho,\delta}(t) = \frac{\delta^{r-1/2}}{\sqrt{G_r}} t^{r-1} e^{i\gamma t}\, v_\rho, \qquad v_\rho = \frac{w_\rho}{\|w_\rho\|_{\mathcal{H}}}.$$

Its norm is one. For $\rho \neq \rho'$, the magnitude of $\langle g_{\rho,\delta}, g_{\rho',\delta} \rangle_\delta$ is at most $(\delta / |\delta - i(\gamma - \gamma')|)^{2r-1}$, again by the gamma integral. Hence the largest eigenvalue of this finite Gram matrix tends to one. By (7.3),

$$\langle F, g_{\rho,\delta} \rangle_\delta = \frac{\delta^{r-1/2}}{\sqrt{G_r}} \left\langle (-1)^{r-1} \frac{\mathrm{d}^{r-1}}{\mathrm{d}s^{r-1}} [M(s) w_s] \Big|_{s=\rho+\delta}, v_\rho \right\rangle_{\mathcal{H}}.$$

The signed derivative of the leading Laurent term $A_\rho w_\rho (s-\rho)^{-r}$ is

$$G_r A_\rho w_\rho (s-\rho)^{-(2r-1)}/(r-1)!.$$

Consequently

$$\lim_{\delta \downarrow 0} \delta^{r-1/2} \langle F, g_{\rho,\delta} \rangle_\delta = \sqrt{G_r}\, \frac{A_\rho}{(r-1)!} \|w_\rho\|_{\mathcal{H}}.$$

The sum of the squared scalar products with the tests is at most $\|F\|_\delta^2$ times their largest Gram eigenvalue. Indeed the operator sending coordinate vectors to the tests has squared norm equal to that eigenvalue, and the analysis map taking scalar products is its adjoint. Multiply by $\delta^{2r-1}$, let $\delta \downarrow 0$ using (7.4), and divide by $G_r$. This proves (7.1). □

**Corollary 7.5** (A critical-line lower bound)**.** *For every finite set $\mathcal{Z}$ of simple critical-line zeros of $\zeta$,*

$$\limsup_{N\to\infty} \frac{Q_\lambda(N)}{N} \geq \frac{2}{3|\zeta(1/2)|^2} + \sum_{\rho\in\mathcal{Z}} \left(2 - 2\,\mathrm{Re}\,\frac{1}{1+\rho}\right) \left|\frac{\zeta(2\rho)}{\rho\zeta'(\rho)}\right|^2. \tag{7.5}$$

*The left side is infinite if RH fails or if any critical-line zero has multiplicity greater than one.*

*Proof.* At $s = 1/2$ the residue of $M$ is $1/\zeta(1/2)$, and $K(1/2) = 2/3$. At a simple critical zero $\rho$, its residue is $\zeta(2\rho)/(\rho\zeta'(\rho))$. The numerator is nonzero by the classical nonvanishing of $\zeta$ on $\mathrm{Re}\, s = 1$. Apply Proposition 7.4 with $\beta = 1/2$, $r = 1$ and $\mathcal{P} = \{1/2\} \cup \mathcal{Z}$. Any right-hand zero, or critical zero of higher multiplicity, would give an excluded pole in the case of a finite limit superior. The functional equation reflects any off-line zero to the right of the critical line if necessary. □

The verified simple zeros of [9] give strictly positive terms in (7.5). Each finite set of such zeros gives a lower bound, and adding further zeros increases it. The estimate concerns the lower side of the energy question. The square-root-scale upper bound with an arbitrary power loss remains open.

Two certified numerical lower bounds. Using the first 1000 positive simple critical zeros and their conjugates in (7.5) gives a right side strictly above 45769/100000. The calculation uses outward-rounded ball arithmetic at 128 bits. The supplied verification script isolates the zeros, checks that the ordinate enclosures are disjoint, and encloses the derivative and residue at each zero. The `python-flint` 0.9.0 interface calls the rigorous FLINT zeta routines described in its documentation. No zero outside the stated set enters the sum.

A positive weighting in the logarithmic variable yields the stronger certified bound

$$\limsup_{N\to\infty} \frac{Q_\lambda(N)}{N} > \frac{3257}{5000} = 0.6514. \tag{7.6}$$

To prove it, let $C$ denote the limit superior in (7.6). If $C < \infty$, the proof of Proposition 7.4 gives, for $F(t) = e^{-t/2}U_{e^t}$, $\limsup_{t\to\infty} \|F(t)\|_{\mathcal{H}} = \sqrt{C}$. Let $\rho_1 = 1/2 + i\gamma_1$ be the first positive critical zero, $A_1 = \zeta(2\rho_1)/(\rho_1\zeta'(\rho_1))$, and $a = -1/\zeta(1/2) > 0$. For $m = 100$, put

$$P(t) = \frac{1}{m+1}\left|\sum_{j=0}^{m} e^{ij(\gamma_1 t-\theta)}\right|^2, \qquad e^{i\theta}A_1 = -|A_1|.$$

The trigonometric polynomial is nonnegative and its constant coefficient is one, so $\delta\int_0^\infty e^{-\delta t}P(t)\,\mathrm{d}t \to 1$. For each $\varepsilon > 0$, the norm of $F(t)$ is at most $\sqrt{C}+\varepsilon$ outside a fixed bounded interval. The integral on that interval vanishes after multiplication by $\delta$. Thus the limit of $\delta\int_0^\infty e^{-\delta t}P(t)F(t)\,\mathrm{d}t$, when it exists, has norm at most $\sqrt{C}$.

For $2 \le k \le 100$, ball arithmetic verifies $\zeta(1/2 + ik\gamma_1) \neq 0$ by enclosing its value over the whole ordinate enclosure. Equation (7.3) then shows that the terms at those frequencies and their conjugates vanish in the limit. The constant term contributes $-aw_{1/2}$. The two terms at frequencies $\pm\gamma_1$ give $-c\,\mathrm{Re}\,w_{\rho_1}$ with $c = 2m|A_1|/(m+1)$, since the first Fourier coefficient of $P$ has modulus $m/(m+1)$. The limit therefore exists and

$$C \ge \|aw_{1/2} + c\,\mathrm{Re}\,w_{\rho_1}\|_{\mathcal{H}}^2.$$

Writing $v = \mathrm{Re}\,w_{\rho_1}$, direct integration gives

$$\begin{aligned}
\|w_{1/2}\|_{\mathcal{H}}^2 &= \frac{2}{3},\\
\|v\|_{\mathcal{H}}^2 &= \frac{3}{2} + \mathrm{Re}\,\frac{1}{1+2\rho_1} - 2\,\mathrm{Re}\,\frac{1}{1+\rho_1},\\
\langle w_{1/2}, v\rangle_{\mathcal{H}} &= \frac{1}{3} - \mathrm{Re}\,\frac{1}{1+\rho_1} + 2\,\mathrm{Re}\,\frac{1}{3/2+\rho_1}.
\end{aligned}$$

The same script encloses the squared norm and verifies that its lower endpoint exceeds the exact rational $3257/5000$. This proves (7.6), including the immediate case $C = \infty$.

### 7.2. **A two-frequency positive weight.**

**Proposition 7.6.** *The signed Liouville Fejér energy satisfies the unconditional bound*

$$\limsup_{N\to\infty} \frac{Q_\lambda(N)}{N} > \frac{1811}{2500} = 0.7244.$$

*Proof.* Put $C = \limsup_{N\to\infty} Q_\lambda(N)/N$. It suffices to consider $C < \infty$. In $\mathcal{H} = L^2((0,1);\mathbb{C}^2)$, put

$$U_X(u) = (L(uX), L(X)-L(uX)), \quad w_s(u) = (u^s, 1-u^s), \quad F(t) = e^{-t/2}U_{e^t}.$$

Here $L(X) = 0$ for $0 < X < 1$. For $X = N + a$ with $0 \le a < 1$, the prefix and suffix formula gives

$$\|U_X\|_{\mathcal{H}}^2 = \frac{NQ_\lambda(N) + a|L(N)|^2}{X}, \qquad |L(N)|^2 \le 2Q_\lambda(N).$$

Consequently $\limsup_{t\to\infty} \|F(t)\|_{\mathcal{H}} = \sqrt{C}$. Partial summation and the substitution $v = uX$ give

$$\int_0^\infty F(t)e^{-(s-1/2)t}\,\mathrm{d}t = M(s)w_s, \qquad M(s) = \frac{\zeta(2s)}{s\zeta(s)}.$$

The identity holds first for $\operatorname{Re} s > 1$, and then for $\operatorname{Re} s > 1/2$ by holomorphy of the integral under the bound on $F$. Thus $M(s)w_s$ has no pole in that open half-plane, and its poles on the boundary have order at most one.

Let $\rho_j = 1/2 + i\gamma_j$, $j = 1, 2$, be the first two positive critical zeros. They are simple. Put

$$A_j = \frac{\zeta(2\rho_j)}{\rho_j\zeta'(\rho_j)}, \qquad a = -\frac{1}{\zeta(1/2)} > 0, \qquad m = 30.$$

Choose real phases $\theta_j$ satisfying $e^{i\theta_j}A_j = -|A_j|$, and take

$$P(t) = \prod_{j=1}^{2} \frac{1}{m+1}\left|\sum_{\ell=0}^{m} e^{i\ell(\gamma_j t - \theta_j)}\right|^2.$$

This is a nonnegative trigonometric polynomial. Its frequencies are $n_1\gamma_1 + n_2\gamma_2$, with $-30 \le n_1, n_2 \le 30$. The accompanying ball-arithmetic certificate checks that all 3721 frequency enclosures are pairwise disjoint. In particular only $(n_1, n_2) = (0, 0)$ has zero frequency, and the constant coefficient of $P$ is one. It follows that

$$\delta\int_0^\infty P(t)e^{-\delta t}\,\mathrm{d}t \longrightarrow 1.$$

For every $\varepsilon > 0$, one has $\|F(t)\|_{\mathcal{H}} \le \sqrt{C} + \varepsilon$ for all sufficiently large $t$. The contribution of a fixed bounded interval vanishes after multiplication by $\delta$. Nonnegativity therefore gives

$$\limsup_{\delta\downarrow 0} \left\|\delta\int_0^\infty F(t)P(t)e^{-\delta t}\,\mathrm{d}t\right\|_{\mathcal{H}} \le \sqrt{C}.$$

Outside the origin and the four fundamental frequencies, the certificate verifies

$$\zeta\bigl(1/2 + i(n_1\gamma_1 + n_2\gamma_2)\bigr) \ne 0.$$

There are 3716 such tests. The vector Mellin identity shows that their terms tend to zero in the last integral. The origin contributes $-aw_{1/2}$. Each pair $\pm\gamma_j$ contributes $-c_j \operatorname{Re} w_{\rho_j}$, where $c_j = 60|A_j|/31$. Indeed the coefficient of $e^{i\gamma_j t}$ in $P$ is $(30/31)e^{-i\theta_j}$, and its Mellin argument is $1/2 + \delta - i\gamma_j$. Pairing it with the negative frequency gives $2(30/31)\operatorname{Re}(e^{i\theta_j}A_j w_{\rho_j})$. Thus the vector limit exists and

$$C \ge \left\|aw_{1/2} + c_1 \operatorname{Re} w_{\rho_1} + c_2 \operatorname{Re} w_{\rho_2}\right\|_{\mathcal{H}}^2.$$

The squared norm is computed using the exact identity

$$\langle \operatorname{Re} w_z, \operatorname{Re} w_w \rangle_{\mathcal{H}} = 1 - \operatorname{Re} \frac{1}{1+z} - \operatorname{Re} \frac{1}{1+w} + \operatorname{Re} \frac{1}{1+z+w} + \operatorname{Re} \frac{1}{1+z+\overline{w}}.$$

The certificate uses `python-flint` 0.9.0 with 128-bit Arb/Acb arithmetic. It isolates the two simple zeros, verifies that both derivative enclosures exclude zero, proves strict separation of every frequency enclosure, and evaluates all 3716 regular zeta values over those enclosures. It obtains the squared-norm enclosure

$$0.72445959780354945892543005712377997 \pm 7.55 \cdot 10^{-36},$$

whose lower endpoint exceeds the exact rational 1811/2500. The FLINT routines return rigorous enclosures, as documented in the library's zeta-zero reference. This proves the assertion. □

The frequency checks concern this finite polynomial. A sequence of weights yielding an unbounded lower bound would require control of the additional integer combinations and their pole contributions at every stage.

## 8. The energy and the construction

Write $\lambda(n) = (-1)^{\Omega(n)}$. For a sign function $f$ and a real $x \geq 0$, put $L_f(x) = \sum_{n \leq x} f(n)$, with $L_f(x) = 0$ for $x < 1$. For integers $N \geq 1$, its signed Fejér energy is

$$Q_f(N) = \frac{1}{N} \sum_{t \in \mathbb{Z}} \left| \sum_{\substack{t < n \leq t+N \\ 1 \leq n \leq N}} f(n) \right|^2 .$$

This is a finite sum. A pair of indices at distance $h$ belongs to exactly $N - h$ windows. Thus the expansion weights the signed pairs at that distance by $2(1 - h/N)$.

**Lemma 8.1.** *For every real sign function and every $N \geq 1$,*

$$Q_f(N) = \frac{|L_f(N)|^2}{N} + \frac{1}{N} \sum_{j=1}^{N-1} \left( |L_f(j)|^2 + |L_f(N) - L_f(j)|^2 \right), \tag{8.1}$$

$$\frac{1}{2} |L_f(N)|^2 \leq Q_f(N) \leq 5 \max_{0 \leq j \leq N} |L_f(j)|^2. \tag{8.2}$$

*Proof.* The window beginning at $t = 0$ contains the full truncation. Each nonempty proper prefix occurs once at $t = j - N$, and its complementary suffix occurs once at $t = j$. All other windows have zero sum. This gives (8.1). The identity

$$|u|^2 + |v - u|^2 = 2|u - v/2|^2 + |v|^2/2$$

gives the lower bound. If $M = \max_{j\le N} |L_f(j)|$, the summand in (8.1) is at most $M^2 + 4M^2$, which gives the upper bound. □

Let $\Theta$ be the supremum of the real parts of the nontrivial zeros of $\zeta$. For a fixed cutoff $y_0 \ge 2$, put

$$c_{y_0}(s) = \prod_{p\le y_0} \frac{1-p^{-s}}{1+p^{-s}}.$$

This is a nonvanishing holomorphic function on $\operatorname{Re} s > 0$. For the Dirichlet series $F_f$ of $f$, write $H_f(s) = F_f(s)/(\zeta(s)c_{y_0}(s))$.

**Theorem 8.2.** *For every $\beta \in (1/2, 1)$ and $y_0 \ge 2$, there is one completely multiplicative $f : \mathbb{N} \to \{-1, 1\}$ with the following properties.*

1. *$f(p) = \lambda(p)$ for $p \le y_0$ and $\sum_{p:f(p)=1} p^{-1} < \infty$.*
2. *For every $A > 0$, there is $C_{A,f}$ such that $|L_f(x)| \le C_{A,f}x/(\log x)^A$ for all $x \ge 2$.*
3. *The Dirichlet series $F_f(s) = \sum_{n\ge1} f(n)n^{-s}$ extends meromorphically to $\operatorname{Re} s > 1/2$ as*

$$F_f(s) = \frac{\zeta(2s)\zeta(s+1-\beta)}{\zeta(s)} R_f(s), \tag{8.3}$$

   *where $R_f$ is holomorphic and nonvanishing there and positive on the real axis. The function $F_f$ has a simple zero at 1, and $H_f$ has a double zero there. Both have a simple pole at $\beta$.*
4. *The energy satisfies*

$$\limsup_{N\to\infty} \frac{\log(1+Q_f(N))}{\log N} = 2\max(\Theta, \beta). \tag{8.4}$$

   *In particular, $Q_f(N)/N^{1+\varepsilon}$ is unbounded for every $0 < \varepsilon < 2\beta - 1$.*

The same $f$ works for all $A$ and $\varepsilon$ in the theorem.[1] The construction in the next section rounds prime weights by taking integer parts of their cumulative sum. The proof uses classical logarithmic cancellation for Liouville and the standard summatory estimate associated with a fixed zero-free half-plane. Related multiplicative-function models appear in [52, 53].

## 9. Rounding prime weights and computing the Euler product

For each prime $p > y_0$, set $q_p = \frac{1}{2}p^{\beta-1}$ and define

$$S(t) = \sum_{y_0<p\le t} q_p, \qquad I_p = \lfloor S(p)\rfloor - \lfloor S(p) - q_p\rfloor.$$

Each $I_p$ is either zero or one, since $0 < q_p < 1$. Set $f(p) = -1$ at every prime at most $y_0$, and $f(p) = 2I_p - 1$ at larger primes. Extend completely

[1] The construction is one fixed completely multiplicative function after choosing $\beta$ and the initial prime cutoff. It is not a different comparison sequence at each value of $N$.

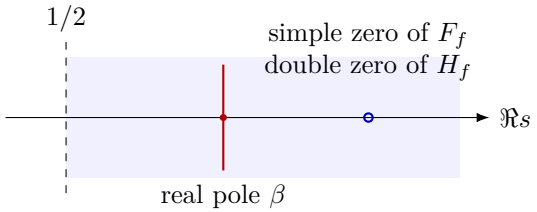



FIGURE 5. Analytic structure of the deterministic multiplicative construction. The pole at the prescribed real point $\beta$ is present independently of the location of unknown zeta zeros.

multiplicatively. Let $\mathcal{S} = \{p > y_0 : I_p = 1\}$ be the set of primes where the sign changes. Telescoping the floors gives, for every real $t$,

$$E(t) = \sum_{y_0 < p \le t} (I_p - q_p) = \lfloor S(t) \rfloor - S(t) \in (-1, 0]. \tag{9.1}$$

Define the centered prime series $A(s) = \sum_{p>y_0} (I_p - q_p) p^{-s}$. Summation by parts and (9.1) give

$$A(s) = s \int_1^\infty E(t) t^{-s-1} \, \mathrm{d}t \qquad (\mathrm{Re}\, s > 0). \tag{9.2}$$

The integral converges locally uniformly in that half-plane and defines a holomorphic function. For every real $\sigma > \beta$ it follows that

$$\sum_{p>y_0} I_p p^{-\sigma} = \frac{1}{2} \sum_{p>y_0} p^{-(\sigma+1-\beta)} + A(\sigma) < \infty. \tag{9.3}$$

In particular $\sum_{p \in \mathcal{S}} 1/p < \infty$, and the sum $\sum_p (1 - f(p)\lambda(p))/p$ equals $2 \sum_{p \in \mathcal{S}} 1/p$.

Consider the Euler product

$$G(s) = \prod_{p \in \mathcal{S}} \frac{1 + p^{-s}}{1 - p^{-s}}.$$

It converges absolutely for $\mathrm{Re}\, s > \beta$ by (9.3). On $\mathrm{Re}\, s > 1$, comparison of local factors gives $F_f(s) = \zeta(2s) G(s) / \zeta(s)$. To factor $G$, subtract the logarithmic Euler series of $\zeta(s+1-\beta)$ from that of $G$. The result on $\mathrm{Re}\, s > 1$ is

$$\begin{aligned} C_f(s) = 2A(s) &- \sum_{p \le y_0} p^{-(s+1-\beta)} \\ &+ 2 \sum_{p>y_0} I_p \sum_{\substack{k \ge 3 \\ k \,\mathrm{odd}}} \frac{p^{-ks}}{k} - \sum_p \sum_{k \ge 2} \frac{p^{-k(s+1-\beta)}}{k}. \end{aligned} \tag{9.4}$$

The first term is holomorphic by (9.2). The last two sums converge absolutely and locally uniformly for $\operatorname{Re} s > 1/2$. For a lower bound $\sigma > 1/2$ on the real part, their sums of absolute values are bounded by constants times $\sum_p p^{-3\sigma}$ and $\sum_p p^{-2(\sigma+1-\beta)}$, respectively. Both exponents exceed one. Thus $R_f = \exp C_f$ is holomorphic and nonvanishing on $\operatorname{Re} s > 1/2$, and is positive at real points there. Exponentiating (9.4) gives $G(s) = \zeta(s+1-\beta)R_f(s)$ on $\operatorname{Re} s > 1$, proving (8.3) by continuation.

The simple pole of $\zeta$ at 1 now gives

$$\frac{F_f(s)}{\zeta(s)c_{y_0}(s)} = \frac{\zeta(2)\zeta(2-\beta)R_f(1)}{c_{y_0}(1)}(s-1)^2 + O((s-1)^3). \tag{9.5}$$

The coefficient is positive, and the zero of $F_f$ itself at 1 is simple. At $s = \beta$, one has $\zeta(\beta) < 0$, by the alternating series formula on $0 < \beta < 1$. The shifted zeta factor contributes a genuine pole with residue

$$\operatorname*{Res}_{s=\beta} F_f(s) = \frac{\zeta(2\beta)R_f(\beta)}{\zeta(\beta)} \neq 0. \tag{9.6}$$

The other denominator in (9.5) is holomorphic and nonzero at $\beta$, so that quotient also has a simple pole there.

The local factors of $H_f$ at $p \le y_0$ are one. At a larger prime in $\mathcal{S}$ the factor is also one. At a larger prime outside $\mathcal{S}$ it is

$$\frac{1-p^{-s}}{1+p^{-s}} = 1 + 2\sum_{j\ge1}(-1)^j p^{-js}.$$

Thus the coefficients are $2^{\omega(n)}\lambda(n)$ on integers whose prime factors exceed $y_0$ and are outside $\mathcal{S}$, and zero elsewhere. Here $\omega(n)$ counts distinct prime factors.

## 10. Summatory bounds and the exact energy exponent

Let $h$ be the multiplicative function with $h(1) = 1$ and with $h(p^j) = 2$ for $p \in \mathcal{S}$, $j \ge 1$, and zero at other prime powers. Its coefficients are nonnegative and its Dirichlet series is $G$. Local factors give the exact convolution

$$f = \lambda * h. \tag{10.1}$$

Choose $\sigma_0 \in (\beta, 1)$, so that $G(\sigma_0) < \infty$. The classical Davenport estimate at additive frequency zero gives $L_\lambda(x) \ll_A x/(\log(2x))^A$ for every $A > 0$. The Liouville form is stated in [55, §1, after Lemma 1.4], with attribution to [54]. In the convolution sum $L_f(x) = \sum_{d\le x} h(d)L_\lambda(x/d)$, use the logarithmic estimate for $d \le \sqrt{x}$ and the bound $|L_\lambda(x/d)| \le x/d$ for the remaining terms. This gives

$$\begin{aligned} |L_f(x)| &\le C_A G(1)\frac{x}{(\log x)^A} + x\sum_{d>\sqrt{x}}\frac{h(d)}{d} \\ &\le C_A G(1)\frac{x}{(\log x)^A} + G(\sigma_0)x^{(1+\sigma_0)/2}. \end{aligned}$$

Since $(1+\sigma_0)/2 < 1$, the second term is absorbed in the first for each fixed $A$, with a constant valid for all $x \geq 2$. The same estimate holds uniformly for the additive twist $\sum_{n\leq x} f(n)e^{2\pi i n\alpha}$. In the convolution sum the inner frequency becomes $d\alpha$, so Davenport's uniform estimate applies with the same constant. Lemma 8.1, separating prefixes below $\sqrt{N}$ from larger ones, also gives $Q_f(N) \ll_{A,f} N^2/(\log N)^A$ for every $A$.

Suppose $Q_f(N) = O(N^{2\alpha})$ for some $\alpha < \beta$. Equation (8.2) then gives $L_f(x) = O(x^\alpha)$, and partial summation makes

$$s\int_1^\infty L_f(x)x^{-s-1}\,\mathrm{d}x$$

holomorphic for $\operatorname{Re} s > \alpha$. It equals $F_f(s)$ initially on $\operatorname{Re} s > 1$. The pole (9.6) rules out this bound. The same argument applies with any pole $\rho$ of $F_f$ to the right of $\alpha$. If the energy limit superior in (8.4) were below $2\operatorname{Re}\rho$, choose $\alpha$ between half of that limit superior and $\operatorname{Re}\rho$. This would give the forbidden power bound. Every zero $\rho$ of $\zeta$ with $\operatorname{Re}\rho > \beta$ is such a pole. Indeed both $2\rho$ and $\rho+1-\beta$ have real part greater than one, so neither zeta factor in the numerator of (8.3) vanishes. Taking real parts of these zeros arbitrarily close to $\Theta$ when $\Theta > \beta$ gives the lower exponent $2\max(\Theta,\beta)$.

For the upper exponent, use the standard consequence of a fixed zero-free half-plane that $L_\lambda(x) = O_\mu(x^\mu)$ for every $\mu > \Theta$. The reciprocal-zeta contour argument is classical [26, Ch. XIV]. One may pass from its Möbius form to Liouville using $\lambda(n) = \sum_{d^2|n} \mu_{\mathrm{Mob}}(n/d^2)$ and the convergence of $\sum_d d^{-2\mu}$ for $\mu > 1/2$. For $\mu > \max(\Theta,\beta)$, (10.1) therefore gives

$$|L_f(x)| \leq C_\mu x^\mu \sum_{d\leq x} \frac{h(d)}{d^\mu} \leq C_\mu x^\mu G(\mu).$$

Apply Lemma 8.1 and let $\mu$ decrease to $\max(\Theta,\beta)$. This proves (8.4) and finishes Theorem 8.2.

The same pole forbids a square-root bound for the exponential sum $B_f(X) = \sum_{n\geq 1} f(n)e^{-n/X}$. On $\operatorname{Re} s > 1$, absolute convergence and Euler's gamma integral give

$$\int_0^\infty B_f(X)X^{-s}\frac{\mathrm{d}X}{X} = \Gamma(s)F_f(s).$$

If $B_f(X) = O(X^{1/2+\varepsilon})$ with $1/2+\varepsilon < \beta$, the left side would be holomorphic to the right of $1/2+\varepsilon$, since $B_f(X)$ decays exponentially as $X \downarrow 0$. The nonzero gamma factor at $\beta$ preserves the pole on the right.

## 11. The Abel zero and its leading coefficient

**Proposition 11.1.** *Let $g$ be a completely multiplicative sign function whose Dirichlet series is holomorphic near 1 and satisfies $F_g(1) = 0$. Then the zero is simple, $F_g'(1) \geq \zeta(2)$, and equality holds exactly for $g = \lambda$. Consequently the quotient $F_g/(\zeta c_{y_0})$ has a double zero at 1. Its leading coefficient equals $\zeta(2)/c_{y_0}(1)$ exactly when $g = \lambda$.*

*Proof.* For real $\sigma > 1$, the Euler products give

$$\frac{F_g(\sigma)}{F_\lambda(\sigma)} = \prod_{p:g(p)=1} \frac{1+p^{-\sigma}}{1-p^{-\sigma}} \geq 1.$$

Since $F_\lambda(1+\delta) \sim \zeta(2)\delta$, a zero of order greater than one would contradict this inequality. Dividing by $\delta$ and letting $\delta \downarrow 0$ proves the derivative bound. If $g(p) = 1$ at any prime $p$, the ratio has limit inferior at least $(p+1)/(p-1) > 1$. Thus equality of derivatives forces every prime sign to be $-1$, and complete multiplicativity gives $g = \lambda$. The quotient assertion follows by dividing by the pole of $\zeta$ and by the nonzero value $c_{y_0}(1)$. □

The constructed function attains the maximal zero order, while its slope is $\zeta(2)G(1) > \zeta(2)$. The strict inequality follows from the presence of a changed prime. In fact $S(t) \to \infty$, since the sum of reciprocal primes diverges and $p^{\beta-1} \geq p^{-1}$. Thus complete multiplicativity, agreement at any prescribed finite prime prefix, every logarithmic saving, and maximal Abel vanishing together permit energy growth above $N^{1+\varepsilon}$. The exact Abel coefficient singles out Liouville within the sign class. The square-root upper bound for Liouville itself remains open.

## 12. Quantitative limits of local Abel data

The construction in (8.3) has a stronger approximation property than agreement of the Abel zero order. Its normalized Abel derivatives can approximate those of Liouville to any prescribed finite accuracy, while a real pole still prevents the critical energy estimate. This distinguishes an exact arithmetic identity from an approximation to it. None of the results below excludes a zeta zero.

Fix $1/2 < \beta < 1$ and an integer $P \geq 2$. For primes $p > P$, put

$$q_p = \tfrac{1}{2}p^{\beta-1}, \qquad S(t) = \sum_{P<p\leq t} q_p, \qquad I_p = \lfloor S(p) \rfloor - \lfloor S(p) - q_p \rfloor.$$

Set $f_P(p) = -1$ for $p \leq P$, set $f_P(p) = 2I_p - 1$ otherwise, and extend completely multiplicatively. Write

$$\mathcal{S}_P = \{p > P : I_p = 1\}, \qquad G_P(s) = \prod_{p\in\mathcal{S}_P} \frac{1+p^{-s}}{1-p^{-s}}.$$

In particular $f_P(n) = \lambda(n)$ for every integer $n \leq P$. The finite cutoff and $\beta$ specify one fixed function; they do not change as the scale of an energy estimate tends to infinity.

**Lemma 12.1** (An explicit Euler-factor bound)**.** *For every real $\sigma > \beta$, the logarithm specified by its Euler series is holomorphic on $\operatorname{Re} s > \beta$ and satisfies*

$$\sup_{\operatorname{Re} s\geq\sigma} |\log G_P(s)| \leq \Delta(P,\sigma), \qquad \Delta(P,\sigma) = \frac{P^{\beta-\sigma}}{(\sigma-\beta)(1-P^{-2\sigma})}. \tag{12.1}$$

*Consequently $|G_P(s) - 1| \leq e^{\Delta(P,\sigma)} - 1$ in that half-plane.*

*Proof.* The centered cumulative discrepancy has a sign:

$$E(t) = \sum_{P<p\le t} (I_p - q_p) = \lfloor S(t) \rfloor - S(t) \in (-1, 0].$$

For $T > P$, partial summation gives

$$\sum_{P<p\le T} (I_p - q_p)p^{-\sigma} = E(T)T^{-\sigma} + \sigma \int_P^T E(t)t^{-\sigma-1}\,\mathrm{d}t \le 0.$$

It follows, by monotone convergence of the nonnegative sums, that

$$\begin{aligned}\sum_{p\in\mathcal{S}_P} p^{-\sigma} &\le \tfrac{1}{2} \sum_{p>P} p^{-(1+\sigma-\beta)} \\ &\le \tfrac{1}{2} \sum_{n>P} n^{-(1+\sigma-\beta)} \le \frac{P^{\beta-\sigma}}{2(\sigma-\beta)}.\end{aligned}$$

The last inequality compares a decreasing sum with its integral from $P$ to infinity. Thus the following series converges absolutely and locally uniformly for $\operatorname{Re} s > \beta$:

$$\log G_P(s) = 2 \sum_{p\in\mathcal{S}_P} \sum_{\substack{j\ge 1 \\ j \text{ odd}}} \frac{p^{-js}}{j}.$$

For $\operatorname{Re} s \ge \sigma$, its absolute value is at most

$$\frac{2}{1 - P^{-2\sigma}} \sum_{p\in\mathcal{S}_P} p^{-\sigma} \le \Delta(P, \sigma).$$

Exponentiating proves the final assertion. □

The quotient called $H_f$ in the preceding construction includes a finite Euler factor $c_P$. Here it is convenient to cancel that common factor and normalize the double zero at one. Define

$$\mathcal{J}_\lambda(s) = \frac{\zeta(2s)}{((s-1)\zeta(s))^2}, \qquad \mathcal{J}_P(s) = \mathcal{J}_\lambda(s) G_P(s). \tag{12.2}$$

For $f = f_P$, this is precisely $c_P(s)H_f(s)/(s-1)^2$; for $f = \lambda$, the analogous identity holds with the same cutoff. Both functions have their removable values assigned at one. [2]

**Theorem 12.2** (Quantitative approximation of every finite Abel jet)**.** *Let $0 < r < 1 - \beta$ and put*

$$C_r = \frac{\zeta(2-2r)}{(1-2r)^2}.$$

*The functions in (12.2) are holomorphic on a neighborhood of $|s-1| \le r$. For every integer $k \ge 0$,*

$$\left| \mathcal{J}_P^{(k)}(1) - \mathcal{J}_\lambda^{(k)}(1) \right| \le k! r^{-k} C_r (e^{\Delta(P,1-r)} - 1). \tag{12.3}$$

[2] A finite Abel jet of order $K$ consists of these derivatives at one for $0 \le k \le K$. It specifies local data.

*For each fixed $r$ and $K$, the right side tends to zero uniformly for $0 \le k \le K$ as $P \to \infty$.*

*Proof.* Euler summation, first for $\operatorname{Re} s > 1$ and then by continuation to $\operatorname{Re} s > 0$, gives

$$\zeta(s) = \frac{s}{s-1} - sI(s), \qquad I(s) = \int_1^\infty \{x\} x^{-s-1}\,\mathrm{d}x.$$

The integral is locally uniformly convergent in this half-plane and $|I(s)| \le 1/\operatorname{Re} s$. On $|s-1| \le r < 1/2$, therefore,

$$\begin{aligned} |(s-1)\zeta(s)| &= |s|\,|1-(s-1)I(s)| \\ &\ge (1-r)\left(1 - \frac{r}{1-r}\right) = 1-2r > 0. \end{aligned}$$

The removable value at one is one. The numerator $\zeta(2s)$ is holomorphic here and has absolute value at most $\zeta(2-2r)$ by its absolutely convergent series. Hence

$$\sup_{|s-1|\le r} |\mathcal{J}_\lambda(s)| \le C_r.$$

Since $1 - r > \beta$, Lemma 12.1 bounds $|G_P - 1|$ on the same disk. Apply the Cauchy integral formula to $\mathcal{J}_P - \mathcal{J}_\lambda$ on its boundary to obtain (12.3). The exponent $\beta - 1 + r$ in $\Delta(P, 1-r)$ is negative, which proves convergence. □

**Corollary 12.3** (A completely explicit cutoff)**.** *Fix $K \ge 0$ and $\eta > 0$. Set $\sigma = 1 - r$, $d = \sigma - \beta > 0$ and $M = C_r \max_{0\le k\le K} k! r^{-k}$. Any integer $P$ with*

$$P \ge \max\left\{2, \left[\frac{1}{d(1-2^{-2\sigma})\log(1+\eta/M)}\right]^{1/d}\right\} \tag{12.4}$$

*satisfies $|\mathcal{J}_P^{(k)}(1) - \mathcal{J}_\lambda^{(k)}(1)| \le \eta$ for all $0 \le k \le K$.*

*Proof.* For $P \ge 2$, (12.1) is at most $P^{-d}/[d(1-2^{-2\sigma})]$. The stated cutoff makes this at most $\log(1+\eta/M)$. Substitute in (12.3). □

**Theorem 12.4** (Failure of the critical local energy for the same functions)**.** *Put $B_P(X) = \sum_{n\ge1} f_P(n) e^{-n/X}$ and*

$$\mathcal{H}_{B_P}(X) = \int_{X/2}^{2X} |B_P(Y)|^2 \frac{\mathrm{d}Y}{Y}.$$

*For every fixed $P$, the function $f_P$ has all the logarithmic summatory and additive-twist savings proved in the preceding construction. Nevertheless, for every $0 < \varepsilon < 2\beta - 1$,*

$$\mathcal{H}_{B_P}(X) \ne O(X^{1+\varepsilon}). \tag{12.5}$$

*This holds for the very same function whose Abel jets satisfy (12.3), not for a different function at each scale.*

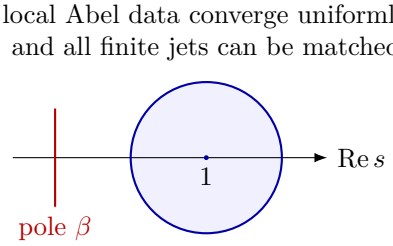



FIGURE 6. The approximation and the pole coexist. The illustration uses $\beta = 3/4$ and $r = 1/8$; the disk is drawn schematically.

*Proof.* The factorization (8.3) has a simple pole at the real point $\beta$: $\zeta(s+1-\beta)$ supplies that pole, the remaining factor $R_f$ is nonzero, and $\zeta(2\beta)/\zeta(\beta) \neq 0$. For the denominator, the alternating zeta series shows $\zeta(\beta) < 0$ for $0 < \beta < 1$.

Suppose instead that the local energy is $O(X^{1+\varepsilon})$. For $\sigma > (1+\varepsilon)/2$, Cauchy–Schwarz on each dyadic interval gives

$$\int_{2^j}^{2^{j+1}} |B_P(Y)| Y^{-\sigma} \frac{\mathrm{d}Y}{Y} \leq (\log 2)^{1/2} 2^{-j\sigma} \mathcal{H}_{B_P}(2^j)^{1/2} \ll 2^{j((1+\varepsilon)/2-\sigma)}.$$

The sum over $j \geq 0$ converges locally uniformly to the right of $(1+\varepsilon)/2$. The integral from zero to one is entire, by exponential decay and $|f_P(n)| = 1$. Thus

$$\int_0^\infty B_P(Y) Y^{-s} \frac{\mathrm{d}Y}{Y}$$

is holomorphic on $\operatorname{Re} s > (1+\varepsilon)/2$. Initially for $\operatorname{Re} s > 1$ it equals $\Gamma(s) F_{f_P}(s)$ by absolute convergence. Continuation would then remove its genuine pole at $\beta$, since $(1+\varepsilon)/2 < \beta$ and $\Gamma(\beta) \neq 0$. This is a contradiction. The logarithmic estimates follow from the earlier nonnegative convolution $f_P = \lambda * h$ and require no RH input. □

For a numerical tolerance with an exact certificate, take $\beta = 3/4$, $r = 1/8$, $K = 3$ and $P = 2^{320}$. Since $\zeta(7/4) \leq 1 + \int_1^\infty t^{-7/4}\,\mathrm{d}t = 7/3$, we may replace $C_r$ by $112/27$. Thus $M \leq 114688/9$, and

$$\Delta(P, 7/8) = \frac{8\,2^{-40}}{1 - 2^{-560}} < 1, \qquad \frac{114688}{9} \frac{\Delta(P, 7/8)}{1 - \Delta(P, 7/8)} < 10^{-6}.$$

The last inequality is rational arithmetic. The bound $e^x - 1 \leq x/(1-x)$ for $0 \leq x < 1$ follows by comparing the exponential series with the geometric series. The first four Abel derivatives are therefore within $10^{-6}$ of those of Liouville, and every value $f_P(n)$ up to $2^{320}$ agrees exactly with $\lambda(n)$. The fixed function still violates every local-energy bound $O(X^{1+\varepsilon})$ with $0 < \varepsilon < 1/2$.

The check `anc/abel-jet-bounds.py` verifies the displayed rational constants; the infinite construction and its pole are proved analytically above.

Exact agreement is different. The earlier coefficient-rigidity result shows that equality of the normalized leading Abel coefficient in the completely multiplicative sign class forces $f = \lambda$. The present theorem does not contradict it: here $G_P(1) > 1$ at every finite cutoff, although $G_P(1) \to 1$. Approximate equalities, at any fixed precision and finite derivative order, cannot replace the exact arithmetic condition. Nor does the exact condition itself prove an upper bound for the actual Liouville energy.

12.1. **The unresolved upper estimate and the Mellin step.** For the actual Liouville function, the bound still needed is

$$\mathcal{H}_B(X) \ll_\varepsilon X^{1+\varepsilon} \qquad \text{for every } \varepsilon > 0 \text{ and all } X \geq 1. \tag{12.6}$$

The dyadic argument in the last proof shows directly that this bound would make the actual Mellin integral holomorphic on $\operatorname{Re} s > 1/2$. It would coincide there with $\Gamma(s)\zeta(2s)/\zeta(s)$ and exclude every right-hand zero. Conversely, Proposition 14.1 under RH bounds $B(Y)$ by $Y^{1/2+\varepsilon/2}$ and gives (12.6) by integration. Hence this is an equivalence, not a proved upper estimate.

In particular, using Plancherel on a line $\operatorname{Re} s > 1/2$ for the actual $B$ requires the corresponding Mellin convergence or $L^2$ control first. Meromorphic continuation of the quotient by itself supplies neither. Moving a contour past a hypothetical zero also introduces its residue; it does not discard the corresponding large mode. The signed off-diagonal in (17.3) remains unbounded at the required power. No new zero-free half-plane or proof of RH is claimed in this revision.

## 13. What the recent results give

The recent advances concern different quantities. Anthropic's result controls a lower proportion of simple zeros on the critical line through Weil's form and pair correlation [8]. Lamzouri's independent proof obtains more than 67.25% of zeros that are both simple and on the line [16]. Guth and Maynard's large-value estimates bound the number of zeros to the right of a specified line as a function of height [12]. A count of this type can allow an exceptional zero. Corollary 7.2 shows that even one such zero affects the energy exponent. A rigorous computation of Platt and Trudgian checks all positive ordinates through $3 \cdot 10^{12}$ [9]. The height cutoff, the proportion and the horizontal threshold are separate quantities.

The OpenAI quasi-Riemann manuscript states that $\zeta(s) \neq 0$ for $\operatorname{Re} s > 7/8$, with a Lean formalization of the stated zero-free result in the accompanying repository [22, 23]. Taking this zero-free statement as input in Theorem 7.1 gives

$$L(x) \ll_\varepsilon x^{7/8+\varepsilon}, \qquad Q_\lambda(N) \ll_\varepsilon N^{7/4+\varepsilon}.$$

The difference between 7/4 and 1 is precisely the remaining power in this application. The quasi-Riemann input is used only for this specialization,

and the proofs of the energy identities, the equivalence, the counterexamples and the endpoint obstruction do not use it.

The proof attempt in this revision proceeds by retaining the signs through (6.2). At $H = N$, the desired estimate is exactly an upper bound $O_\varepsilon(N^{1+\varepsilon})$ for the diagonal $N$ plus the weighted signed correlation sum. The termwise absolute-value route would require control of a different quantity, and Sections 4 and 5 show why naive random-sign or fixed-shift estimates do not provide it. Additive Fourier cancellation controls the energy through (6.1), but an $O(N^{1/2+\varepsilon})$ uniform Fourier bound already includes the estimate for $L(N)$ at $\alpha = 0$. Using it as an input would therefore assume the required strength. The remaining estimate is the signed energy bound itself, or another estimate that excludes all zeros with $1/2 < \operatorname{Re} s \le 7/8$ when the quasi-Riemann input is used.

13.1. **Refinements after the seven-eighths manuscript.** Liu's revised arXiv preprint [17] reports the threshold

$$b_1 = \frac{18\xi + 7}{18\xi + 9} = 0.874957019420098946\ldots,$$
$$0 = 1188\xi^3 - 2160\xi^2 + 171\xi + 190,$$

with $\xi \in (0.38858, 0.38859)$. A later preprint [18] reports a general-number-field threshold $b_2 = 0.874886647039964\ldots$, specified by a quartic equation. The version 3 PDF of [19], Theorem 1.1, reports

$$b_{\mathrm{tw}} = 0.87405850379185900499\ldots$$

for all finite-order Hecke $L$-functions over all number fields, by a further unit-twisted comparison. These are external research preprints. The conditional specializations below use their stated zero-free theorems, whose full analytic proofs are not reproduced or independently verified in this revision.

For any of these thresholds $b$, taking the corresponding zero-free statement as input in Theorem 7.1 and Proposition 14.1 gives

$$L(x) \ll_\varepsilon x^{b+\varepsilon}, \qquad B(X) \ll_\varepsilon X^{b+\varepsilon}, \qquad Q_\lambda(N) \ll_\varepsilon N^{2b+\varepsilon}.$$

The same input for all Dirichlet characters specializes Theorem 15.1. These deductions follow directly from the proved transfer criteria, and are attributed to their zero-free input.

The comparison in [19], Section 25, improves a starting boundary through a recurrence $b_{j+1} = b_{\mathrm{tw}} + (b_j - b_{\mathrm{tw}})/50$. Its exact solution is $b_j = b_{\mathrm{tw}} + (b_0 - b_{\mathrm{tw}})/50^j$, by induction. Thus that iteration converges to $b_{\mathrm{tw}}$. Section 27 gives a limitation for its stated comparison system even under an improved width input. Neither calculation gives an iteration to $1/2$.

## 14. Proof attempts without Chowla

The logarithmic correlation theorem was used for the obstruction in Section 5. The two constructions below use only one-variable Dirichlet series and Mellin transforms. They give different sufficient estimates for RH.

14.1. **A single smooth weight that detects every right-hand zero.** Let $B(X) = \sum_{n\ge1} \lambda(n) e^{-n/X}$ for $X > 0$. This sum is absolutely convergent. For $\operatorname{Re} s > 1$, absolute convergence and the substitution $y = n/X$ give

$$\int_0^\infty B(X) X^{-s} \frac{\mathrm{d}X}{X} = \Gamma(s) \frac{\zeta(2s)}{\zeta(s)}. \tag{14.1}$$

The integral over $0 < X \le 1$ is entire because $|B(X)| \le e^{-1/X}/(1 - e^{-1/X})$.

**Proposition 14.1** (A smoothed bypass)**.** *For $1/2 \le \theta \le 1$, the bound $B(X) \ll_{\varepsilon,\theta} X^{\theta+\varepsilon}$ for all $\varepsilon > 0$ is equivalent to the zero-free condition $\zeta(s) \ne 0$ for $\operatorname{Re} s > \theta$.*

*Proof.* Under the bound, the integral in (14.1) is holomorphic on $\operatorname{Re} s > \theta$. The identity extends meromorphically, and neither $\Gamma(s)$ nor $\zeta(2s)$ vanishes there. For the converse use Theorem 7.1 and the partial summation identity $B(X) = X^{-1} \int_1^\infty L(u) e^{-u/X}\, \mathrm{d}u$. Changing $u = Xv$ gives the required bound for $X \ge 1$. □

The nonvanishing of the Mellin weight in (14.1) matters. In the recent work of Li, Sun and Wang [7], a particularly effective weight is built from $W(x) = \sum_{m\ge1} V(mx)$, where $V$ is smooth, compactly supported in $(0, \infty)$, and $\int_0^\infty V = 0$. At each $X > 0$ the sums below are finite. Complete multiplicativity gives

$$\sum_{n\ge1} \lambda(n) W(n/X) = \sum_{r\ge1} V(r/X) \sum_{n\mid r} \lambda(n) = \sum_{j\ge1} V(j^2/X).$$

Indeed $\sum_{k=0}^{\ell} (-1)^k$ is 1 when $\ell$ is even and 0 otherwise, which proves the divisor identity prime by prime. The sum on the right is $O_V(\sqrt{X})$ for $X \ge 1$, by the support of $V$. For $\operatorname{Re} s > 1$, however, the Mellin transform of $W$ is $\zeta(s)\widehat{V}(s)$, where $\widehat{V}(s) = \int_0^\infty V(x) x^{s-1}\, \mathrm{d}x$. The equality continues to the right half-plane by Poisson summation. To see the convergence at zero, extend $V$ by zero to $\mathbb{R}$ and write $\sum_{m\in\mathbb{Z}} V(mx) = x^{-1} \sum_{k\in\mathbb{Z}} \widehat{V}_{\mathrm{F}}(k/x)$, with $\widehat{V}_{\mathrm{F}}$ the Fourier transform. The $k = 0$ term vanishes and the others decay faster than any fixed power as $x \downarrow 0$. The factor $\zeta(s)$ cancels the very poles that the Mellin test needs to detect.[3]

[3] Li, Sun and Wang also construct a different weight adapted to a specified zero in their Appendix A. The pole cancellation here concerns the divisor-convolution weight only.

14.2. **Reciprocal dilates with an explicit error rate.** The Nyman–Beurling criterion and its discrete strengthening [1, 2] suggest working with reciprocal dilates. For $n \geq 1$, define the balanced functions

$$g_n(x) = \{1/(nx)\} - n^{-1}\{1/x\}, \qquad x > 0.$$

They vanish for $x > 1$. For $1/2 \leq \sigma < 1$, set

$$\|u\|_\sigma^2 = \int_0^1 |u(x)|^2 x^{2\sigma-1}\,\mathrm{d}x, \qquad d_{M,\sigma} = \inf_{c_1,\dots,c_M \in \mathbb{C}} \left\| 1 - \sum_{n \leq M} c_n g_n \right\|_\sigma .$$

The norm at $\sigma = 1/2$ is the ordinary $L^2(0,1)$ norm. The functions $\{nx\}$ considered in [5] form a different approximation problem. Appendix B solves that finite problem and explains the difference.

**Lemma 14.2** (Exact Mellin transform)**.** *For $0 < \operatorname{Re} s < 1$,*

$$\int_0^1 g_n(x) x^{s-1}\,\mathrm{d}x = -(n^{-s} - n^{-1})\frac{\zeta(s)}{s}.$$

*Proof.* After substituting $y = 1/(nx)$ in the integral of the unbalanced function over $x > 0$, the transform is $n^{-s}\int_0^\infty \{y\} y^{-s-1}\,\mathrm{d}y$. On $0 < y < 1$ the integral is $1/(1-s)$. On $y > 1$ it is $1/(s-1) - \zeta(s)/s$ by the Euler summation identity in Appendix A. The two rational terms cancel. Subtract the corresponding $n^{-1}$ multiple of the $n = 1$ transform and use the support of $g_n$. □

**Theorem 14.3** (Convergence in shifted norms)**.** *Suppose $1/2 \leq \theta < \sigma < 1$ and $\zeta$ has no zero in $\operatorname{Re} s > \theta$. For each $0 < \delta < \sigma - \theta$, the explicit choice*

$$c_n = -\mu(n)(1 - n/M) \qquad (n \leq M)$$

*satisfies $\|1 - \sum_{n \leq M} c_n g_n\|_\sigma \ll_{\sigma,\theta,\delta} M^{-\delta}$. Conversely, for every nontrivial zero $\rho$ with $\operatorname{Re}\rho > \sigma$,*

$$d_{M,\sigma}^2 \geq \frac{2(\operatorname{Re}\rho - \sigma)}{|\rho|^2} \quad \text{for every } M.$$

*Proof.* Let $D_M(s) = \sum_{n \leq M} \mu(n)(1 - n/M)n^{-s}$. For $s = \sigma + it$, absolute convergence on $\operatorname{Re} w = 2$ gives

$$D_M(s) = \frac{1}{2\pi i}\int_{\operatorname{Re} w = 2} \frac{M^w}{w(w+1)\zeta(s+w)}\,\mathrm{d}w.$$

The scalar kernel identity follows by closing left when $M/n > 1$, with residues 1 and $-n/M$, and closing right when $M/n < 1$. At $n = M$ the kernel is zero by continuity, using absolute convergence on $\operatorname{Re} w = 2$.

Move the contour to $\operatorname{Re} w = -\delta$. There is only the residue $1/\zeta(s)$ at $w = 0$. There are no poles from zeros because $\sigma - \delta > \theta$, and $-\delta > -1$. Lemma A.1 and the Euler product to its right show that the horizontal integrals tend to zero for each fixed $t$. On the new vertical line the integral is

$$O_{\sigma,\theta,\delta,\eta}\left(M^{-\delta}\int_{\mathbb{R}} \frac{(1+|t+v|)^\eta}{(1+|v|)^2}\,\mathrm{d}v\right) = O_{\sigma,\theta,\delta,\eta}(M^{-\delta}(1+|t|)^\eta)$$

for $0 < \eta < 1$. The last equality follows from $1 + |t + v| \le (1 + |t|)(1 + |v|)$. The same argument at $s = 1$ gives $D_M(1) = O(M^{-\delta})$ because $1/\zeta(1) = 0$ in the removable extension.

By Lemma 14.2, the residual has Mellin transform $[1 - \zeta(s)(D_M(s) - D_M(1))]/s$. The proof of Lemma A.1 bounds $|\zeta(s)|$ by $O((1 + |t|)^{\eta})$ on $\operatorname{Re} s = \sigma$ as well, using $+\operatorname{Re} G$ instead of $-\operatorname{Re} G$. Thus the residual transform is $O(M^{-\delta}(1 + |t|)^{2\eta-1})$. For $\eta < 1/4$ its square integrates. Apply Plancherel to $e^{-\sigma y}u(e^{-y})\mathbf{1}_{y>0}$ with $y = -\log x$. Its squared norm is $\|u\|_\sigma^2$, and its Fourier transform in angular frequency is the Mellin transform on $\operatorname{Re} s = \sigma$. This proves the upper bound.

If $\operatorname{Re}\rho > \sigma$, the residual evaluates to $1/\rho$ at $\rho$. Cauchy–Schwarz in the weighted space bounds the absolute value of the evaluation by $\|u\|_\sigma(2(\operatorname{Re}\rho - \sigma))^{-1/2}$. This proves the lower bound uniformly in the coefficients. □

Taking the published quasi-Riemann manuscript as input gives convergence for $\sigma > 7/8$. The proof loses its positive choice of $\delta$ as $\sigma \downarrow 7/8$. For a proof of RH, it would suffice to find any sequence of balanced reciprocal-dilate polynomials for which the residual tends to zero at $\sigma = 1/2$, by the lower bound in Theorem 14.3 and the functional equation.[4] The explicit construction proves the convergence away from the zero-free boundary. It leaves the critical norm unbounded by a quantity tending to zero.

One can quantify the cost of trying to reach the critical norm by interpolation. For $0 < \sigma_0 < 1/2 < \sigma_1 < 1$, Hölder's inequality gives

$$\|u\|_{1/2} \le \|u\|_{\sigma_0}^{1-t}\|u\|_{\sigma_1}^{t}, \qquad t = \frac{1/2 - \sigma_0}{\sigma_1 - \sigma_0}.$$

For the explicit residual $u_M = 1 - \sum c_n g_n$, the trivial coefficient bound gives $\|u_M\|_{\sigma_0} \ll_{\sigma_0} M^{1-\sigma_0}$. Indeed, write $x = 1/y$. For $n \le y$, $|g_n(1/y)| \le 2$, and for $n > y$, $|g_n(1/y)| = \lfloor y \rfloor/n$. Consequently $|u_M(1/y)| \ll y(1 + \log(M/y))$ when $1 \le y \le M$, and $|u_M(1/y)| \ll M$ otherwise. Integrating the square with weight $y^{-2\sigma_0-1}$ gives the stated bound, using $y = Mv$ in the first integral. Even with $\delta$ at its limiting value $\sigma_1 - \theta$, the numerator of the interpolated exponent is

$$(\sigma_1 - \tfrac{1}{2})(1 - \sigma_0) - (\tfrac{1}{2} - \sigma_0)(\sigma_1 - \theta) = \tfrac{1}{2}(\sigma_1 - \tfrac{1}{2}) + (\tfrac{1}{2} - \sigma_0)(\theta - \tfrac{1}{2}) > 0.$$

Thus this interpolation with the available lower-norm estimate cannot prove decay at 1/2.

14.3. **Why the explicit family stops at the critical norm.** The failed interpolation does not settle whether the same coefficients might converge by a different argument. The following rescaling calculation answers that question. The obstruction is present even if all zeta zeros lie on the line. For related failures of natural approximants, see Báez-Duarte [14].

[4] A boundary point on $\operatorname{Re} s = \sigma$ is not a bounded evaluation in the weighted Hilbert space. The argument for a hypothetical zero strictly to the right therefore gives no pointwise evaluation bound on the line itself.

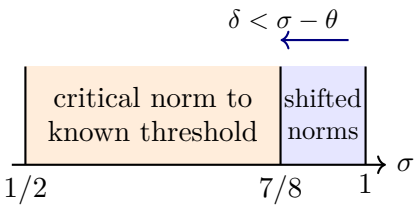



FIGURE 7. The shifted-norm construction with the 7/8 input. The rate in Theorem 14.3 depends on the distance to the right of the zero-free boundary.

**Theorem 14.4** (A critical-norm obstruction for the linear Riesz family)**.** *For $M \geq 2$ let*

$$u_M(x) = 1 + \sum_{n \leq M} \mu(n)(1 - n/M) g_n(x).$$

*Then* $\limsup_{M\to\infty} \|u_M\|_{1/2} > 0$. *The infimum over coefficients defining $d_{M,1/2}$ is a separate problem.*

*Proof.* Put $b_m = \sum_{n|m} n\mu(n) = \prod_{p|m}(1-p)$ and $B_1(y) = \sum_{m\leq y} b_m$. For $1 \leq y \leq M$,

$$u_M(1/y) = \lfloor y \rfloor D_M(1) + M^{-1} B_1(y). \tag{14.2}$$

Indeed $g_n(1/y) = \lfloor y \rfloor / n - \lfloor y/n \rfloor$. In the second term the unweighted sum $\sum_{n\leq M} \mu(n)\lfloor y/n \rfloor$ equals 1, by the divisor identity $\sum_{n|m} \mu(n) = \mathbf{1}_{m=1}$. The remaining weighted sum is $B_1(y)/M$. Formula (14.2) holds for real $M$ as well as integers.

Suppose that $\|u_M\|_{1/2} \to 0$ along the integers. The same would hold for real $M$. To see this, put $m = \lfloor M \rfloor$. Only the coefficients at $n \leq m$ contribute to the difference, and their differences have absolute value at most $n/m^2$. As $|g_n| \leq 2$, the differences of the functions are uniformly bounded. At fixed $x = 1/y \in (0,1)$, for $n > y$ one has $g_n(x) = \lfloor y \rfloor / n$, so the difference tends to zero pointwise. Dominated convergence gives the claim.

Choose a nontrivial zero $\rho = 1/2 + i\gamma$ on the line, whose existence is classical, and put $s_0 = 1 + \rho$. Choose $W \in C_c^\infty(0,1)$ with

$$\int_0^1 vW(v)\,\mathrm{d}v = 0, \qquad \int_0^1 v^{s_0} W(v)\,\mathrm{d}v \neq 0.$$

Such a weight can be constructed from a nonnegative bump $\psi$ supported on a closed interval $[a,b] \subset (0,1)$. Set $W = \psi(\overline{v^{s_0}} - cv)$ with $c = \int \psi v \,\overline{v^{s_0}} / \int \psi v^2$. Strict Cauchy–Schwarz gives the second condition because $v^{s_0}$ is not proportional to $v$ on an open interval.

Write $G(X) = \int_0^1 B_1(Xv) W(v)\,\mathrm{d}v$. Applying (14.2) on the support of $W$ for $X \geq 1/a$ gives

$$X^{-3/2} G(X) = X^{-1/2} \int_0^1 u_X(1/(Xv)) W(v)\,\mathrm{d}v + O_W(X^{-1/2}|D_X(1)|).$$

The vanishing moment eliminates $Xv$ from the floor term. Since $|D_X(1)| \ll \log(2X)$, the error tends to zero. In the main term, substitute $x = 1/(Xv)$ and apply Cauchy–Schwarz. This bounds its absolute value by

$$\|vW\|_{L^2(0,1)} \|u_X\|_{1/2}.$$

Thus $G(X) = o(X^{3/2})$ would follow.

For $\operatorname{Re} s > 2$, absolute convergence and partial summation give

$$\int_0^\infty G(X) X^{-s-1} \, \mathrm{d}X = \left( \int_0^1 v^s W(v) \, \mathrm{d}v \right) \frac{\zeta(s)}{s\,\zeta(s-1)}.$$

In detail, the Mellin integral of $B_1$ is $s^{-1} \sum b_m m^{-s}$. Its Dirichlet series is $\zeta(s)/\zeta(s-1)$ by the divisor formula for $b_m$. Substitution $y = Xv$ gives the factor in parentheses. The assumed little-oh bound makes the left side holomorphic on $\operatorname{Re} s > 3/2$. It also implies that its value at $s_0 + \epsilon$, multiplied by $\epsilon$, tends to zero as $\epsilon \downarrow 0$. Split the integral at a fixed large $X_0$ and bound the tail using $|G(X)| \le o_{X_0 \to \infty}(1) X^{3/2}$ for $X \ge X_0$.

The right side has a pole at $s_0$ by the choice of $W$, the Euler product at $\operatorname{Re} s_0 = 3/2$, and the zero of $\zeta(s_0 - 1)$. The meromorphic identity theorem gives a contradiction to the last limit, regardless of the multiplicity of $\rho$. This excludes the assumed convergence. □

For a simple $\rho$, the same calculation gives the lower bound

$$\limsup_M \|u_M\|_{1/2} \ge \frac{|\zeta(s_0) \int_0^1 v^{s_0} W(v) \, \mathrm{d}v|}{|s_0 \zeta'(\rho)| \, \|vW\|_{L^2(0,1)}}.$$

Indeed, finite limit superior of the norms bounds the limit superior of $|G(X)| X^{-3/2}$ by that of the norms times $\|vW\|_2$. The Abel integral bounds the absolute residue by the same quantity. If $\rho$ is multiple, finite limit superior would give only an $O(\epsilon^{-1})$ bound at $s_0 + \epsilon$, incompatible with its higher-order pole.

**Theorem 14.5** (Every fixed taper of bounded variation)**.** *Let $w : [0,1] \to \mathbb{C}$ have bounded variation and $w(0+) = 1$. Define*

$$U_M(x) = 1 + \sum_{n \le M} \mu(n) w(n/M) g_n(x).$$

*For each nontrivial zero $\rho$ with $\beta = \operatorname{Re} \rho \ge 1/2$,*

$$\limsup_{M \to \infty} M^{1/2-\beta} \|U_M\|_{1/2} > 0.$$

*This limit superior is infinite if $\rho$ is multiple. In particular the residuals cannot converge to zero in the critical norm, even under RH. The result includes every fixed real order $r > 0$ of Riesz smoothing, $w(t) = (1-t)^r$.*

*Proof.* Extend $w$ by zero on $t > 1$, and set $f(t) = 1 - w(t)$ for $t > 0$. Thus $f$ is bounded, with limits zero at zero and one at infinity. For $X \ge 2$ real let

$D_{X,w}(1) = \sum_{n\le X} \mu(n)w(n/X)/n$. The divisor identity used in (14.2) now gives, for $Xv \ge 1$,

$$U_X(1/(Xv)) = \lfloor Xv\rfloor D_{X,w}(1) + \sum_{n\le Xv} \mu(n) f(n/X)\lfloor Xv/n\rfloor. \tag{14.3}$$

There is no restriction $v \le 1$ here, since $w$ has been extended by zero.

Passing from integer $m = \lfloor X\rfloor$ to real $X$ changes the residual by $o(1)$ in the critical norm. Indeed $n/X \in [(n-1)/m, n/m]$ for $n \le m$, so $\sum_{n\le m} |w(n/X) - w(n/m)|$ is at most the total variation of $w$. The residual difference is uniformly bounded by $2\,\mathrm{Var}(w)$. At fixed $y \ge 1$, the finitely many terms $n \le K$ tend to zero as $m \to \infty$. For $K > y$, the remaining terms contribute at most $\lfloor y\rfloor\,\mathrm{Var}(w)/K$, because $g_n(1/y) = \lfloor y\rfloor/n$ for $n > y$. Let $K \to \infty$ and use dominated convergence. The passage to real $X$ preserves the scaled limit superior for $\beta \ge 1/2$.

Take a test function $W \in C_c^\infty(0,\infty)$ with $\int vW(v)\,\mathrm{d}v = 0$ and write

$$K_W(t) = f(t)\int_0^\infty W(v)\lfloor v/t\rfloor\,\mathrm{d}v.$$

The integral factor equals $-\int W(v)\{v/t\}\,\mathrm{d}v$ by the first moment condition. It is bounded at zero and vanishes for $t > \sup\operatorname{supp} W$. Therefore $\widehat{K_W}(s) = \int_0^\infty K_W(t)t^{s-1}\,\mathrm{d}t$ is holomorphic on $\operatorname{Re} s > 0$. With $H(X) = \sum_{n\ge1}\mu(n)K_W(n/X)$, equation (14.3) and the rescaling of the norm give

$$|H(X)| \ll_W \sqrt{X}\,\|U_X\|_{1/2} + \log(2X)$$

for large $X$. The logarithmic term comes from replacing $\lfloor Xv\rfloor$ by $Xv$ and using $D_{X,w}(1) = O(\log(2X))$. For $\operatorname{Re} s > 1$, termwise integration gives

$$\int_0^\infty H(X)X^{-s-1}\,\mathrm{d}X = \frac{\widehat{K_W}(s)}{\zeta(s)}. \tag{14.4}$$

We show that $W$ can be chosen with $\widehat{K_W}(\rho) \ne 0$. By Fubini,

$$\widehat{K_W}(\rho) = \int_0^\infty W(v)J_\rho(v)\,\mathrm{d}v, \qquad J_\rho(v) = -\int_0^\infty f(t)\{v/t\}t^{\rho-1}\,\mathrm{d}t.$$

The integral is absolutely convergent because $0 < \beta < 1$, and the substitution $t = v/u$ bounds $|J_\rho(v)| \ll v^\beta$. It is continuous by dominated convergence on compact $v$-intervals, since for fixed $v$ only a null set of $t$ lies at jumps of $\{v/t\}$. If every test of first moment zero annihilated it, it would be proportional to $v$ as a distribution and hence as a continuous function. For instance, subtract a multiple of a fixed smooth test of first moment one from an arbitrary test. The growth bound at infinity then forces $J_\rho \equiv 0$.

To see why this is impossible, put $F(r) = f(e^r)$ and $k_\rho(u) = \{e^u\}e^{-\rho u}$ for $r, u \in \mathbb{R}$. Substitution gives $e^{-\rho r}J_\rho(e^r) = -(k_\rho * F)(r)$. The kernel $k_\rho$ is integrable, with exponential tails. Its Fourier transform with convention $e^{-i\xi u}$ is $-\zeta(\rho + i\xi)/(\rho + i\xi)$, by the fractional-part integral proved in Lemma 14.2. Its real zeros are isolated. The distributional derivative $\nu = \mathrm{d}F$ is a finite complex measure, since $F$ has bounded variation. Its total mass is $F(+\infty) -$

$F(-\infty) = 1$. If $k_\rho * F = 0$, differentiation gives $k_\rho * \nu = 0$.[5] Fourier transformation gives $\widehat{k}_\rho\,\widehat{\nu} = 0$. But $\widehat{\nu}$ is continuous and equals 1 at zero, so it is nonzero on a neighborhood of zero. This would force $\widehat{k}_\rho$ to vanish on that neighborhood, a contradiction. Thus a detecting test exists.

With that choice of $W$, the right side of (14.4) has a pole at $\rho$. If the scaled limit superior were zero, then $H(X) = o(X^\beta)$. The Mellin integral would extend holomorphically to $\operatorname{Re} s > \beta$, and multiplication by $\epsilon$ at $\rho + \epsilon$ would give a limit of zero by the Abel estimate in Theorem 14.4. This contradicts the pole. If the scaled limit superior were finite, the same argument with $H(X) = O(X^\beta)$ would give at most $O(\epsilon^{-1})$ at $\rho + \epsilon$, which excludes a higher-order pole. □

**Corollary 14.6** (Exact growth of the linear residual)**.** *For the linear Riesz family of Theorem 14.4,*

$$\limsup_{M\to\infty} \frac{\log(1 + \|u_M\|_{1/2})}{\log M} = \Theta - \tfrac{1}{2},$$

*where $\Theta$ is as in Corollary 7.2. Consequently RH is equivalent to $\|u_M\|_{1/2} \ll_\varepsilon M^\varepsilon$ for every $\varepsilon > 0$, although these norms do not tend to zero.*

*Proof.* Theorem 14.5 gives the lower bound by taking zeros with real part approaching $\Theta$. For $\Theta = 1/2$ the lower bound is automatic. For the upper bound take $s = 1/2 + it$ in the Riesz formula for $D_M(s)$ and shift the line of $w$ to $\operatorname{Re} w = \Theta - 1/2 + \delta > 0$, with $0 < \delta < 1/4$. There is no residue. On that line the reciprocal has the bound of Lemma A.1 when $\Theta < 1$, or the absolutely convergent Euler bound if $\Theta = 1$. The kernel estimate used in Theorem 14.3 gives

$$|D_M(1/2 + it)| \ll_{\delta,\eta} M^{\Theta-1/2+\delta}(1 + |t|)^\eta.$$

Also $|D_M(1)| \ll \log(2M)$. The classical convexity bound $|\zeta(1/2 + it)| \ll_\eta (1+|t|)^{1/4+\eta}$ [26, Ch. V] makes the Mellin residual $[1-\zeta(s)(D_M(s)-D_M(1))]/s$ square integrable with $L^2$ norm $O_{\delta,\eta}(M^{\Theta-1/2+\delta})$, using $\eta < 1/8$. Plancherel yields the same bound in the critical norm. Let $\delta \downarrow 0$. □

The obstruction applies to a profile held fixed as $M$ changes. It leaves the unrestricted finite-dimensional projection problem open. The logarithmic taper of Bettin, Conrey and Farmer [3] changes with $M$ and lies outside the theorem. Its conditional asymptotic uses RH and additional information about reciprocal zeta derivatives.

14.4. **A coefficient cost for adaptive approximants.** The fixed-profile obstruction leaves open coefficients that change shape with $M$. The following necessary condition applies to every finite choice of coefficients and does not depend on a choice of profile.

[5] One may differentiate after pairing with a compactly supported smooth test. Fubini is justified by $k_\rho \in L^1$, boundedness of $F$, and finiteness of the variation measure. The convolution $k_\rho * \nu$ is in $L^1$.

**Theorem 14.7** (Harmonic coefficient cost)**.** *Let $U = 1 + \sum_{n=2}^{M} a_n g_n$, let $d = \|U\|_{1/2}$, and put $a_n = 0$ for $n > M$. For every integer $K \geq 2$,*

$$\left(\sum_{n\geq 2} \frac{|a_n|^2}{n}\right)^{1/2} \geq \left(\left(\sum_{2\leq n\leq K} \frac{\mu(n)^2}{n}\right)^{1/2} - 2d\sqrt{K} \sum_{m\leq K} \frac{1}{m}\right)_+ .$$

*Consequently, along any family of approximants with $d \downarrow 0$,*

$$\sum_{n\geq 2} \frac{|a_n|^2}{n} \geq \left(\frac{12}{\pi^2} - o(1)\right) \log(1/d).$$

*Proof.* The coefficientwise necessity has a precedent in Báez-Duarte [14, Lemma 4.2]. We give the quantitative estimate. The function $U(1/y)$ is constant on $k \leq y < k+1$, $k \geq 1$. Write this value as $V_k$, and define $V_0 = 0$. With $C = \sum_{n\geq 2} a_n/n$, the floor formula gives

$$V_k = 1 + kC - \sum_{n\geq 2} a_n \lfloor k/n \rfloor, \qquad \Delta V_r = C + \mathbf{1}_{r=1} - \sum_{\substack{n|r \\ n\geq 2}} a_n.$$

Möbius inversion therefore yields $a_n - \mu(n) = -\sum_{r|n} \mu(n/r)\Delta V_r$ for every $n \geq 2$. Also $d^2 = \sum_{k\geq 1} |V_k|^2/[k(k+1)]$. Thus

$$\sum_{r\leq R} \frac{|\Delta V_r|^2}{r} \leq 2 \sum_{r\leq R} \frac{|V_r|^2}{r} + 2 \sum_{r\leq R} \frac{|V_{r-1}|^2}{r} \leq 4Rd^2.$$

The first sum is at most $2(R+1)d^2$, and the second at most $2(R-1)d^2$, using $V_0 = 0$. Minkowski's inequality applied to the divisor sum, with $n = mr$, now gives

$$\begin{aligned}\left(\sum_{2\leq n\leq K} \frac{|a_n - \mu(n)|^2}{n}\right)^{1/2} &\leq \sum_{m\leq K} \frac{1}{\sqrt{m}} \left(\sum_{r\leq K/m} \frac{|\Delta V_r|^2}{r}\right)^{1/2} \\ &\leq 2d\sqrt{K} \sum_{m\leq K} \frac{1}{m}.\end{aligned}$$

The triangle inequality proves the finite bound.

For completeness, $\sum_{n\leq K} \mu(n)^2/n = (6/\pi^2)\log K + O(1)$. Indeed $\mu(n)^2 = \sum_{r^2|n} \mu(r)$ gives a sum of $\mu(r)r^{-2}H_{\lfloor K/r^2 \rfloor}$ for $r \leq \sqrt{K}$. Insert $H_\ell = \log \ell + \gamma + O(1/\ell)$. The series $\sum \mu(r)/r^2 = 1/\zeta(2)$ and $\sum \mu(r) \log r/r^2$ converge absolutely, and the tail and floor errors are $O((1+\log K)/\sqrt{K})$. Euler's evaluation $\zeta(2) = \pi^2/6$ gives the coefficient. Finally take $K = \lfloor d^{-2}/(\log(1/d))^4 \rfloor$. The error in the finite bound is $O(1/\log(1/d))$, while $\log K = 2\log(1/d) + O(\log\log(1/d))$. Squaring proves the claim. □

A uniformly bounded harmonic coefficient norm therefore excludes convergence even for adaptive coefficients. This is a necessary constraint on a possible construction. It gives no lower bound for the unrestricted distance by itself.

**Corollary 14.8** (Explicit coefficient bound)**.** *For $0 < d \le e^{-4}$, the coefficients in Theorem 14.7 satisfy*

$$\sum_{n\ge 2} \frac{|a_n|^2}{n} \ge \frac{12}{\pi^2}\log(1/d) - \frac{24}{\pi^2}\log\log(1/d) - 15.$$

*Proof.* Write $c = 6/\pi^2$. The squarefree identity gives $\left|\sum_{n\le t}\mu(n)^2 - ct\right| \le 3\sqrt{t}$ for $t \ge 1$, by bounding the floor errors by $\sqrt{t}$ and the tail of $\sum r^{-2}$ by $2/\sqrt{t}$. Partial summation then gives $\sum_{2\le n\le K}\mu(n)^2/n \ge c\log K - 7$. Put $L = \log(1/d) \ge 4$ and $K = \lfloor e^{2L}/L^4\rfloor$. Then $\log K \ge 2L - 4\log L - \log 2$, and $2d\sqrt{K}H_K \le (4L+2)/L^2 \le 4.5/L$. Apply the finite bound with $A = (c\log K - 7)_+$, use $(\sqrt{A} - e)_+^2 \ge A - 2\sqrt{A}\,e$ and $\sqrt{A} \le \sqrt{2cL}$. The cross term is at most $9\sqrt{2c}/\sqrt{L} < 6.4$, and $7 + c\log 2 < 8$. These bounds give the stated constant 15. □

14.5. **A scale-dependent logarithmic taper.** A logarithmic taper is one way to leave the class in Theorem 14.5. For $M > 1$ set

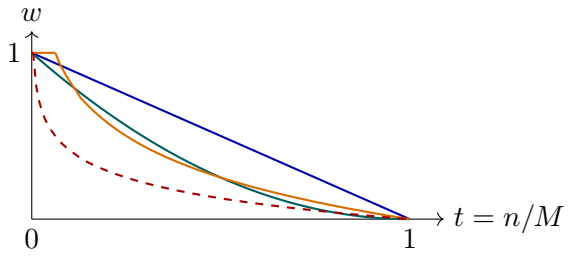
$$D_M^{\log}(s) = \sum_{n\le M}\mu(n)\left(1 - \frac{\log n}{\log M}\right)n^{-s},$$

$$u_M^{\log} = 1 + \sum_{n\le M}\mu(n)\left(1 - \frac{\log n}{\log M}\right)g_n.$$

Suppose again that $\zeta$ has no zeros on $\operatorname{Re} s > \theta$, and fix $\theta < \sigma < 1$. Then $(\log M)u_M^{\log}$ converges in the norm $\|\cdot\|_\sigma$ to the function with Mellin transform

$$\frac{\zeta(s) + \zeta'(s)/\zeta(s)}{s} \qquad (\operatorname{Re} s = \sigma). \tag{14.5}$$

The norm error is $O(M^{-\delta})$ for every $0 < \delta < \sigma - \theta$. The limit function is $\lfloor 1/x\rfloor - \psi(1/x)$, where $\psi(y) = \sum_{n\le y}\Lambda(n)$.




$1 - t$ $(1-t)^2$ logarithmic, $M = 16$ logarithmic, $M = 256$

FIGURE 8. Fixed profiles and the changing logarithmic profile. The latter is drawn as $\min(1, -\log t/\log M)$, which agrees with the coefficient taper on $1/M \le t \le 1$. For each fixed $t > 0$ it tends to zero as $M$ grows.

To prove this, replace the Riesz kernel by $M^w/(w^2 \log M)$. The scalar inverse transform is $(\log(M/n))_+/\log M$, by the double residue at zero. Shift the line to $\operatorname{Re} w = -\delta$ as in Theorem 14.3. For $F = 1/\zeta$ this gives

$$D_M^{\log}(s) = F(s) + \frac{F'(s)}{\log M} + O_{\sigma,\delta,\eta}\left(\frac{M^{-\delta}(1+|t|)^{\eta}}{\log M}\right)$$

on $\operatorname{Re} s = \sigma$. At $s = 1$, the same calculation gives $D_M^{\log}(1) = 1/\log M + O(M^{-\delta}/\log M)$ since $F(1) = 0$ and $F'(1) = 1$. Insert these formulas in $[1 - \zeta(s)(D_M^{\log}(s) - D_M^{\log}(1))]/s$. The leading transform after multiplication by $\log M$ is (14.5). Lemma A.1, Cauchy's estimate for $F'$ on a fixed substrip, and the corresponding subpower bound for $\zeta(s)$ give square integrability. Plancherel bounds the error by $O(M^{-\delta})$.

For $1 \le y < M$ the formula in the $x$-variable is exact:

$$(\log M) u_M^{\log}(1/y) = (\log M) D_M^{\log}(1) \lfloor y \rfloor - \psi(y).$$

To verify it, expand the floors in $g_n(1/y)$ and use $\sum_{n|m} \mu(n) \log n = -\Lambda(m)$. The latter follows by taking the logarithmic derivative of the Euler product, or directly at a prime power and by cancellation over the subsets of distinct prime factors. Since the coefficient of $\lfloor y \rfloor$ tends to one, this identifies the pointwise limit and hence the shifted $L^2$ limit. At the critical line the proposed limiting transform has a pole at each zero on the line. The scaled estimate therefore does not extend to that norm. It gives no conclusion about unscaled convergence there.

**Proposition 14.9** (An interior criterion for the logarithmic taper)**.** *For integer $M \ge 2$, put*

$$\widetilde{u}_M = u_M^{\log} + M\big(1/\log M - D_M^{\log}(1)\big) g_M.$$

*Then RH is equivalent to boundedness of $\|\widetilde{u}_M\|_{L^2(1/M,1)}$. It is also equivalent to convergence of this norm to zero. Under RH its square is $O(1/\log M)$.*

*Proof.* For $1 \le y < M$, $g_M(1/y) = \lfloor y \rfloor / M$. The last coefficient in the definition replaces $(\log M) D_M^{\log}(1)$ by one in the exact interior identity. Thus

$$\|\widetilde{u}_M\|^2_{L^2(1/M,1)} = \frac{1}{(\log M)^2} \int_1^M |\psi(y) - \lfloor y \rfloor|^2 \frac{\mathrm{d}y}{y^2}.$$

Under RH, Cramér's mean-square estimate gives $\int_Y^{2Y} |\psi(y) - y|^2 \, \mathrm{d}y \ll Y^2$, as recalled in [4]. Replacing $y$ by $\lfloor y \rfloor$ changes this upper bound by at most a fixed factor and $O(Y)$. Each dyadic block therefore contributes $O(1)$ to the weighted integral. Summing gives the asserted rate.

Conversely, bounded interior norms give a weighted integral $O((\log(2M))^2)$. On every dyadic block $[Y, 2Y]$, Cauchy–Schwarz then bounds $\int_Y^{2Y} |\psi(y) - \lfloor y \rfloor| y^{-\sigma-1} \, \mathrm{d}y$ by $O_\sigma((\log(2Y)) Y^{1/2-\sigma})$. Thus the Mellin transform of $\psi(y) - \lfloor y \rfloor$ is holomorphic on $\operatorname{Re} s > 1/2$. For $\operatorname{Re} s > 1$ it equals $(-\zeta'(s)/\zeta(s) - \zeta(s))/s$, by the absolutely convergent Dirichlet series of $\Lambda$ and of the constant

sequence. A zero in $\operatorname{Re} s > 1/2$ would give a pole there, and the functional equation completes the implication to RH. Convergence implies boundedness, which closes the equivalence. □

14.6. **The tail and quantitative tests at a zero.** Write $\ell = \log M$, $E(y) = \lfloor y \rfloor - \psi(y)$ and $q_M = M(1 - \ell D_M^{\log}(1))$. The balanced residual has the exact formula

(14.6)
$$\ell \widetilde{u}_M(1/y) = E(y) - \sum_{M<n\le y} \mu(n)\log(n/M)\lfloor y/n\rfloor - q_M\lfloor y/M\rfloor, \qquad y \ge 1.$$

Indeed the untruncated sum with weight $\mu(n)\log(M/n)\lfloor y/n\rfloor$ is $\ell + \psi(y)$. On removing its terms with $n > M$, expanding the balanced dilates, and inserting the coefficient of $g_M$, one obtains (14.6). The tail norm squared on $0 < x < 1/M$ is the integral of the squared right-hand side against $\ell^{-2}y^{-2}\,\mathrm{d}y$ on $y > M$. The terms must be kept together in this expression.

For any sequence $u_M$ in $L^2(0,1)$ with $\|u_M\|_\sigma \to 0$ at one $\sigma > 1/2$, convergence in the critical norm holds if and only if

$$\lim_{r\downarrow 0} \limsup_{M\to\infty} \int_0^r |u_M(x)|^2\,\mathrm{d}x = 0.$$

The necessity follows by bounding the tail by the whole integral. For sufficiency, the part on $[r,1]$ is at most $r^{1-2\sigma}\|u_M\|_\sigma^2$. For instance, $u_M = \sqrt{M}\mathbf{1}_{(0,1/M)}$ has critical norm one and shifted norm squared $M^{1-2\sigma}/(2\sigma)$. Convergence at every fixed shifted exponent thus allows concentration at the endpoint. For the explicit linear taper, splitting at $r = M^{-a}$ and using $\|u_M\|_\sigma \ll M^{-\delta}$ and $|u_M| \ll M$ gives the upper bound $O(M^{a(2\sigma-1)-2\delta} + M^{2-a})$ for the critical norm squared. These estimates require both $a > 2$ and $a < \delta/(\sigma - 1/2) \le 1$ to give decay.

**Proposition 14.10** (Growth forced by an off-line zero)**.** *Let $\rho$ be a zeta zero of order $m$ with $\beta = \operatorname{Re}\rho > 1/2$, and set $I(Y) = \int_1^Y |E(y)|^2\,\mathrm{d}y/y^2$. Then*

$$\limsup_{X\to\infty} \frac{|B(X)|}{X^\beta(\log X)^{m-1}} \ge \frac{m|\Gamma(\rho)\zeta(2\rho)|}{|\zeta^{(m)}(\rho)|},$$
$$\limsup_{Y\to\infty} \frac{I(Y)}{Y^{2\beta-1}} \ge \frac{m^2}{(2\beta-1)|\rho|^2}.$$

*Proof.* If the first limit superior is finite, denote it by $A$. For any $a > A$, the large-$X$ portion of the integral in (14.1) is holomorphic for $\operatorname{Re} s > \beta$. At $s = \rho + \varepsilon$, multiplying it by $\varepsilon^m$ gives a modulus bounded by $a(m-1)!$, using $\int_1^\infty X^{-\varepsilon}(\log X)^{m-1}\,\mathrm{d}X/X = (m-1)!/\varepsilon^m$. The bounded-$X$ portion tends to zero after multiplication. By continuation from $\operatorname{Re} s > 1$, the limit is the leading Laurent coefficient $m!\Gamma(\rho)\zeta(2\rho)/\zeta^{(m)}(\rho)$. Let $a \downarrow A$.

For the second inequality put $\alpha = 2\beta - 1 > 0$. If the second limit superior is $C < \infty$, choose $c > C$ so that $I(y) \le cy^\alpha$ for all sufficiently large $y$. For

$\varepsilon > 0$, integration by parts gives

$$\int_1^\infty y^{-\alpha-\varepsilon}\,\mathrm{d}I(y) \le O(1) + c\frac{\alpha+\varepsilon}{\varepsilon} \qquad (0 < \varepsilon \le 1).$$

Cauchy–Schwarz applied to the Mellin transform $F_E(s) = \int_1^\infty E(y)y^{-s-1}\,\mathrm{d}y$ now gives

$$|F_E(\rho+\varepsilon)|^2 \le \left(\int_1^\infty y^{-\alpha-\varepsilon}\,\mathrm{d}I(y)\right)\left(\int_1^\infty y^{-2\beta-\varepsilon+\alpha}\,\mathrm{d}y\right) = \frac{1}{\varepsilon}\int_1^\infty y^{-\alpha-\varepsilon}\,\mathrm{d}I(y).$$

The same bound on absolute integrals is locally uniform for $\operatorname{Re} s > \beta$, so the transform is holomorphic there. Its continuation from $\operatorname{Re} s > 1$ is $(\zeta(s)+\zeta'(s)/\zeta(s))/s$, as in Proposition 14.9. The residue at $\rho$ is $m/\rho$. Taking $\varepsilon \downarrow 0$ yields $m^2/|\rho|^2 \le c\alpha$, and then $c \downarrow C$ proves the bound. Infinite limit superiors require no estimate. □

Since $I$ is nondecreasing, the second lower bound also holds with $Y$ restricted to integers. By Proposition 14.9, it gives a positive lower limit superior for $M^{1-2\beta}(\log M)^2\|\widetilde{u}_M\|^2_{L^2(1/M,1)}$. At a high zero the Gamma factor in the first inequality is small, but the exponent $\beta$ still exceeds $1/2$. A zero-density estimate allows isolated zeros in this range. To exclude them by either route requires an upper estimate that contradicts the corresponding lower bound.

**Corollary 14.11** (Power growth of the balanced logarithmic taper)**.** *If $\Theta$ is the supremum of the real parts of the nontrivial zeros, then*

$$\limsup_{M\to\infty} \frac{\log(1+\|\widetilde{u}_M\|_{1/2})}{\log M} = \Theta - \frac{1}{2}.$$

*Proof.* For $\Theta > 1/2$, Proposition 14.10 gives a lower bound $\beta - 1/2$ for each zero with $\beta > 1/2$. The added one makes the lower bound zero automatic if $\Theta = 1/2$.

For the upper bound take $s = 1/2 + it$ in the Mellin integral for $D_M^{\log}(s)$ and move the $w$-line from two to $\Theta - 1/2 + \delta > 0$, for $0 < \delta < 1/4$. The rectangle has no poles. Lemma A.1 bounds the reciprocal on the new line by $O_{\delta,\eta}((1+|t+\operatorname{Im} w|)^\eta)$ for every $\eta > 0$ if $\Theta < 1$, and absolute convergence suffices if $\Theta = 1$. As for the linear taper, the kernel's quadratic decay gives

$$|D_M^{\log}(1/2+it)| \ll_{\delta,\eta} \frac{M^{\Theta-1/2+\delta}(1+|t|)^\eta}{\log M}.$$

The Mellin formula for $u_M^{\log}$, convexity on the critical line, and Plancherel give $\|u_M^{\log}\|_{1/2} \ll_\delta M^{\Theta-1/2+\delta}$, since $D_M^{\log}(1) = O(\log M)$ and one may take $\eta < 1/8$.

The bound $\|g_M\|_{1/2} \ll M^{-1/2}$ follows from $|g_M| \le 2$ on $(0, 1/M)$ and $|g_M(x)| \le 1/(Mx)$ on $[1/M, 1]$. If $\Theta < 1$, the double-residue calculation at $s = 1$ with $w$-line $\Theta - 1 + \delta$, for $0 < \delta < 1 - \Theta$, gives

$$D_M^{\log}(1) = 1/\log M + O_\delta(M^{\Theta-1+\delta}/\log M).$$

The balancing term thus has norm $O_\delta(M^{\Theta-1/2+\delta}/\log M)$. For $\Theta = 1$ the elementary bound for $D_M^{\log}(1)$ instead gives $O(\sqrt{M}\log M)$. Combining the bounds and sending $\delta \downarrow 0$ gives the upper exponent. In the case $\Theta = 1/2$, this exponent alone gives no critical-norm convergence. □

14.7. **What the quasi-Riemann proof transfers.** In Theorem 1.1 of the September 30 manuscript [22], the zero-free statement is proved first for the finite-order Hecke family over $\mathbb{Q}(\sqrt{-3})$, and then transferred to Dirichlet $L$-functions by quadratic factorization. The underlying continuation step compares two estimates for the same smoothed inverse signal. To describe its logic, write $b$ for a hypothetical supremum of the real parts of the zeros in the family and $\theta_0 < b$ for the proposed threshold. Suppose a smoothed Mellin signal with a nonvanishing multiplier has a comparison $J(X)$ satisfying

$$J(X) = O(X^{\theta_0+c+\omega}),$$
$$J(X) - \text{signal}(X) = O(X^{b+c-\tau}), \qquad 0 < \omega < b - \theta_0, \quad \tau > 0.$$

Taking the triangle inequality gives $O(X^{b+c-\min(b-\theta_0-\omega,\tau)})$. If the exponent saving is uniform over the family, Mellin continuation of the type in (14.1), with the shift $c$, rules out a zero arbitrarily close to $b$. The arithmetic work is in obtaining the two estimates with a uniform positive margin. Part I, Theorem 3.1, uses a completed cubic-theta sum and gives 11/12. Part II uses prime compensation, asymmetric scales and two additional moment estimates to get 7/8. Section 20 combines those moment estimates with the row counts.[6]

The final factorization can be seen directly at each unramified prime. Let $\chi_{-3}$ be the quadratic character of $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$. At a split prime $p$, the norm lift of $\chi$ has two factors $(1-\chi(p)p^{-s})^{-1}$. At an inert prime, the factor is $(1-\chi(p)^2p^{-2s})^{-1}$, which equals the product of the factors with $\chi(p)$ and $-\chi(p)$. These are precisely the two Dirichlet factors of $L(s,\chi)L(s,\chi\chi_{-3})$. Excluding the finitely many ramified and conductor primes gives the identity for incomplete products in $\operatorname{Re} s > 1$, then by meromorphic continuation. Removed Euler factors are nonzero on $\operatorname{Re} s > 0$. This transfers nonvanishing away from the principal pole. At $s = 1$, use the classical nonvanishing of nonprincipal Dirichlet $L$-functions.

The Pólya–Vinogradov v3 paper [6] uses the 11/12 input in Theorem 1.5. Its Proposition 3.7 applies a Riesz kernel to prime sums in a rectangle, and its Theorem 1.1 combines this with additive dissection and harmonic weights. We use the Riesz kernel in Theorem 14.3, where the object is an inverse zeta polynomial. The character-sum bound in [6] does not transfer to the shifted Liouville products. For instance $a(n) = \lambda(n)\lambda(n+1)$ has $a(3) = -1$, $a(7) = 1$ and $a(21) = 1$. It fails the multiplicative identity on the coprime

[6]The Mellin division step is safe only when the multiplier is nonvanishing in the region being tested. Proposition 2.1 of [22] uses a Gaussian and a holomorphic factor within 1/2 of one.

pair $3, 7$. Nor does the fixed $11/12$ or $7/8$ continuation estimate provide the needed uniform margin at $1/2$.

## 15. The character version and GRH

For a Dirichlet character $\chi$, including the primitive character of conductor 1, put $a_\chi(n) = \lambda(n)\chi(n)$. Define $L^\lambda_\chi(N) = \sum_{n\le N} a_\chi(n)$ and

$$Q_\chi(N) = \frac{1}{N}\sum_{t\in\mathbb{Z}} \left| \sum_{t<n\le t+N} a_\chi(n)\mathbf{1}_{[1,N]}(n) \right|^2 .$$

The boundary identity (6.4) holds with complex moduli in each term. Thus $\frac{1}{2}|L^\lambda_\chi(N)|^2 \le Q_\chi(N) \le 5\max_{j\le N}|L^\lambda_\chi(j)|^2$. The Fourier identity holds as well, with $C_h = \sum_{n\le N-h} a_\chi(n+h)\overline{a_\chi(n)}$ and the real part of its weighted sum.

**Theorem 15.1** (Signed energy for Dirichlet characters)**.** *Let $1/2 \le \theta \le 1$. The following conditions are equivalent.*

1. *Every Dirichlet $L$-function has no zero in $\operatorname{Re} s > \theta$.*
2. *For every fixed character $\chi$ and every $\varepsilon > 0$, $L^\lambda_\chi(N) \ll_{\chi,\varepsilon} N^{\theta+\varepsilon}$.*
3. *For every fixed character $\chi$ and every $\varepsilon > 0$, $Q_\chi(N) \ll_{\chi,\varepsilon} N^{2\theta+\varepsilon}$.*

*At $\theta = 1/2$ these are equivalent to Dirichlet GRH.*

*Proof.* The boundary inequalities give the equivalence of the second and third conditions. Under the second condition partial summation extends

$$F_\chi(s) = \sum_{n\ge1} \lambda(n)\chi(n)n^{-s} = \frac{L(2s,\chi^2)}{L(s,\chi)}$$

holomorphically to $\operatorname{Re} s > \theta$. The numerator has an absolutely convergent nonvanishing Euler product there. This proves the first condition.

For the reverse implication fix $\chi$. The proof of Lemma A.1 applies to $L(s,\chi)$. For a nonprincipal character, the partial sums of $\chi$ are bounded by its modulus, and partial summation gives $L(s,\chi) = s\int_1^\infty (\sum_{n\le x}\chi(n))x^{-s-1}\,dx$ on $\operatorname{Re} s > 0$. This gives the polynomial growth on the disks used in that lemma. The Euler product normalizes the logarithm at the center. For a principal character use $\zeta(s)$ times its finitely many Euler factors. Each is nonzero on $\operatorname{Re} s > 0$. The three-circles argument gives reciprocal subpower growth for this fixed $\chi$, uniformly in each closed strip to the right of $\theta$. The numerator $L(2s,\chi^2)$ is bounded on such a strip because $2\operatorname{Re} s > 1$. Apply the truncated Perron proof of Proposition A.2, using $|\lambda(n)\chi(n)| \le 1$. This proves the second condition. At $\theta = 1$, it is trivial and the Euler product gives the first. The functional equations for primitive characters complete the final assertion. □

No uniform bound on the dependence of the constants on the conductor is assumed here. To obtain GRH, the estimates must nevertheless hold for every fixed character and at all sufficiently large $N$. The quasi-Riemann

input specializes this criterion to $\theta = 7/8$, giving $Q_\chi(N) \ll_{\chi,\varepsilon} N^{7/4+\varepsilon}$. The desired exponent is $1+\varepsilon$.

## 16. Smoothing across Dirichlet conductors

For $X > 0$, define the smoothed twists and their weighted mean square by

$$B_\chi(X) = \sum_{n\ge 1} \lambda(n)\chi(n)e^{-n/X},$$

$$A_*(X) = \sum_{q\ge 1} \frac{1}{q^2\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |B_\chi(X)|^2.$$

The star restricts to primitive characters. The character of conductor one is included. Absolute convergence follows from $|B_\chi(X)| \le X$, since $\sum_{n\ge 1} e^{-n/X} = (e^{1/X}-1)^{-1} \le X$.

**Theorem 16.1** (Large conductors)**.** *For $X > 0$ and $Q \ge 1$,*

$$\sum_{q\ge Q} \frac{1}{q^2\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |B_\chi(X)|^2 \ll X^2Q^{-3} + XQ^{-1}.$$

*The constant is absolute. In particular, for $X \ge 1$ the contribution from $q \ge X^{1/3}$ is $O(X)$. The same bound holds if one includes imprimitive characters with primitive conductor at least $Q$ and weights each character by its modulus.*

*Proof.* Put $T_\chi(y) = \sum_{n\le y} \lambda(n)\chi(n)$, interpreted as zero for $y < 1$. On a block $R \le q < 2R$, use $1/(q^2\varphi(q)) \le R^{-3}q/\varphi(q)$. The multiplicative large sieve [21] gives

$$\sum_{R\le q<2R} \frac{1}{q^2\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |T_\chi(y)|^2 \ll R^{-3}(y+4R^2)y.$$

The square norm of the coefficient sequence in the sieve is at most $y$. Partial summation writes $B_\chi(X) = X^{-1}\int_0^\infty T_\chi(y)e^{-y/X}\,\mathrm{d}y$. Minkowski's integral inequality in the weighted Euclidean norm bounds the square root of the block contribution by

$$\frac{C}{XR^{3/2}} \int_0^\infty (y + 2R\sqrt{y})e^{-y/X}\,\mathrm{d}y \ll XR^{-3/2} + \sqrt{X/R}.$$

Squaring and summing over $R = 2^jQ$, $j \ge 0$, gives the assertion for primitive characters. This argument applies to all sequences of modulus at most one.

For a character modulo $rk$ induced by a primitive character $\chi_*$ modulo $r$, inclusion-exclusion and complete multiplicativity give

$$B_\chi(X) = \sum_{d\mid k} \mu(d)\lambda(d)\chi_*(d)B_{\chi_*}(X/d).$$

The factors at primes dividing $r$ vanish automatically. For fixed $k$ there is one lift per primitive character, and $\varphi(rk) \ge \varphi(r)\varphi(k)$ by the prime-power

formula. Apply the triangle inequality to the weighted norm over all $r \geq Q$ in this layer. Its upper bound is

$$\frac{C\tau(k)}{k\sqrt{\varphi(k)}}(XQ^{-3/2} + \sqrt{X/Q}),$$

using the primitive bound at $X/d > 0$ and $|\mu(d)\lambda(d)\chi_*(d)| \leq 1$. Squaring and summing over $k$ is allowed because $\sum_{k\geq 1} \tau(k)^2/(k^2\varphi(k)) < \infty$, by the elementary divisor bound. □

For the uniform estimate over completely multiplicative signs, the term $X^2/Q^3$ is necessary. Choose an odd prime $r \asymp Q$, $r \geq Q$, and extend the Legendre character $\chi_r$ to a sign function $f$ by setting $f(r) = 1$ and keeping complete multiplicativity. The $\chi_r$ twist of $f$ is the indicator of integers coprime to $r$, so its smoothed sum is $\gg X$ for $X \geq 2$. That single primitive term contributes $\gg X^2/Q^3$.

**Theorem 16.2** (Exact exponent across conductors)**.** *Let $\Theta_D$ be the supremum of real parts of nontrivial zeros over primitive Dirichlet L-functions, including $\zeta$. Then*

$$\limsup_{X\to\infty} \frac{\log(1 + A_*(X))}{\log X} = 2\Theta_D.$$

*Proof.* Write $r$ for the limit superior. Every fixed character has positive weight, so $B_\chi(X) \ll_{\chi,\varepsilon} X^{r/2+\varepsilon}$. For $\operatorname{Re} s > 1$, the Mellin identity is

$$\int_0^\infty B_\chi(X)X^{-s}\frac{\mathrm{d}X}{X} = \Gamma(s)\frac{L(2s,\chi^2)}{L(s,\chi)}.$$

The integral near zero is entire by exponential decay. At a zero $\rho$ with $\operatorname{Re}\rho > 1/2$, the numerator has a nonvanishing Euler product and Gamma has no zeros. If $r/2 < \operatorname{Re}\rho$, take $\varepsilon$ smaller than the difference. The growth bound then extends the integral holomorphically across $\rho$, a contradiction. For the conductor-one character, the quotient also has a pole at $s = 1/2$ from $\zeta(2s)$, since $\zeta(1/2) \neq 0$ by the alternating-series formula. This gives $r \geq 1$ even when $\Theta_D = 1/2$.

If $\Theta_D = 1$, the bound $A_*(X) \ll X^2$ gives the upper estimate. Otherwise fix $\Theta_D < \sigma < 1$. For every $\eta > 0$ there is a constant independent of $q$ and $t$ with

$$|L(u + it, \chi)^{-1}| \ll_{\sigma,\Theta_D,\eta} (q(2 + |t|))^\eta \qquad (\sigma \leq u \leq 2).$$

For a nonprincipal character the sums over complete periods vanish and the partial sums have modulus at most $q$. Partial summation gives $|L(s,\chi)| \leq q|s|/\operatorname{Re} s$ on $\operatorname{Re} s > 0$. To get the reciprocal bound, use the disks of Lemma A.1 centered at $2 + it$, with radii $1/2 < 2 - \sigma < R_2 < R < 2 - \Theta_D$. The logarithm normalized at the center by the Euler product has real part $O(\log(q(2 + |t|)))$ from above on radius $R$. Borel–Carathéodory gives the same bound on its modulus on radius $R_2$, and the Euler product bounds it absolutely on radius $1/2$. Three circles at radius $2 - \sigma$ give a bound by

a fixed power less than one of $\log(q(2+|t|))$. Exponentiating yields the displayed estimate. For conductor one use Lemma A.1, with the reciprocal extended by zero at the principal pole.

Inverting the exponential Mellin transform and shifting from $\operatorname{Re} s = 2$ to $\operatorname{Re} s = \sigma$ gives $|B_\chi(X)| \ll_{\sigma,\Theta_D,\eta} q^\eta X^\sigma$. The quotient is holomorphic on this strip and its numerator is bounded there by $\zeta(2\sigma)$. Stirling's formula supplies exponential decay on the horizontal edges and integrability on the new line. Choose $0 < \eta < 1/2$ and sum over conductors. Since $\sum_q q^{-2+2\eta} < \infty$, we obtain $A_*(X) \ll X^{2\sigma}$. Let $\sigma \downarrow \Theta_D$. □

By the functional equations, Dirichlet GRH holds if and only if $\Theta_D = 1/2$. The two theorems therefore reduce this approach to

$$\sum_{q<X^{1/3}} \frac{1}{q^2\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |B_\chi(X)|^2 \ll_\varepsilon X^{1+\varepsilon} \qquad (\varepsilon > 0).$$

The estimate remains open. Conductor one alone contributes $|B_{\chi_1}(X)|^2$, so positive averaging retains the original RH obstruction.

16.1. **Orthogonality and square divisors.** Write $A(X)$ for the sum over all characters. Orthogonality on the reduced residue classes yields

$$A(X) = \sum_{n,m\ge1} \lambda(n)\lambda(m)e^{-(n+m)/X} \sum_{\substack{q\mid n-m\\(q,n)=1}} q^{-2}.$$

All positive $q$ divide zero in the diagonal term. The innermost sum is bounded by $\zeta(2)$, which justifies rearrangement. The diagonal equals

$$\zeta(2)\sum_{d\ge1} \frac{\mu(d)}{d^2(e^{2d/X}-1)} = \frac{\zeta(2)}{2\zeta(3)}X + O(1).$$

Indeed, inclusion-exclusion gives $\sum_{(q,n)=1} q^{-2} = \zeta(2)\sum_{d\mid n}\mu(d)d^{-2}$. Summing over multiples of $d$ gives the geometric denominator. Use $(e^y-1)^{-1} = y^{-1} + O(1)$ uniformly for $y > 0$, and sum the absolutely convergent series. The signed off-diagonal includes $|B_{\chi_1}(X)|^2 - \sum_n e^{-2n/X}$.

The prime-power identities

$$\lambda(n) = \sum_{d^2\mid n} \mu(n/d^2), \qquad \mu(n) = \sum_{d^2\mid n} \mu(d)\lambda(n/d^2)$$

give two vector relations. Let $M_\chi(X) = \sum_{n\ge1}\mu(n)\chi(n)e^{-n/X}$, and let $U_d$ act on the $\chi$-coordinate by multiplication by $\chi(d)^2$. Then

$$B(X) = \sum_{d\ge1} U_d M(X/d^2), \qquad M(X) = \sum_{d\ge1} \mu(d)U_d B(X/d^2).$$

The series are absolutely convergent in the weighted norm because the smoothed sums are bounded absolutely by $X/d^2$. Each $U_d$ is a contraction. A bound $O_\varepsilon(X^{1/2+\varepsilon})$ in either weighted norm transfers to the other by Minkowski and $\sum_d d^{-1-2\varepsilon} < \infty$. Exponential decay gives the same upper

bound for $0 < X < 1$ with a changed constant. Both factorizations leave the small-conductor cancellation in the $d = 1$ term.

16.2. **Why increasing the modulus retains conductor one.** The large-conductor estimate is indexed by primitive conductor, which is different from the modulus of an induced character. The distinction is exact for the principal character $\chi_{0,p}$ modulo a prime $p$:

$$B_{\chi_{0,p}}(X) = B(X) + B(X/p). \tag{16.1}$$

Indeed the removed multiples of $p$ contribute $\lambda(p)B(X/p) = -B(X/p)$. Thus, if $X \geq 2$ and $p \geq 2X \log X$,

$$|B_{\chi_{0,p}}(X) - B(X)| \leq (e^{p/X} - 1)^{-1} \leq (X^2 - 1)^{-1} \leq 2X^{-2}.$$

This large-modulus twist still has primitive conductor one, and is excluded from the large-primitive-conductor tail in Theorem 16.1. On the Mellin side the multiplier is $1 + p^{-s}$, which is nonzero for $\operatorname{Re} s > 0$ and preserves every right-hand pole. Formula (16.1) is also checked coefficientwise in `anc/check_primitive_identities.py`.

There is a separate obstruction to a coefficient-uniform reverse inequality. For a bounded sequence $a$, put

$$\operatorname{Tail}_a(X, Q) = \sum_{q \geq Q} \frac{1}{q^2 \varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} \left| \sum_{n \geq 1} a_n e^{-n/X} \chi(n) \right|^2 .$$

The primitive-character proof of Theorem 16.1 applies to every $|a_n| \leq 1$, and gives $\operatorname{Tail}_a(X, X^{1/3}) \ll X$. If a uniform reconstruction inequality

$$\left| \sum_{n \geq 1} a_n e^{-n/X} \right|^2 \leq K(X) \operatorname{Tail}_a(X, X^{1/3})$$

were available, then necessarily $K(X) \gg X$ whenever it holds. To see this, use the completely multiplicative sequence $a_n = 1$, whose untwisted sum is $(e^{1/X} - 1)^{-1} \sim X$. Its square is of order $X^2$, whereas the tail is $O(X)$. Thus no such uniform reconstruction can have $K(X) = O(X^\kappa)$ with $\kappa < 1$. Even a hypothetical reconstruction with the smallest compatible scale $K(X) \asymp X$ would give only an $O(X^2)$ upper estimate when combined with the tail bound; a larger coefficient would weaken it further. A stronger reconstruction for Liouville would need additional arithmetic cancellation.

## 17. A direct local-energy construction

We now attack the single smoothed sum $B$ of Proposition 14.1. The construction in this section controls point evaluation by a neighboring-scale mean square without assuming a zero-free half-plane. It then separates a computable diagonal from the signed arithmetic term still requiring an estimate. The results of this section do not establish that estimate.

17.1. **Reconstruction with a logarithmic cost.** For any complex sequence with $|a_n| \le 1$, put

$$F(X) = \sum_{n\ge 1} a_n e^{-n/X}, \qquad \mathcal{H}_F(X) = \int_{X/2}^{2X} |F(Y)|^2 \frac{\mathrm{d}Y}{Y}.$$

Both expressions converge absolutely for $X > 0$.

**Theorem 17.1** (Explicit analytic localization)**.** *For $X \ge 1$ and $m = \lceil 8\log_2(2X)\rceil$,*

$$|F(X)|^2 \le 10(m+2)\mathcal{H}_F(X) + X^{-4}. \tag{17.1}$$

*There is an explicit degree-$m$ polynomial kernel $K_m(0,u)$ on $[-1/4, 1/4]$ such that*

$$F(X) = \int_{-1/4}^{1/4} K_m(0,u) F\Big(\frac{X}{1+u}\Big)\,\mathrm{d}u + \mathcal{R}_X, \qquad |\mathcal{R}_X| \le \frac{\sqrt{2(m+2)}}{128X^7}.$$

*Proof.* The holomorphic function

$$G_X(z) = \sum_{n\ge 1} a_n e^{-n(1+z)/X}$$

is defined for $\operatorname{Re} z > -1$. On $|z| \le 1/2$,

$$|G_X(z)| \le (e^{1/(2X)} - 1)^{-1} \le 2X.$$

Let $P_m$ be its Taylor polynomial at zero. Cauchy's estimate gives

$$\sup_{|u|\le 1/4} |G_X(u) - P_m(u)| \le 2X \sum_{j>m} (2|u|)^j \le 2X\, 2^{-m} =: \delta.$$

In particular $P_m(0) = G_X(0) = F(X)$ exactly.

Let $L_j$ denote the Legendre polynomial normalized by $L_j(1) = 1$. Its norm on $[-1,1]$ is $2/(2j+1)$. For completeness these identities follow from Rodrigues' formula

$$L_j(t) = \frac{1}{2^j j!}\frac{\mathrm{d}^j}{\mathrm{d}t^j}(t^2-1)^j.$$

Integration by parts $j$ times proves orthogonality to polynomials of degree less than $j$. Its leading coefficient is $(2j)!/(2^j(j!)^2)$; the recurrence $\int_{-1}^1 (1-t^2)^j\,\mathrm{d}t = \frac{2j}{2j+1}\int_{-1}^1 (1-t^2)^{j-1}\,\mathrm{d}t$ then gives the stated norm. Thus the evaluation kernel on the shorter interval is

$$K_m(0,u) = 2\sum_{j=0}^{m} (2j+1) L_j(0) L_j(4u).$$

Expanding a polynomial in this orthogonal basis gives

$$P(0) = \int_{-1/4}^{1/4} K_m(0,u) P(u)\,\mathrm{d}u, \qquad |P(0)|^2 \le K_m(0,0)\int_{-1/4}^{1/4} |P(u)|^2\,\mathrm{d}u. \tag{17.2}$$

These are classical reproducing-kernel identities [27, §18.2(v)]; their constants have been derived here. [7]

Rodrigues' formula also gives

$$L_{2r+1}(0) = 0, \quad L_{2r}(0) = (-1)^r \binom{2r}{r} 4^{-r}, \quad \left( \binom{2r}{r} 4^{-r} \right)^2 \le \frac{1}{r+1}.$$

The last inequality follows by induction, since $(2r+1)^2(r+2) \le 4(r+1)^3$. Consequently

$$K_m(0,0) = 2 \sum_{r \le m/2} (4r+1) \left( \binom{2r}{r} 4^{-r} \right)^2 \le 4(m+2).$$

Apply (17.2) to $P_m$ and use $|P_m|^2 \le 2|G_X|^2 + 2\delta^2$:

$$|F(X)|^2 \le 8(m+2) \int_{-1/4}^{1/4} |G_X(u)|^2 \, \mathrm{d}u + 4(m+2)\delta^2.$$

The substitution $Y = X/(1+u)$ bounds this integral by

$$\int_{4X/5}^{4X/3} |F(Y)|^2 \frac{X}{Y^2} \, \mathrm{d}Y \le \frac{5}{4} \mathcal{H}_F(X).$$

Also $\delta \le (128X^7)^{-1}$ and $m+2 \le 16X$ for $X \ge 1$. Hence $4(m+2)\delta^2 \le X^{-4}$, proving (17.1). Finally $\int |K_m(0,u)| \, \mathrm{d}u \le \sqrt{K_m(0,0)/2}$ applied to $G_X - P_m$ gives the reconstruction formula and error. □

In particular, $\mathcal{H}_B(X) \ll_\varepsilon X^{1+\varepsilon}$ would imply the requested pointwise square-root estimate. Only the analytic transfer has been proved; it supplies no upper bound for $\mathcal{H}_B$.

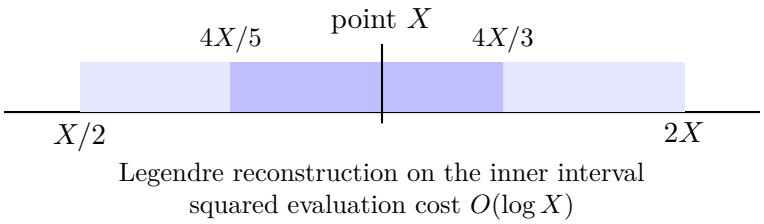



FIGURE 9. Localization in scale, shown schematically. The kernel recovers the point from nearby values. Arithmetic cancellation is still required in their mean square.

[7] The polynomial approximation is applied to the analytic function of reciprocal scale, not to the arithmetic coefficients. It therefore does not assume cancellation of $\lambda$.

17.2. **The exact signed term.** Define the positive kernel

$$K(v)=\int_{1/2}^{2}e^{-v/t}\frac{\mathrm{d}t}{t}=\int_{1/2}^{2}e^{-vu}\frac{\mathrm{d}u}{u}.$$

Absolute convergence gives

$$\begin{aligned}\mathcal{H}_B(X)&=\mathcal{D}(X)+\mathcal{O}(X),\\ \mathcal{D}(X)&=\int_{X/2}^{2X}\frac{1}{e^{2/Y}-1}\frac{\mathrm{d}Y}{Y},\\ \mathcal{O}(X)&=2\sum_{h\ge1}\sum_{n\ge1}\lambda(n)\lambda(n+h)K((2n+h)/X).\end{aligned}\tag{17.3}$$

The symbol $\mathcal{O}(X)$ in this display names a function, not a big-$O$ estimate.

**Proposition 17.2** (A diagonal with an explicit error)**.** *For every $X>0$,*

$$0<\mathcal{D}(X)-\left(\frac{3X}{4}-\log 2\right)<\frac{1}{4X}.$$

*Proof.* For $t>0$,

$$0<\frac{1}{e^t-1}-\frac{1}{t}+\frac{1}{2}<\frac{t}{12}.$$

The lower inequality follows by differentiating $(t-2)e^t+t+2$ twice. For the upper inequality, multiplication by $12t(e^t-1)$ reduces it to

$$(t^2-6t+12)(e^t-1)-12t=\sum_{j\ge5}\frac{(j-3)(j-4)}{j!}t^j>0.$$

Substitute $t=2/Y$ and integrate. The first two terms give $3X/4-\log 2$, and the error is less than $\frac{1}{6}\int_{X/2}^{2X}Y^{-2}\,\mathrm{d}Y=1/(4X)$. □

Therefore a signed upper bound $\mathcal{O}(X)\ll_\varepsilon X^{1+\varepsilon}$ would suffice. Replacing its terms by absolute values only gives order $X^2$. Positivity of $\mathcal{H}_B$ gives a lower bound for $\mathcal{O}$, not the missing upper bound.

Complete multiplicativity reorganizes the same energy as

$$\mathcal{H}_B(X)=\sum_{k\ge2}c(k)\int_{X/2}^{2X}\frac{1}{e^{k/Y}-1}\frac{\mathrm{d}Y}{Y},\qquad c(k)=\sum_{\substack{1\le a<k\\(a,k)=1}}\lambda(a)\lambda(k-a).$$

Indeed write the two indices as $da, db$ with $(a,b)=1$; their signs multiply to $\lambda(a)\lambda(b)$. The term $c(2)=1$ is exactly the diagonal. The remaining terms are coprime additive correlations. The identity leaves their cancellation to be proved.

17.3. **One primitive correlation and a one-sided target.** The coprime coefficients in the preceding expansion also admit an exponential weight without the local-energy integral. Put

$$C_\lambda(X) = \sum_{\substack{a,b\ge 1\\(a,b)=1}} \lambda(a)\lambda(b)e^{-(a+b)/X} = \sum_{k\ge 2} c(k)e^{-k/X}.$$

The letter $C_\lambda$ here denotes this smoothed primitive correlation, not a correlation at a fixed shift. It need not be nonnegative.

**Proposition 17.3** (Exact primitive decomposition)**.** *For every $X > 0$,*

$$B(X)^2 = \sum_{d\ge 1} C_\lambda(X/d), \qquad C_\lambda(X) = \sum_{d\ge 1} \mu(d)B(X/d)^2. \tag{17.4}$$

*Both series converge absolutely.*

*Proof.* Partition the ordered pairs $(m, n)$ in $B(X)^2$ by their greatest common divisor $d$, and write $m = da$, $n = db$. Complete multiplicativity gives $\lambda(da)\lambda(db) = \lambda(a)\lambda(b)$ because $\lambda(d)^2 = 1$. This proves the first identity. To obtain the second, insert

$$\mathbf{1}_{(m,n)=1} = \sum_{d\mid m,\ d\mid n} \mu(d)$$

in the defining series for $C_\lambda$ and use the same identity for the signs. All rearrangements are justified by the finite majorant

$$\sum_{d\ge 1}\left(\sum_{a\ge 1} e^{-da/X}\right)^2 = \sum_{d\ge 1}(e^{d/X}-1)^{-2} < \infty.$$

For instance, its tail is bounded by a geometric series after $d \ge X$. The same bound proves absolute convergence of the two series in (17.4). □

**Theorem 17.4** (A one-sided primitive criterion)**.** *The following assertions are equivalent:*

1. *RH;*
2. *$B(X) \ll_\varepsilon X^{1/2+\varepsilon}$ for every $\varepsilon > 0$ and $X \ge 1$;*
3. *for every $\varepsilon > 0$ there exists $K_\varepsilon$ such that $C_\lambda(X) \le K_\varepsilon X^{1+\varepsilon}$ for every $X \ge 1$.*

*Only an upper bound is required in the third assertion.*

*Proof.* Proposition 14.1 gives the equivalence of the first two assertions. For the implication from the third to the second, first enlarge $K_\varepsilon$ so that its bound holds for all $X > 0$. This is possible because

$$|C_\lambda(X)| \le (e^{1/X}-1)^{-2}$$

decays faster than any fixed power as $X \downarrow 0$. The first identity in (17.4) now yields

$$0 \le B(X)^2 \le K_\varepsilon \zeta(1+\varepsilon)X^{1+\varepsilon}. \tag{17.5}$$

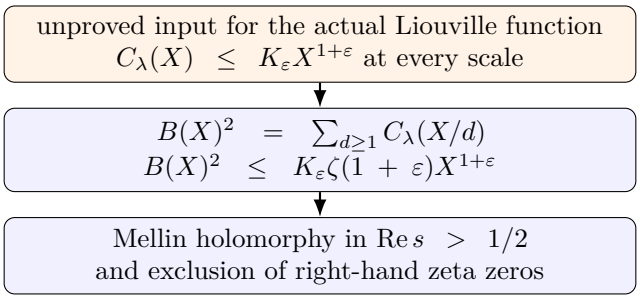



FIGURE 10. Proved implications from an unproved upper estimate. The primitive decomposition retains the signs; it does not prove its input.

Taking $\varepsilon = 2\delta$ proves the second assertion with exponent $1/2+\delta$, for every $\delta > 0$.

Conversely, use the second assertion with exponent $1/2+\varepsilon/2$, and extend the resulting bound to $0 < X < 1$ by exponential decay. Write the squared bound as $|B(X)|^2 \le A_\varepsilon X^{1+\varepsilon}$. The second identity in (17.4) then gives

$$|C_\lambda(X)| \le A_\varepsilon \frac{\zeta(1+\varepsilon)}{\zeta(2+2\varepsilon)} X^{1+\varepsilon}. \tag{17.6}$$

Here the squarefree Euler product gives $\sum_{d\ge1} |\mu(d)| d^{-1-\varepsilon} = \zeta(1+\varepsilon)/\zeta(2+2\varepsilon)$, with absolute convergence. This proves the third assertion. □

The theorem isolates a single weighted upper estimate. No bound for each individual $c(k)$ or each fixed additive shift is needed. However, the upper estimate for $C_\lambda$ has not been proved. The exact identities do not supply it, and positivity of $B(X)^2$ does not bound $C_\lambda$ from above. [8]

For exact finite controls, put

$$A(k) = \sum_{a=1}^{k-1} \lambda(a)\lambda(k-a), \qquad A(1) = c(1) = 0.$$

Coefficientwise the same proof gives

$$A(k) = \sum_{d\mid k} c(k/d), \qquad c(k) = \sum_{d\mid k} \mu(d) A(k/d). \tag{17.7}$$

The checker `anc/check_primitive_identities.py` compares these divisor identities with direct greatest-common-divisor enumeration for $1 \le k \le 1024$. Its scope is finite identities, not the asymptotic assertion in Theorem 17.4.

[8] The constant and all sufficiently large scales are part of the third assertion. Checking finitely many coefficients or scales does not establish it.

17.4. **Why a prime Gram argument does not close the estimate.** The logarithmic identity obtained from complete multiplicativity is

$$(\lambda \log)(n) = (\lambda * (\lambda\Lambda))(n),$$
$$\lambda(d)\Lambda(d) = 2\mathbf{1}_{d=\square}\Lambda(\sqrt{d}) - \Lambda(d).$$

Checking prime powers proves the second line. Smoothing gives

$$\sum_{n\geq 1} \lambda(n) \log n \, e^{-n/X} = 2\sum_{d\geq 1} \Lambda(d) B(X/d^2) - \sum_{d\geq 1} \Lambda(d) B(X/d). \tag{17.8}$$

All sums converge absolutely at fixed $X$. However, for distinct primes $p, q$,

$$\sum_{n\leq N} \lambda(pn)\lambda(qn) = N.$$

Thus prime dilations are maximally correlated in this Gram calculation. Treating them as orthogonal would discard a full off-diagonal. The identity (17.8) is not a contraction estimate.[9]

17.5. **The sharp norm of the signed dilation operator.** The coefficients in (17.8) can be grouped by prime powers. The exact identity is

$$\sum_{n\geq 1} \lambda(n) \log n \, e^{-n/X} = \sum_{p^k\geq 2} (-1)^k \log p \, B(X/p^k). \tag{17.9}$$

Indeed $\lambda \log = \lambda * (\lambda\Lambda)$, and $\lambda(p^k)\Lambda(p^k) = (-1)^k \log p$. Absolute convergence at fixed $X$ follows by comparison with $\sum_n n \log(2n) e^{-n/X}$.

Consider a finite truncation on a weighted Hilbert space:

$$\mathcal{K}_\sigma = L^2((0,\infty), X^{-2\sigma}\, \mathrm{d}X/X), \qquad T_R f(X) = \sum_{p^k\leq R} (-1)^k \log p \, f(X/p^k).$$

The following calculation retains every signed prime-power term.

**Theorem 17.5** (Exact dilation norm)**.** *For every real $\sigma$ and finite $R \geq 2$,*

$$\|T_R\|_{\mathcal{K}_\sigma \to \mathcal{K}_\sigma} = \sum_{p^k\leq R} (\log p) p^{-k\sigma}. \tag{17.10}$$

*For each fixed $1/2 < \sigma < 1$, as $R \to \infty$,*

$$\|T_R\| \sim \frac{R^{1-\sigma}}{1-\sigma}, \qquad \frac{\|T_R\|}{\log R} \longrightarrow \infty. \tag{17.11}$$

*Proof.* The map $Uf(t) = e^{-\sigma t} f(e^t)$ is an isometry onto $L^2(\mathbb{R})$. Under it, $f(X/d)$ becomes $d^{-\sigma}(Uf)(t - \log d)$. With the Fourier convention using $e^{-it\tau}$, the multiplier of $T_R$ is

$$m_{R,\sigma}(\tau) = \sum_{p^k\leq R} (-1)^k (\log p) p^{-k\sigma} e^{-ik\tau \log p}.$$

[9] Almost-all short-interval results compare short and long averages [30]. That comparison does not supply the square-root size of the long average required here. We do not use a short-interval estimate with this additional strength as an unstated premise.

Plancherel identifies the operator norm with its essential supremum; see [28, §1.14] for the transform conventions. The triangle inequality proves the upper bound in (17.10).

The logarithms of the finitely many primes at most $R$ are linearly independent over $\mathbb{Q}$, by unique factorization. Thus the continuous phase orbit $(e^{-i\tau \log p})_p$ is dense in the finite product of unit circles. Here is a direct justification. Every nonconstant integer Fourier character of this orbit has the form $e^{-i\tau \sum_p m_p \log p}$ with a nonzero frequency. Its average over $[-T,T]$ tends to zero. Trigonometric-polynomial approximation therefore gives the Haar average for each continuous function on the torus. A nonnegative continuous bump supported in any nonempty open set has positive Haar integral, so the orbit meets that set.

In particular, all prime phases can approach $-1$ simultaneously. Then $(-1)^k e^{-ik\tau \log p}$ approaches 1 for every term of the finite sum. The multiplier approaches the positive sum in (17.10). Its continuity implies the same lower bound for the essential supremum, proving equality.

For the asymptotic, the prime number theorem [29, §27.12] and partial summation give

$$\sum_{p\le R} (\log p) p^{-\sigma} \sim \frac{R^{1-\sigma}}{1-\sigma}.$$

The higher prime powers contribute a bounded amount because

$$\sum_p \sum_{k\ge 2} (\log p) p^{-k\sigma} \le \frac{1}{1-2^{-\sigma}} \sum_{n\ge 2} (\log n) n^{-2\sigma} < \infty.$$

This proves (17.11). □

The size in this theorem is not confined to very high logarithmic frequencies.

**Proposition 17.6** (A low-frequency sign change)**.** *Fix $1/2 < \sigma < 1$, and put*

$$P_R = \sum_{p\le R} (\log p) p^{-\sigma}, \qquad \tau_R = \frac{\pi}{\log R}.$$

*Then*

$$m_{R,\sigma}(\tau_R) = P_R + O_\sigma\left(\frac{R^{1-\sigma}}{\log R}\right), \tag{17.12}$$

$$m_{R,\sigma}(0) = -P_R + O_\sigma(1). \tag{17.13}$$

*If $\psi$ is a fixed continuous smoothing multiplier near zero with $\psi(0) \neq 0$, then $|\psi(\tau_R) m_{R,\sigma}(\tau_R)| \sim |\psi(0)| P_R$.*

*Proof.* The prime part of $m_{R,\sigma}(\tau_R)$ is

$$-\sum_{p\le R} (\log p) p^{-\sigma} e^{-i\tau_R \log p} = \sum_{p\le R} (\log p) p^{-\sigma} e^{i\pi \log(R/p)/\log R}.$$

Its difference from $P_R$ has absolute value at most

$$\frac{\pi}{\log R}\sum_{p\le R}(\log p)p^{-\sigma}\log(R/p).$$

Chebyshev's upper bound and partial summation give $P_u \ll_\sigma u^{1-\sigma}$. Interchanging a finite sum and an integral consequently gives

$$\sum_{p\le R}(\log p)p^{-\sigma}\log(R/p)=\int_2^R P_u\frac{\mathrm{d}u}{u}\ll_\sigma R^{1-\sigma}.$$

The higher prime powers contribute $O_\sigma(1)$, as in the last proof. This proves (17.12), and setting $\tau=0$ proves (17.13). Finally use $\tau_R\to 0$, continuity of $\psi$, and the asymptotic for $P_R$. □

Thus this signed operator supplies neither a universal contraction in $\mathcal{K}_\sigma$ nor a uniform sign for its real Fourier multiplier. Even a fixed nonvanishing smoothing factor near zero does not remove its growing size. These are statements about a generic operator estimate. They do not assert that the functions testing its norm are Liouville sums, or that the particular $B$ belongs to $\mathcal{K}_\sigma$. [10] Moreover, the left side of (17.9) cannot be replaced by $(\log X)B(X)$ without bounding their difference. The renewal identity remains exact; the Liouville-specific upper estimate remains missing.

17.6. **An explicit finite-prime model.** Let $f_y$ be completely multiplicative, equal to $-1$ at primes at most $y$ and to $+1$ at larger primes. Write

$$B_y(X)=\sum_{n\ge1}f_y(n)e^{-n/X},\qquad c_y=\prod_{p\le y}\frac{p-1}{p+1},\qquad G_y(\sigma)=\prod_{p\le y}\frac{1+p^{-\sigma}}{1-p^{-\sigma}}.$$

**Theorem 17.7** (Finite-prime main term)**.** *For $X>0$, $y\ge2$, and $0<\sigma<1$,*

$$|B_y(X)-c_yX|\le X^\sigma G_y(\sigma). \tag{17.14}$$

*In particular, for $y=(\log X)^2$ as $X\to\infty$,*

$$B_y(X)=c_yX+O(X^{1/2+o(1)}),\qquad B_y(X)\ge\frac{X}{2(\log X)^4}$$

*for all sufficiently large $X$.*

*Proof.* Define $g_y$ multiplicatively by $g_y(p^j)=2(-1)^j$ for $p\le y$, $j\ge1$, and zero on powers of larger primes. Prime by prime $f_y=\mathbf{1}*g_y$. Hence

$$B_y(X)=\sum_{d\ge1}\frac{g_y(d)}{e^{d/X}-1},\quad \sum_{d\ge1}\frac{g_y(d)}{d}=c_y,\quad \sum_{d\ge1}\frac{|g_y(d)|}{d^\sigma}=G_y(\sigma).$$

Only finitely many primes support $g_y$, so these Dirichlet sums converge absolutely for $\sigma>0$. Also

$$0<\frac1t-\frac{1}{e^t-1}\le\min(1/2,1/t)\le t^{-\sigma}.$$

[10] The latter membership at every $\sigma>1/2$ is itself an arithmetic assertion. It cannot be inserted as an unproved premise in a proof of the critical energy bound.

Subtract $X\sum g_y(d)/d$ and apply this term by term to obtain (17.14).

The elementary Chebyshev bound gives $\pi(t) \ll t/\log t$, and partial summation then gives

$$\log G_y(1/2) \ll \sum_{p\le y} p^{-1/2} \ll \frac{\sqrt{y}}{\log y}.$$

The elementary details are recorded in Appendix D. For $y = (\log X)^2$, this is $o(\log X)$, so $G_y(1/2) = X^{o(1)}$. If $r = \lfloor y \rfloor$, positivity of the factors gives

$$c_y \ge \prod_{n=2}^{r} \frac{n-1}{n+1} = \frac{2}{r(r+1)} \ge y^{-2}.$$

The error $X^{1/2+o(1)}$ is eventually less than $X/(2(\log X)^4)$, proving the last assertion. □

This is a bound for an explicitly constructed comparison function, not for $\lambda$. It shows that complete multiplicativity plus agreement at all primes through $(\log X)^2$ cannot yield a uniform square-root bound over the resulting class of functions.

17.7. **The large-prime cancellation that remains.** Put $P^-(1) = \infty$, and define

$$h_y(d) = \begin{cases} 2^{\omega(d)}\lambda(d), & P^-(d) > y, \\ 0, & \text{otherwise.} \end{cases}$$

At a prime greater than $y$, convolution by its local coefficients

$$1 + 2\sum_{j\ge1}(-1)^j p^{-js} = \frac{1-p^{-s}}{1+p^{-s}}$$

changes the local signs of $f_y$ to those of $\lambda$. Therefore $\lambda = f_y * h_y$, and

$$B(X) = B_y(X) + T_y(X), \qquad T_y(X) = \sum_{\substack{d>1 \\ P^-(d)>y}} h_y(d) B_y(X/d). \tag{17.15}$$

Exponential decay justifies this identity absolutely at every fixed $X$. With $y = (\log X)^2$, the desired estimate would follow from

$$T_y(X) = -c_y X + O_\varepsilon(X^{1/2+\varepsilon}). \tag{17.16}$$

The comparison-kernel error in Theorem 17.7 is already of square-root size with a subpower loss. The signed estimate (17.16) remains unproved.

**Proposition 17.8** (Prime-tail cancellation to logarithmic order)**.** *For $y = (\log X)^2$, and every fixed $A > 0$,*

$$T_y(X) = -c_y X + O_A\left(\frac{X}{(\log X)^A}\right).$$

*In particular, $T_y(X)/(c_y X) \to -1$.*

*Proof.* The classical uniform additive estimate of Davenport, in the Liouville form recalled in Section 3 and quoted in [20], gives at additive frequency zero

$$L(t) \ll_A \frac{t}{(\log(2t))^A} \qquad (t \geq 1).$$

Partial summation gives the absolutely convergent identity

$$B(X) = \frac{1}{X}\int_1^\infty L(t)e^{-t/X}\,\mathrm{d}t.$$

For $X \geq 4$, the contribution from $1 \leq t \leq \sqrt{X}$ has modulus at most $X^{-1}\int_1^{\sqrt{X}} t\,\mathrm{d}t \leq 1/2$. For $t \geq \sqrt{X}$, use $\log(2t) \geq \frac{1}{2}\log X$ to bound the remaining part by

$$\frac{C_A 2^A}{X(\log X)^A}\int_0^\infty te^{-t/X}\,\mathrm{d}t = \frac{C_A 2^A X}{(\log X)^A}.$$

Consequently $B(X) \ll_A X/(\log X)^A$. By Theorem 17.7 and (17.15),

$$T_y(X) + c_y X = B(X) - (B_y(X) - c_y X) = O_A\Big(\frac{X}{(\log X)^A}\Big) + O(X^{1/2+o(1)}).$$

For each fixed $A$, the last error is absorbed in the first. Finally $c_y \geq (\log X)^{-4}$ gives $T_y(X)/(c_y X) = -1 + O_A((\log X)^{4-A})$. Take any $A > 4$. □

This proves the main-term cancellation in the actual exponential kernel. It is a consequence of classical additive cancellation, not a new zero-free theorem. Its error is larger than the square-root error requested in (17.16).

**Proposition 17.9** (The precise power-saving endpoint)**.** *For $y = (\log X)^2$, the estimate*

$$T_y(X) + c_y X \ll_\varepsilon X^{1/2+\varepsilon} \quad \textit{for every } \varepsilon > 0$$

*is equivalent to RH.*

*Proof.* The exact identity and Theorem 17.7 give

$$T_y(X) + c_y X - B(X) = O(X^{1/2+o(1)}).$$

Hence the asserted family of bounds holds if and only if $B(X) \ll_\varepsilon X^{1/2+\varepsilon}$ for every $\varepsilon > 0$. The equivalence of the latter condition with RH is Proposition 14.1. □

There is a cancellation theorem in a different, convergent weight. For real $s > 1$, Euler products give

$$H_y(s) := \sum_{d\geq 1}\frac{h_y(d)}{d^s} = \frac{\zeta(2s)}{\zeta(s)^2 c_y(s)}, \qquad c_y(s) = \prod_{p\leq y}\frac{1-p^{-s}}{1+p^{-s}}. \tag{17.17}$$

Uniformly for $0 < \delta \leq \delta_0$ and $\delta \log y \leq 1$,

$$H_y(1+\delta) = \frac{\zeta(2)\delta^2}{c_y}\left(1 + O(\delta(1+\log y))\right). \tag{17.18}$$

Indeed the pole of $\zeta$ at one gives $\zeta(2+2\delta)/\zeta(1+\delta)^2 = \zeta(2)\delta^2(1+O(\delta))$, while

$$\frac{\mathrm{d}}{\mathrm{d}s}\log c_y(s) = 2\sum_{p\le y}\frac{(\log p)p^{-s}}{1-p^{-2s}} \ll \log y \quad (s\ge 1)$$

by Chebyshev and partial summation. Integrating this derivative proves the uniform factor in (17.18). For fixed $y$, the tail without $d=1$ tends to $-1$. This is an actual all-prime cancellation in the Abel weight.

The distinction between the two kernels is essential. [11] Equation (17.18) concerns $d^{-1-\delta}$, whereas (17.15) contains the scale-dependent kernel $B_y(X/d)$. No bound for their signed discrepancy has been obtained. Treating their cancellation as interchangeable would assume the unresolved step.

For clarity, the double zero in (17.18) does not exclude nonreal poles. For fixed $y$, each finite Euler factor of $c_y(s)$ is holomorphic and nonzero on $\operatorname{Re} s > 0$. If $\rho$ were a zeta zero of order $r$ with $\operatorname{Re}\rho > 1/2$, then (17.17) would have the leading Laurent term

$$H_y(s) = \frac{(r!)^2\zeta(2\rho)}{c_y(\rho)(\zeta^{(r)}(\rho))^2}(s-\rho)^{-2r} + O((s-\rho)^{1-2r}).$$

The coefficient is nonzero. The local cancellation at one therefore supplies no contradiction to such a zero; excluding it requires the global power-saving estimate.

## 18. A weighted energy certificate and its boundary cost

For the smoothed Liouville sum $B$ of (14.1), define

$$J(\sigma) = \int_1^\infty |B(X)|^2 X^{-2\sigma}\frac{\mathrm{d}X}{X}, \qquad H(s) = \int_0^1 B(X)X^{-s}\frac{\mathrm{d}X}{X}.$$

The second integral is entire, by the exponential decay at zero. The first is an extended nonnegative real number.

**Theorem 18.1** (Finite weighted energy excludes right-hand zeros)**.** *If $J(\sigma) < \infty$ for some $\sigma > 1/2$, then $\zeta(s) \neq 0$ on $\operatorname{Re} s > \sigma$. On that half-plane,*

$$\left|\Gamma(s)\frac{\zeta(2s)}{\zeta(s)} - H(s)\right|^2 \le \frac{J(\sigma)}{2(\operatorname{Re} s - \sigma)}.$$

*Consequently RH is equivalent to $J(\sigma) < \infty$ for every $\sigma > 1/2$.*

*Proof.* The integral $T(s) = \int_1^\infty B(X)X^{-s}\,\mathrm{d}X/X$ converges absolutely on $\operatorname{Re} s > \sigma$, by Cauchy–Schwarz, and gives the displayed inequality for $T$. It is holomorphic there. For example, the same argument with factors $(\log X)^k$ bounds all differentiated integrals locally. Initially on $\operatorname{Re} s > \max(1,\sigma)$, (14.1) gives $\zeta(s)(T(s)+H(s)) = \Gamma(s)\zeta(2s)$. The identity extends meromorphically. Its right side is holomorphic and nonzero on $\operatorname{Re} s > \sigma > 1/2$. A zero of $\zeta$

[11] The positive absolute-coefficient series is $\sum |h_y(d)|d^{-s} = \zeta(s)^2 c_y(s)/\zeta(2s)$; for fixed $y$ it has a double pole at one. Thus absolute comparison does not preserve the small signed value in (17.18).

there is therefore impossible. If the energies are finite at every $\sigma > 1/2$, these half-planes cover $\operatorname{Re} s > 1/2$; the functional equation gives RH. Conversely RH gives $B(X) \ll_\varepsilon X^{1/2+\varepsilon}$ by Proposition 14.1. For a fixed $\sigma > 1/2$, choose $0 < \varepsilon < \sigma - 1/2$ and integrate its square. □

**Proposition 18.2** (A pole forces a quantitative energy cost)**.** *Suppose that $\rho$ is a zero of $\zeta$ of order $m$ with $\beta = \operatorname{Re}\rho > 1/2$, and put*

$$a_\rho = \frac{m!\,\Gamma(\rho)\zeta(2\rho)}{\zeta^{(m)}(\rho)}.$$

*Then*

$$\liminf_{u\downarrow 0} u^{2m-1} J(\beta + u/2) \geq |a_\rho|^2.$$

*The statement uses the extended value $+\infty$ when an energy diverges. Independently of RH,*

$$\liminf_{u\downarrow 0} uJ(1/2 + u/2) \geq \left|\frac{\Gamma(1/2)}{2\zeta(1/2)}\right|^2 > 0.$$

*Proof.* At $\rho$, the quotient in Theorem 18.1 has leading term $a_\rho(s-\rho)^{-m}$, since $\Gamma(\rho)\zeta(2\rho) \neq 0$. Subtracting $H$ preserves it. When $J(\beta + u/2)$ is finite, evaluate the inequality at $s = \rho + u$, where $2(\operatorname{Re} s - \beta - u/2) = u$. This gives $u^{2m-1}J(\beta + u/2) \geq |u^m(\Gamma(s)\zeta(2s)/\zeta(s) - H(s))|^2$. The right side tends to $|a_\rho|^2$. When the energy is infinite the inequality is automatic. For the final assertion use the pole of $\zeta(2s)$ at $s = 1/2$, whose residue in the quotient is $\Gamma(1/2)/(2\zeta(1/2))$. The alternating zeta series shows that $\zeta(1/2) < 0$.[12] □

These are explicit tests for a proposed energy estimate. No upper bound for $J(\sigma)$ in all $\sigma > 1/2$ is established here. Proving that finiteness remains the arithmetic step.

$$J(\sigma) < \infty \xrightarrow{\text{Cauchy–Schwarz}} T(s) \text{ holomorphic} \downarrow \zeta(s) \neq 0 \text{ for } \Re s > \sigma$$

FIGURE 11. The proved implication from weighted energy to a zero-free half-plane. The hypothesis in the left box remains an arithmetic requirement for each $\sigma > 1/2$.

[12]This real pole is present even when RH holds. It gives the unavoidable boundary cost of the energy criterion.

## Appendix A. From a fixed zero-free half-plane to summatory cancellation

We give the reverse implication used in Theorem 7.1. Suppose $1/2 \le \theta < 1$ and $\zeta$ has no zeros in $\operatorname{Re} s > \theta$. We use the classical Euler product, the meromorphic continuation and the standard truncated Perron formula as in [26].

**Lemma A.1** (Subpower growth of the reciprocal)**.** *For $0 < \delta < 1/4$ and $\eta > 0$,*

$$\left|\frac{1}{\zeta(\sigma+it)}\right| \ll_{\theta,\delta,\eta} (1+|t|)^{\eta} \qquad (\theta+\delta \le \sigma \le 2),$$

*where $1/\zeta$ is extended by zero at $s = 1$.*

*Proof.* Take $|t| \ge 4$ and use the disk centered at $2 + it$ of radius $R = 2 - \theta - \delta/4$. The disk lies in the zero-free half-plane and misses the pole. Choose a holomorphic branch $G = \log \zeta$ there, normalized at the center by the absolutely convergent Euler product. On the subdisk of radius $r_0 = 1/2$, that product bounds $|G|$ uniformly.

For $\operatorname{Re} s > 0$, away from the pole, the identity

$$\zeta(s) = \frac{s}{s-1} - s\int_1^{\infty} \{x\} x^{-s-1}\,\mathrm{d}x$$

follows from partial summation for $\operatorname{Re} s > 1$ and then continuation. It gives a polynomial bound on $|\zeta|$ on the closed disk of radius $R$, since its real parts are at least $\theta+\delta/4$. Thus $\max \operatorname{Re} G \ll \log(2+|t|)$ there. Borel–Carathéodory with $r_2 = 2-\theta-\delta/2 < R$ gives $\max_{|s-(2+it)|\le r_2} |G(s)| \ll_{\theta,\delta} \log(2+|t|)$. The branch is holomorphic on a neighborhood of that disk because the entire closed disk of radius $R$ stays away from zeros and the pole.

The three-circles theorem at $r_1 = 2-\theta-\delta$, between $r_0$ and $r_2$, bounds $|G|$ by $C(\log(2+|t|))^{\tau}$ for some fixed $\tau < 1$ on the disk of radius $r_1$. This covers all points $\sigma + it$ in the stated range. Exponentiating $-\operatorname{Re} G$ gives a bound by $(1+|t|)^{\eta}$ up to a constant. For bounded $t$, use compactness and the removable extension of the reciprocal at the pole. □

**Proposition A.2** (Perron calculation)**.** *Under the zero-free hypothesis of Lemma A.1, $L(x) \ll_{\theta,\varepsilon} x^{\theta+\varepsilon}$ for every $\varepsilon > 0$.*

*Proof.* Take $x \ge 3$ a half-integer, $c = 1 + 1/\log x$, $T = x^2$, and $\sigma_0 = \theta + \delta$ with $0 < \delta < \min(1/4, 1-\theta, \varepsilon/2)$. Write $F(s) = \zeta(2s)/\zeta(s)$. In the strip $\sigma_0 \le \operatorname{Re} s \le c$, $F$ is holomorphic and satisfies $F(\sigma+it) \ll_{\theta,\delta,\eta} (1+|t|)^{\eta}$, since $\zeta(2s)$ is bounded there by an absolutely convergent series and Lemma A.1 bounds the reciprocal. The singularity at $s = 1$ in the denominator gives a removable zero in $F$.

Truncated Perron gives

$$L(x) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} F(s)\frac{x^s}{s}\,\mathrm{d}s + O\left(\sum_{n\ge 1}(x/n)^c \min\{1, (T|\log(x/n)|)^{-1}\}\right).$$

Here $|\lambda(n)| = 1$. On $x/2 < n < 2x$, use $|\log(x/n)| \gg |x-n|/x$, and sum the reciprocal distances from the half-integer $x$. Outside that range, use $|\log(x/n)| \geq \log 2$ and $\zeta(c) \ll \log x$. The error is $O(x \log x/T)$. Move the integral to $\sigma_0$ in the rectangle of height $T$. The two horizontal sides are $O_{\theta,\delta,\eta}(x^c T^{\eta-1})$. The new vertical side is $O_{\theta,\delta,\eta}(x^{\sigma_0}(1+T^{\eta}))$, by integrating $(1+|t|)^{\eta-1}$. Choose $\eta > 0$ with $2\eta < \min(\varepsilon - \delta, 1/2)$. All terms satisfy the required estimate. Every integer prefix is attained at a half-integer $x$, and extending the bound to all real $x \geq 1$ changes only the constant. □

## Appendix B. The ordinary-dilate model has a different limit

For clarity about the basis in [5], define

$$d_M^{\mathrm{ord}} = \inf_c \Big\| 1 - \sum_{k \leq M} c_k \{kx\} \Big\|_{L^2(0,1)}.$$

This is the ordinary integer-dilate problem. The exact formula below is independent of the location of zeta zeros.

**Proposition B.1.** *Let $J_2(d) = d^2 \prod_{p \mid d}(1-p^{-2})$ and $\varphi$ denote Euler's totient. Then*

$$(d_M^{\mathrm{ord}})^2 = \frac{1}{1+3S_M}, \quad S_M = \sum_{d \leq M} \frac{\varphi(d)^2}{J_2(d)}, \qquad S_M \sim M \prod_p \left(1 - \frac{2}{p(p+1)}\right).$$

*Proof.* The centered fractional part has nonzero Fourier coefficients $-1/(2\pi i \ell)$ at $\ell \in \mathbb{Z} \setminus \{0\}$. Parseval therefore gives

$$\int_0^1 \{jx\}\{kx\}\,\mathrm{d}x = \frac{1}{4} + \frac{\gcd(j,k)^2}{12jk}.$$

Indeed, after writing $j = dj_0, k = dk_0$ with $(j_0, k_0) = 1$, the matching nonzero frequencies are $jk_0 r = kj_0 r$. The corresponding coefficient products sum to $(j_0 k_0)^{-1} \sum_{r \neq 0} (4\pi^2 r^2)^{-1} = 1/(12 j_0 k_0)$. The constant Fourier coefficient contributes $1/4$.

Put $u_d = \sum_{d \mid k \leq M} c_k/k$ and $t = \sum_{k \leq M} c_k$. The map from the $c_k$ to the $u_d$ is invertible by descending triangular elimination. The divisor identities $\sum_{d \mid k} \varphi(d) = k$ and $\sum_{d \mid k} J_2(d) = k^2$ follow by multiplicativity and a telescoping sum at each prime power. Thus $t = \sum_d \varphi(d) u_d$, and the squared residual is

$$|1 - t/2|^2 + \frac{1}{12} \sum_{d \leq M} J_2(d) |u_d|^2.$$

For fixed $t$, weighted Cauchy–Schwarz minimizes the last sum to $|t|^2/S_M$ before the factor $1/12$, attained by $u_d = t\varphi(d)/(S_M J_2(d))$. The remaining quadratic in $t$ is minimized at $t = 6S_M/(1+3S_M)$ and has the displayed value.

Finally $a(d) = \varphi(d)^2/J_2(d)$ is multiplicative with $a(p^k) = (p-1)/(p+1)$ for $k \geq 1$. Thus $a = 1 * b$, where $b$ is supported on squarefree integers, $b(p) = -2/(p+1)$, and $b(1) = 1$. Since $\sum |b(d)|/d = \prod_p (1 + 2/[p(p+1)]) < \infty$,

dividing $\sum_{d\le M} b(d)\lfloor M/d\rfloor$ by $M$ and using dominated convergence gives the stated product. Every factor is positive and their deviations from 1 are summable. □

The actual Mellin transform of $\{kx\}$ on $(0,1)$ is

$$\int_0^1 \{kx\}x^{s-1}\,\mathrm{d}x = \frac{k}{s+1} - \frac{k-1}{s} + \frac{k^{-s}}{s}\sum_{r=1}^{k-1} r^s \quad (\operatorname{Re} s > 0).$$

This follows by writing $\lfloor kx\rfloor = \sum_{r=1}^{k-1} \mathbf{1}_{[r/k,1]}(x)$ almost everywhere and integrating. The transform cannot be replaced by a multiple of $k^{-s}\zeta(s)/s$. For example, at $k=1$ it is $1/(s+1)$. Thus the $M^{-1/2}$ distance scale proved above is about the ordinary dilates. The balanced reciprocal functions have the different Mellin image of Lemma 14.2.

## Appendix C. Mellin positivity and one-sided pole detection

The real pole at $s=1/2$ gives a negative bias in the smoothed Liouville sum. It is natural to test whether one can prove an eventual sign and thereby exclude right-hand zeros. This appendix proves the relevant implications, including the positivity principle, without assuming RH. An eventual sign is not proved here.

**Lemma C.1** (The positivity principle)**.** *Let $f:[0,\infty)\to[0,\infty)$ be locally integrable, and suppose its Laplace integral*

$$\mathcal{L}(s) = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$$

*converges for all sufficiently large real $s$. If its abscissa of convergence $a$ is finite, then $\mathcal{L}$ cannot extend holomorphically to a neighborhood of the real point $a$.*

*Proof.* Suppose there is such a continuation, holomorphic in $\operatorname{Re} s > a$ and in a disc of radius $r$ about $a$. Choose $b = a + r/4$. The disc about $b$ of radius $r/2$ lies in the union of those two domains. Differentiation in the convergent integral gives the Taylor coefficients of $\mathcal{L}(b-z)$:

$$\frac{1}{j!}\int_0^\infty t^j f(t)e^{-bt}\,\mathrm{d}t \ge 0.$$

The differentiations are justified by convergence at a real point strictly between $a$ and $b$. For a real $z\in(r/4, r/2)$, the Taylor series converges. By Tonelli its sum is

$$\int_0^\infty f(t)e^{-bt}\sum_{j\ge0}\frac{(zt)^j}{j!}\,\mathrm{d}t = \int_0^\infty f(t)e^{-(b-z)t}\,\mathrm{d}t.$$

But $b-z<a$, contradicting the definition of the abscissa. □

The same principle applies to a nonnegative function $A(X)$ on $[X_0,\infty)$ with Mellin integral $\int_{X_0}^\infty A(X)X^{-s}\,\mathrm{d}X/X$, by putting $X = X_0e^t$. Changing $X_0$ only changes the meromorphic continuation by an entire function.

**Theorem C.2** (Both signs forced by a right-hand zero)**.** *Let $\rho = \beta + i\gamma$ be a nontrivial zeta zero of order $r$ with $\beta > 1/2$. Define*

$$C_\rho = \frac{r|\Gamma(\rho)\zeta(2\rho)|}{|\zeta^{(r)}(\rho)|} > 0.$$

*Then*

$$\limsup_{X\to\infty} \frac{B(X)}{X^\beta(\log X)^{r-1}} \geq C_\rho,$$
$$\liminf_{X\to\infty} \frac{B(X)}{X^\beta(\log X)^{r-1}} \leq -C_\rho.$$

*Proof.* Put $\Phi(s) = \Gamma(s)\zeta(2s)/\zeta(s)$. It is the Mellin transform in (14.1) for $\operatorname{Re} s > 1$. It has no singularity at any real point greater than $1/2$. For $1/2 < s < 1$, this follows from $\zeta(s) = \eta(s)/(1 - 2^{1-s}) < 0$, where the alternating series $\eta(s) > 0$; at one the pole of $\zeta$ gives a removable zero in the quotient. At $\rho$ the leading Laurent coefficient is

$$A_\rho = \frac{r!\Gamma(\rho)\zeta(2\rho)}{\zeta^{(r)}(\rho)}.$$

The numerator is nonzero since $\operatorname{Re}(2\rho) > 1$.

Suppose the first inequality fails. Choose $0 < c < C_\rho$ such that

$$A(X) = cX^\beta(\log X)^{r-1} - B(X) \geq 0$$

for every $X \geq X_0$, with $X_0 > 1$. Its Mellin transform converges for real $s > 1$ and continues meromorphically as

$$\frac{c(r-1)!}{(s-\beta)^r} - \Phi(s) + E(s),$$

where $E$ is entire, accounting for the finite initial interval. The real pole at $\beta$ forces the abscissa of the positive integral to be at least $\beta$. If that abscissa were greater than $\beta$, the displayed continuation would be holomorphic there, contradicting Lemma C.1. Hence it is exactly $\beta$.

Positivity now gives, for $\delta > 0$,

$$\left|\int_{X_0}^\infty A(X)X^{-\beta-\delta-i\gamma}\frac{\mathrm{d}X}{X}\right| \leq \int_{X_0}^\infty A(X)X^{-\beta-\delta}\frac{\mathrm{d}X}{X}.$$

Multiply by $\delta^r$ and let $\delta \downarrow 0$. The complex pole on the left gives $|A_\rho|$; the real pole on the right gives $c(r-1)!$. Therefore $c \geq |A_\rho|/(r-1)! = C_\rho$, a contradiction. For the second inequality apply the identical argument to $A(X) = cX^\beta(\log X)^{r-1} + B(X)$. □

This strengthens the absolute-value pole test of Proposition 14.10 to both one-sided oscillations. It still gives lower bounds, not the square-root upper bound needed to exclude $\rho$.

**Proposition C.3** (The real pole and an eventual sign)**.** *Put*

$$c_0 = -\frac{\sqrt{\pi}}{2\zeta(1/2)} > 0.$$

*Unconditionally,*

$$\liminf_{X\to\infty} \frac{B(X)}{\sqrt{X}} \le -c_0.$$

*If in addition $B(X) \le 0$ for all sufficiently large $X$, then RH holds, every nontrivial zero is simple, and each such zero satisfies*

$$\left|\frac{\Gamma(\rho)\zeta(2\rho)}{\zeta'(\rho)}\right| \le c_0.$$

*Proof.* The function $\Phi$ has a simple real pole at $1/2$ with residue $-c_0$. Suppose the asserted limit inferior were greater than $-c_0$. Choose $0 < c < c_0$ such that $B(X) + c\sqrt{X} \ge 0$ eventually. The Mellin transform of this nonnegative function has continuation

$$\Phi(s) + \frac{c}{s - 1/2} + E(s).$$

Its pole at $1/2$ has negative residue $c - c_0$. The positivity principle and absence of real singularities to the right force its abscissa to be $1/2$. The integral is nonnegative on the real axis there, but the continuation tends to $-\infty$ as $s \downarrow 1/2$, a contradiction.

Now assume $B \le 0$ eventually, and apply the positivity principle to $-B$. Its Mellin continuation is $-\Phi + E$, with positive residue $c_0$ at $1/2$. Its abscissa is exactly $1/2$, by the same real-axis argument. Thus $\Phi$ is holomorphic in $\operatorname{Re} s > 1/2$, which excludes all right-hand zeros. The functional equation gives RH.

At a critical zero of order $r$, the numerator of $\Phi$ is nonzero by nonvanishing of $\zeta$ on the line $\operatorname{Re} s = 1$. Positivity bounds the complex Mellin integral at $s = \rho + \delta$ by its real value at $s = 1/2 + \delta$, the latter being $c_0/\delta + O(1)$. The complex pole has order $r$, so $r \le 1$, and comparison of residues gives the displayed inequality. □

An eventual-sign argument would therefore need more than the square-root estimate with an arbitrary $X^\varepsilon$ loss: it would also establish simplicity. Neither the sign assertion nor the required upper bound for $B$ has been proved. The negative-bias lower bound alone does not control positive excursions or rule out right-hand zeros.

**Proposition C.4** (The real-pole normalization of local energy)**.** *With $c_0$ as above, unconditionally*

$$\limsup_{X\to\infty} \frac{\mathcal{H}_B(X)}{X} \ge \frac{3}{2}c_0^2 = \frac{3\pi}{8\zeta(1/2)^2}.$$

*Proof.* There is nothing to prove if the limit superior is infinite. Otherwise choose $C$ greater than it, so that $\mathcal{H}_B(X) \le CX$ for all sufficiently large $X$. For $\eta > 0$, Tonelli and the substitution $Y = tX$ give

$$\int_0^\infty \mathcal{H}_B(X) X^{-1-\eta} \frac{\mathrm{d}X}{X} = r(1+\eta) P(\eta),$$

where

$$r(s) = \int_{1/2}^{2} t^s \frac{\mathrm{d}t}{t} = \frac{2^s - 2^{-s}}{s}, \qquad P(\eta) = \int_0^\infty |B(Y)|^2 Y^{-1-\eta} \frac{\mathrm{d}Y}{Y}.$$

The small-$X$ part is uniformly finite near $\eta = 0$, by exponential decay. The assumed large-$X$ upper bound gives

$$P(\eta) \le \frac{C/\eta + O(1)}{r(1+\eta)}.$$

For $\delta > 0$, Cauchy–Schwarz on $[1, \infty)$ now gives

$$\left| \int_1^\infty B(Y) Y^{-1/2-\delta} \frac{\mathrm{d}Y}{Y} \right|^2 \le \frac{P(\delta)}{\delta}.$$

The absolute integral is convergent by the same estimate. It agrees with $\Phi(1/2+\delta)$ up to the entire small-$Y$ term, by continuation from $\operatorname{Re} s > 1$. Multiplying by $\delta^2$ and letting $\delta \downarrow 0$ gives $c_0^2 \le C/r(1) = 2C/3$. Let $C$ decrease to the limit superior. □

For the comparison function $B_0(X) = -c_0\sqrt{X}$, the local energy is exactly $(3/2)c_0^2 X$. Thus the constant in this pole comparison is the one dictated by the real residue. This proposition supplies a lower constraint on any proposed upper estimate; it does not establish such an estimate for the actual sum.

## Appendix D. Elementary prime bounds and exact checks

Here are details of the prime estimate used in Theorem 17.7. If $\vartheta(t) = \sum_{p \le t} \log p$, then $\prod_{n < p \le 2n} p$ divides $\binom{2n}{n} \le 4^n$. Summing over dyadic intervals gives $\vartheta(t) \le 4(\log 2)t$ for $t \ge 2$. Separating primes at $\sqrt{t}$ gives

$$\pi(t) \le \sqrt{t} + \frac{2\vartheta(t)}{\log t} \le 7\,\frac{t}{\log t}.$$

The last inequality uses $\log t \le \sqrt{t}$. Partial summation gives

$$\sum_{p \le y} p^{-1/2} = \frac{\pi(y)}{\sqrt{y}} + \frac{1}{2}\int_2^y \frac{\pi(t)}{t^{3/2}}\,\mathrm{d}t \ll \frac{\sqrt{y}}{\log y}.$$

For instance split the integral at $\sqrt{y}$; the first part is $O(y^{1/4})$ and on the second part $\log t \ge \frac{1}{2}\log y$. This proves the asserted order without the prime number theorem or any zero-free input. Similarly $\sum_{p \le y} (\log p)/p \ll \log y$, by partial summation of $\vartheta$, proving the derivative bound used in (17.18).

The new checker `checks/attack_exact.py` uses rational arithmetic for the following independent finite tests: the Legendre reproducing kernel on

polynomial monomials, its evaluation norm bound, the positive coefficients of the diagonal error, both finite-prime convolution identities, the coprime-pair decomposition, and the prime-dilation Gram identity. It also checks an exponential interval certificate for $B(1) > 0$ and $B(3) < -1/20$. The latter rules out assigning one sign to every scale product in (17.8); it is not a test of an eventual sign.

The checker proves none of the infinite arithmetic estimates. The analytic arguments are supplied in the text and have not been formalized in Lean. The original finite Gram formalization retains exactly the scope stated in the reproduction appendix. No new RH proof, zero-free half-plane or novelty priority is asserted in this revision.

## Appendix E. Finite Abel cancellations do not control the exponential sum

Equation (17.18) proves a double zero in an Abel weight at $s = 1$. It does not prove the signed exponential-kernel estimate (17.16). We test the proposed implication in the larger class of bounded real coefficients. The test below is explicit and uses neither RH nor a hypothetical zero of the Riemann zeta function.

**Theorem E.1** (A prescribed finite set of vanishing moments)**.** *Let $k \geq 1$ be an integer, $1/2 < \beta < 1$, and $\gamma > 0$. There is a real sequence $(a_n)$, with $|a_n| \leq 1$, such that:*

(1) *Its Dirichlet series $D_a(s) = \sum_{n\geq 1} a_n n^{-s}$ is absolutely convergent on $\operatorname{Re} s > \beta$, and*

$$D_a^{(j)}(1) = \sum_{n\geq 1} \frac{a_n(-\log n)^j}{n} = 0 \qquad (0 \leq j < k).$$

(2) *It has a meromorphic continuation with simple poles at $\beta + i\gamma$ and $\beta - i\gamma$. The Mellin transform $\Gamma(s)D_a(s)$ still has those poles and has a zero of order at least $k$ at $s = 1$.*

(3) *Its exponentially smoothed sum satisfies*

$$S_a(X) := \sum_{n\geq 1} a_n e^{-n/X} = \varepsilon \operatorname{Re}(\Gamma(\beta + i\gamma)X^{\beta+i\gamma}) + O_{k,\beta,\gamma}(1)$$

*for some explicitly specified $\varepsilon > 0$. Consequently*

$$\limsup_{X\to\infty} \frac{S_a(X)}{X^\beta} = \varepsilon|\Gamma(\beta + i\gamma)|, \qquad \liminf_{X\to\infty} \frac{S_a(X)}{X^\beta} = -\varepsilon|\Gamma(\beta + i\gamma)|.$$

*In particular, these $k$ exact moment cancellations do not imply $S_a(X) = O(X^{1/2+\eta})$ when $0 < \eta < \beta - 1/2$.*

*Proof.* Put $z = \beta + i\gamma$ and $r_n = n^{\beta-1}\cos(\gamma \log n)$. Initially retain this coefficient except on the finite set

$$\mathcal{S} = \{2^m : 0 \leq m < k\}.$$

For $0 \le j < k$ define the absolutely convergent real moments

$$t_j = \sum_{n \notin \mathcal{S}} \frac{r_n (-\log n)^j}{n}.$$

Convergence follows from $|r_n|(\log n)^j/n \le n^{\beta-2}(\log n)^j$. We choose real numbers $b_m$ at the indices $2^m$ by

$$\sum_{m=0}^{k-1} \frac{b_m(-m\log 2)^j}{2^m} = -t_j \qquad (0 \le j < k). \tag{E.1}$$

The matrix is invertible: after nonzero row and column scalings it is the Vandermonde matrix $(m^j)_{j,m=0}^{k-1}$. Its determinant is $\prod_{0\le m<n<k}(n-m) \neq 0$.

There is also a direct finite formula for the solution. Set

$$\ell_m(x) = \prod_{\substack{0\le n<k \\ n\neq m}} \frac{x-n}{m-n} = \sum_{j=0}^{k-1} \ell_{m,j} x^j, \qquad s_j = -\frac{t_j}{(-\log 2)^j}.$$

Then

$$b_m = 2^m \sum_{j=0}^{k-1} \ell_{m,j} s_j.$$

Indeed, applying the moment functional $x^j \mapsto s_j$ to $\ell_m$ inverts the moment system. All the numbers $\ell_{m,j}$ are rational.

Choose

$$\varepsilon = \frac{1}{1 + \max_{0\le m<k} |b_m|} > 0, \qquad a_n = \begin{cases} \varepsilon b_m, & n = 2^m \in \mathcal{S}, \\ \varepsilon r_n, & n \notin \mathcal{S}. \end{cases}$$

Since $|r_n| \le 1$, all coefficients have modulus at most one. The moment equations prove the first assertion, including termwise differentiation at one.

For complex $s$, absolute convergence initially gives the holomorphic identity

$$D_a(s) = \frac{\varepsilon}{2}\left(\zeta(s+1-\beta-i\gamma) + \zeta(s+1-\beta+i\gamma)\right) + \varepsilon \sum_{m=0}^{k-1} (b_m - r_{2^m}) 2^{-ms}. \tag{E.2}$$

The finite sum is entire. The two shifted zeta functions have distinct simple poles at the asserted points, each with residue $\varepsilon/2$. They are regular at one. The moment equations give a zero there of order at least $k$. Multiplication by $\Gamma(s)$, which is nonzero and holomorphic at these points, preserves both assertions. [13]

It remains to justify the smoothed asymptotic without any unproved contour shift. For $X \ge 1$, apply elementary sum–integral comparison to

[13] For complex $s$, the average of the two shifted zeta functions in (E.2) must be used. Writing its real part instead would destroy holomorphy. The displayed poles are constructed poles, not assertions that $\zeta(\beta \pm i\gamma) = 0$.

$f_X(t) = t^{z-1}e^{-t/X}$ on $[1,\infty)$. The integral of its absolute derivative is at most

$$|z-1|\int_1^\infty t^{\beta-2}\,\mathrm{d}t + X^{-1}\int_1^\infty t^{\beta-1}e^{-t/X}\,\mathrm{d}t \le \frac{|z-1|}{1-\beta} + \Gamma(\beta).$$

The omitted integral on $(0,1)$ has modulus at most $1/\beta$. Comparing on each interval $[n,n+1]$ therefore gives the uniform estimate

$$\sum_{n\ge1} n^{z-1}e^{-n/X} = \int_0^\infty t^{z-1}e^{-t/X}\,\mathrm{d}t + O_{\beta,\gamma}(1) = \Gamma(z)X^z + O_{\beta,\gamma}(1).$$

The last identity is Euler's convergent gamma integral [31, §5.9(i)], here applied with $\operatorname{Re} z > 0$. The finite coefficient changes add only $O_{k,\beta,\gamma}(1)$. Taking real parts proves the asymptotic.

Finally choose $\log X_j = (2\pi j - \arg\Gamma(z))/\gamma$ and the analogous sequence with an added $\pi/\gamma$. The two limiting normalized values have opposite signs and the stated modulus. The uniform asymptotic bounds the absolute normalized value by that same modulus, proving the exact limit superior and inferior. □

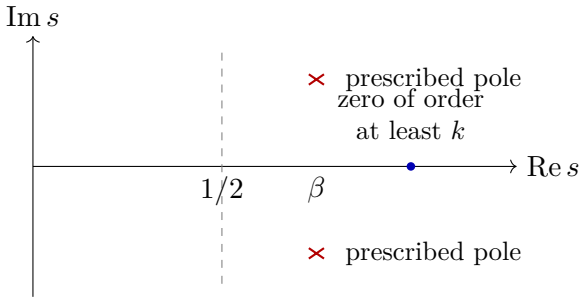



FIGURE 12. A model Dirichlet series in Theorem E.1. Any fixed number of vanishing moments at one can coexist with the two prescribed poles. These are not claimed zeros of the Riemann zeta function.

The construction can also start with $r_n = n^{\beta-1}\cos(\gamma\log n - \arg\Gamma(\beta+i\gamma))$. After the same finite adjustment its phase-aligned leading term is $\varepsilon|\Gamma(\beta+i\gamma)|X^\beta\cos(\gamma\log X)$. Figure 13 plots that leading term after normalization, for $\beta = 3/4$ and $\gamma = 1$.

Theorem E.1 does not rule out a proof that uses the full arithmetic structure of $\lambda$. Its coefficients are not required to be multiplicative. It rules out a generic inference from bounded coefficients and finitely many exact logarithmic moment cancellations to the desired exponential-sum bound. [14] The missing arithmetic estimate in (17.16) remains the point at which the

[14]The coefficients depend on $k$, and the factor $\varepsilon$ need not have a positive lower bound as $k\to\infty$. The theorem constructs no single sequence with every moment vanishing.

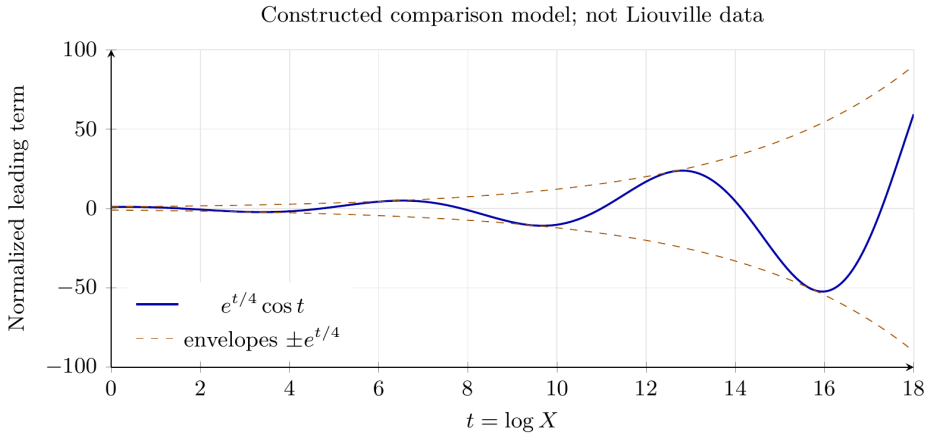



FIGURE 13. The phase-aligned model leading term, divided by $\varepsilon|\Gamma(3/4+i)|\sqrt{X}$, is $e^{t/4}\cos t$, where $t = \log X$. The dashed curves are its envelopes. This is an analytic comparison model, not Liouville data or a numerical test of RH.

current RH argument stops. Adding the proved cancellations at $s = 1$ does not fill that step.

## APPENDIX F. CONTINUITY WITH VERSION 1 AND THE ROLE OF THE LITERATURE

This revision contains the mathematical arguments of the original preprint [10] within the expanded exposition. The square identity is Lemma 2.1; its exact finite-dimensional duality statement uses the corrected extremizer $\beta_h = \overline{C_h}/\|C\|_2$. Proposition 2.2 retains the original terminal-$\eta$ criterion. The maximal argument appears in Theorem 3.1 on a common support, with its $(K+1)^2$ cost displayed before absorbing logarithms into an arbitrarily small power. The padding covers the entire support: for coefficients on $(N, 2N]$, use $2N \le 2^K < 4N$. No assertion about arbitrary translated intervals follows from a spectral bound tied to the size of the indices.

Proposition 2.4 strengthens the original coefficient-uniform failure theorem to an exact maximum. Corollary 2.5 and Figure 1 retain its particular 9/4, 15/8 comparison. Lemma 2.6 retains the square-divisor decomposition and displays every correlation stratum. The original conditional RH conclusion is included, and strengthened to a conditional Dirichlet-GRH implication, in Theorem 2.3. The elementary RH criterion and its reverse direction are given in Theorem 7.1 and Appendix A. These are implications, not proofs of their correlation hypotheses.

F.1. **The analytic context retained from the original paper.** The elementary identities sit within the standard analytic number theory of [33, 34, 26]. The spectral input in Section 3 belongs to the large-sieve

framework of [11, 35, 36, 37, 38]. The present maximal argument is an elementary consequence of a fixed-support Bessel inequality; it does not supply a new spectral input. Work on exceptional eigenvalues [50] addresses a different refinement.

The role of signs is also central in Möbius disjointness [41, 42] and in the correlation theorems of [30, 20, 43, 25, 44, 45, 46]. The related settings of Bohr sets, partial sums and Chowla variants are represented by [47, 48, 49]. These references are retained for context. None is taken here as a proof of the maximal collar or of the square-root prime-tail estimate. Classical zero results [39], finite-height verification [9], and the disproof of the Mertens conjecture [40] likewise have their own scopes. The character-sum viewpoint is represented by [51]; the present character criteria do not import an unstated uniform bound from that theory.

The recent Hexagon paper [32] studies weighted-prime margin criteria, reports finite verification, and analyzes the obstruction to inferring a global bound from averaged information. Its discussion of moving scales and exceptional sets is pertinent to our kernel comparison. We use it as related work, not as an input to any theorem here. In particular, finite verification does not establish the bound at every $X$, and a criterion equivalent to RH does not establish its required inequality.

F.2. **Funding and availability.** This research received no external funding. No empirical dataset is used. The full research package contains the supplied v1 TeX source and its original supplemental scripts under `source-history/v1/`, together with their checksums and a section-by-section content map. The original exponent check and optional plotting script are preserved without alteration. The latter requires NumPy and Matplotlib; it is not needed to compile the manuscript, whose comparison figure is native TikZ/PGFPlots. The new exact checks use Python's standard library.

F.3. **Ethics and interests.** The work is theoretical mathematics and involves no human participants, animal subjects or clinical data requiring ethical approval. The authors declare no conflicts of interest. The computational and AI-assistance statement, including the limited scope of the Lean verification, follows below.

## Appendix G. Exact normalization and rounding controls

The finite-window normalization in the new construction agrees with the signed energy of Section 6. The verification script `anc/check_exact.py` tests the prefix and suffix identity for all 2046 sign strings of lengths one through ten. The infinite construction is controlled by the exact discrepancy identity

$$\sum_{y_0<p\le t} (I_p - q_p) = \lfloor S(t) \rfloor - S(t) \in (-1, 0].$$

Summation by parts, rather than a finite search, gives the holomorphic centered prime series. The sharp leading-coefficient statement uses the Euler-product inequality for every prime where a sign is changed. Finite prime agreement alone does not impose that coefficient.

## Appendix H. Reproduction and provenance

The accompanying exact-integer checks test the finite autocorrelation and window formulas, dyadic partitions, the Rudin–Shapiro construction, the finite-prime examples and the independent-sign variance. These checks complement the analytic proofs. The original preprint is cited as [10]. Version 2 retains its authorship, corrects the endpoint conventions and gives proofs and obstructions for the stronger statements above. Computational and AI-assisted tools were used to develop examples, check finite identities and revise the exposition. This research revision records conditional criteria separately from proved estimates; it does not prove RH or Dirichlet GRH.
AI use declaration. AI tools assisted the development and checking of the quantitative Euler-factor and Abel-jet arguments in Section 12, the primitive-correlation identities in Section 17.3, the dilation-norm arguments in Section 17.5, and the conductor calculations in Section 16.2. They also assisted the finite scripts and exposition. The arguments are written out in the paper; the scripts check only their stated finite identities and constants. Separate AI-agent reviews are computational assistance, not external human refereeing. These additions have not been Lean-formalized. Responsibility for the submitted manuscript and its references remains with the human authors.

The companion file `lean/GramCore.lean` checks the finite matrix inequality in the deterministic obstruction with Lean and Mathlib 4.34.1. It constructs averaged Gram matrices from finite sign vectors and proves the trace and self-pairing bounds. Given the stated cross-pairing bound $16B$, it derives $m \le 32B$ for $2m$ matrices in dimension $m$. The transfer from prefix correlations to the cross-pairing bound, scale counting, and the analytic arguments remain outside this formalization. The full research package includes the toolchain, Mathlib revision and axiom report. The TeX-only archive links to these files in the repository; it does not contain Lean sources.

## References

[1] L. Báez-Duarte, *A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis*, Rend. Lincei Mat. Appl. **14** (2003), 5–11, paper.
[2] J.-F. Burnol, *A lower bound in an approximation problem involving the zeros of the Riemann zeta function*, Adv. Math. **170** (2002), 56–70, doi:10.1006/aima.2001.2066.
[3] S. Bettin, J. B. Conrey and D. W. Farmer, *An optimal choice of Dirichlet polynomials for the Nyman–Beurling criterion*, arXiv:1211.5191.
[4] R. P. Brent, D. J. Platt and T. S. Trudgian, *The mean square of the error term in the prime number theorem*, J. Number Theory **238** (2022), 740–762, doi:10.1016/j.jnt.2021.09.016.

[5] D. Bhattacharjee, P. Nandi, O. Frederick and P. Samal, *Spectral and Analytic Structure of the Nyman–Beurling–Báez–Duarte Approximation*, Preprints.org, version 2, 10 March 2026, doi:10.20944/preprints202506.0772.v2.
[6] D. Bhattacharjee and U. Bhattacharya, *The Pólya–Vinogradov inequality with* $\log\log q$ *for every character*, Hexagon 2610.00256v3, 26 October 2026.
[7] J. Li, Y.-C. Sun and H. Wang, *On the lower bound for the* $L^1$ *norm of Liouville exponential sums. I*, arXiv:2611.03589v1, 2 November 2026.
[8] Claude, *More than two thirds of the zeta zeros are simple and on the critical line*, research manuscript, August 2026. Paper and formalization linked from `https://www.anthropic.com/research/riemann-zeta`.
[9] D. J. Platt and T. S. Trudgian, *The Riemann hypothesis is true up to* $3 \cdot 10^{12}$, Bull. Lond. Math. Soc. **53** (2021), 792–797, doi:10.1112/blms.12460.
[10] D. Bhattacharjee and U. Bhattacharya, *Maximal Large Sieve and Coefficient Uniform Failure for Liouville Correlations*, Preprints.org, 18 June 2026. doi:10.20944/preprints202606.1350.v1.
[11] J.-M. Deshouillers and H. Iwaniec, *Kloosterman sums and Fourier coefficients of cusp forms*, Invent. Math. **70** (1982), 219–288, doi:10.1007/BF01390728.
[12] L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, arXiv:2405.20552v2, 2026.
[13] G. A. Karagulyan, M. T. Lacey and Kh. V. Navoyan, *On maximal Calderón–Zygmund operators and Weyl multipliers*, Sb. Math. **216** (2025), 1375–1392, doi:10.4213/sm10277e.
[14] L. Báez-Duarte, *Arithmetical aspects of Beurling's real variable reformulation of the Riemann hypothesis*, arXiv:math/0011254.
[15] D. J. Katz and C. M. van der Linden, *Peak sidelobe level and peak crosscorrelation of Golay–Rudin–Shapiro sequences*, arXiv:2108.07318.
[16] Y. Lamzouri, *A new proof that more than* $2/3$ *of the zeros of the Riemann zeta function are simple and on the critical line*, arXiv:2609.02882v2, 2026.
[17] B. Liu, *Improved zero-free half-planes for the quasi-Riemann hypothesis*, arXiv:2610.12234v2, 12 October 2026.
[18] B. Liu, *Zero-free half-planes for Hecke L-functions over number fields and Quasi-Riemann Hypothesis*, arXiv:2610.26835v2, 29 October 2026.
[19] B. Liu, *An improved zero-free half-plane for Hecke L-functions over number fields and Quasi-Riemann Hypothesis*, Hexagon 2610.00655v3, version 3 PDF, 2026.
[20] K. Matomäki, M. Radziwiłł and T. Tao, *An averaged form of Chowla's conjecture*, Algebra Number Theory **9** (2015), 2167–2196, arXiv:1503.05121.
[21] H. L. Montgomery and R. C. Vaughan, *The large sieve*, Mathematika **20** (1973), 119–134, doi:10.1112/S0025579300004708.
[22] OpenAI, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\operatorname{Re} s > 7/8$, research manuscript, 30 September 2026, paper.
[23] OpenAI, *The quasi-Riemann hypothesis*, formalization scope for family 003, Lean statement map.
[24] OpenAI, *Ordinary two-point correlations of multiplicative functions*, research manuscript, 24 September 2026, paper.
[25] T. Tao, *The logarithmically averaged Chowla and Elliott conjectures for two-point correlations*, Forum Math. Pi **4** (2016), e8, doi:10.1017/fmp.2016.6.
[26] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, second ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986.
[27] NIST, *Digital Library of Mathematical Functions*, §18.2, General orthogonal polynomials, `https://dlmf.nist.gov/18.2`.
[28] NIST, *Digital Library of Mathematical Functions*, §1.14, Integral transforms, `https://dlmf.nist.gov/1.14`.

[29] NIST, *Digital Library of Mathematical Functions*, §27.12, Asymptotic formulas: primes, https://dlmf.nist.gov/27.12.
[30] K. Matomäki and M. Radziwiłł, *Multiplicative functions in short intervals*, Ann. of Math. **183** (2016), 1015–1056, doi:10.4007/annals.2016.183.3.6.
[31] NIST, *Digital Library of Mathematical Functions*, §5.9, Integral representations of the gamma function, https://dlmf.nist.gov/5.9.
[32] P. Caceres, *The Riemann hypothesis as a margin: a two-sided criterion verified to* $2 \cdot 10^{29}$*, the fourth-moment anatomy of Mobius cancellation, and one constant seen from every side*, Hexagon 2610.00688v2, 2026. Research preprint.
[33] T. M. Apostol, *Introduction to Analytic Number Theory*, Undergraduate Texts in Mathematics, Springer-Verlag, New York, 1976. doi:10.1007/978-1-4757-5579-4.
[34] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications **53**, American Mathematical Society, Providence, RI, 2004. doi:10.1090/coll/053.
[35] H. Iwaniec, *Spectral Methods of Automorphic Forms*, 2nd ed., Graduate Studies in Mathematics **53**, American Mathematical Society/Revista Matemática Iberoamericana, 2002. doi:10.1090/gsm/053.
[36] J. B. Friedlander and H. Iwaniec, *Opera de Cribro*, American Mathematical Society Colloquium Publications **57**, American Mathematical Society, Providence, RI, 2010. doi:10.1090/coll/057.
[37] E. Bombieri, *Le grand crible dans la théorie analytique des nombres*, Astérisque **18**, Société Mathématique de France, 2nd ed., 1987.
[38] H. L. Montgomery, *Ten Lectures on the Interface Between Analytic Number Theory and Harmonic Analysis*, CBMS Regional Conference Series in Mathematics **84**, American Mathematical Society, 1994. doi:10.1090/cbms/084.
[39] A. Selberg, *On the zeros of Riemann's zeta-function*, Skr. Norske Vid. Akad. Oslo I **10** (1942), 1–59.
[40] A. M. Odlyzko and H. J. J. te Riele, *Disproof of the Mertens conjecture*, J. Reine Angew. Math. **357** (1985), 138–160.
[41] P. Sarnak, *Three Lectures on the Möbius Function, Randomness and Dynamics*, lecture notes, Institute for Advanced Study, Princeton, 2011.
[42] J. Bourgain, P. Sarnak and T. Ziegler, *Disjointness of Möbius from horocycle flows*, in *From Fourier Analysis and Number Theory to Radon Transforms and Geometry*, Developments in Mathematics **28**, Springer, New York, 2013, pp. 67–83. doi:10.1007/978-1-4614-4075-8_5.
[43] K. Matomäki, M. Radziwiłł and T. Tao, *Sign patterns of the Liouville and Möbius functions*, Forum Math. Sigma **4** (2016), e14. doi:10.1017/fms.2016.6.
[44] T. Tao and J. Teräväinen, *The structure of logarithmically averaged correlations of multiplicative functions, with applications to the Chowla and Elliott conjectures*, Duke Math. J. **168** (2019), 1977–2027. doi:10.1215/00127094-2019-0002.
[45] O. Klurman, *Correlations of multiplicative functions and applications*, Compos. Math. **153** (2017), 1622–1657. doi:10.1112/S0010437X17007163.
[46] J. Teräväinen, *On binary correlations of multiplicative functions*, Forum Math. Sigma **6** (2018), e10. doi:10.1017/fms.2018.10.
[47] J. Teräväinen and A. Walker, *On a Bohr set analogue of Chowla's conjecture*, Math. Z. **310** (2025), article 75, doi:10.1007/s00209-025-03770-2.
[48] G. Chavez, *Correlations of the Möbius and Liouville functions with their partial sums*, Int. J. Number Theory **22** (2026), 1875–1899, doi:10.1142/S1793042126500934. Preprint under the title *Correlations of multiplicative functions with their partial sums*, arXiv:2409.02106v10.
[49] K. Krishnamoorthy, *On variants of Chowla's conjecture*, Proc. Amer. Math. Soc. **154** (2026), no. 5, 1897–1910. doi:10.1090/proc/17544. Preprint: arXiv:2501.10962v2 (2025).

[50] A. Pascadi, *Large sieve inequalities for exceptional Maass forms and the greatest prime factor of $n^2+1$*, Forum Math. Pi **14** (2026), e8, 1–54, doi:10.1017/fmp.2026.10025.
[51] A. Granville and K. Soundararajan, *Large character sums: pretentious characters and the Pólya–Vinogradov theorem*, J. Amer. Math. Soc. **20** (2007), 357–384. doi:10.1090/S0894-0347-06-00536-4.
[52] M. Aymone, *On multiplicative functions that are small on average and zero-free regions for the Riemann zeta function*, Lithuanian Math. J. **62** (2022), 146–149. DOI.
[53] M. Aymone and V. Sidoravicius, *Partial sums of biased random multiplicative functions*, J. Number Theory **172** (2017), 343–382. DOI.
[54] H. Davenport, *On some infinite series involving arithmetical functions (II)*, Quart. J. Math. Oxford Ser. **8** (1937), 313–320. DOI.
[55] K. Matomäki, M. Radziwiłł and T. Tao, *An averaged form of Chowla's conjecture*, Algebra Number Theory **9** (2015), 2167–2196. DOI.

Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India

*Current address*: Formerly, University of Calcutta, Senate House, 87/1 College Street, Kolkata 700073, West Bengal, India

*Email address*: d.bhattacharjee@erl-forschung.de
*Email address*: itsdeep@live.com
*Email address*: egspl@proton.me

Formerly, National Taiwan University, Taipei 106319, Taiwan
*Email address*: rai.bhattacharya.in@gmail.com